

Final

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Problem 1

Summary

For the series $Y_t = X_t$ I chose a SARIMA(4,0,1)x(0,1,2)[12] model excluding the intercept term. The coefficients shown below are from a model fit on X_t after having been non seasonally differenced and seasonally differenced with lag 12.

Coefficients:

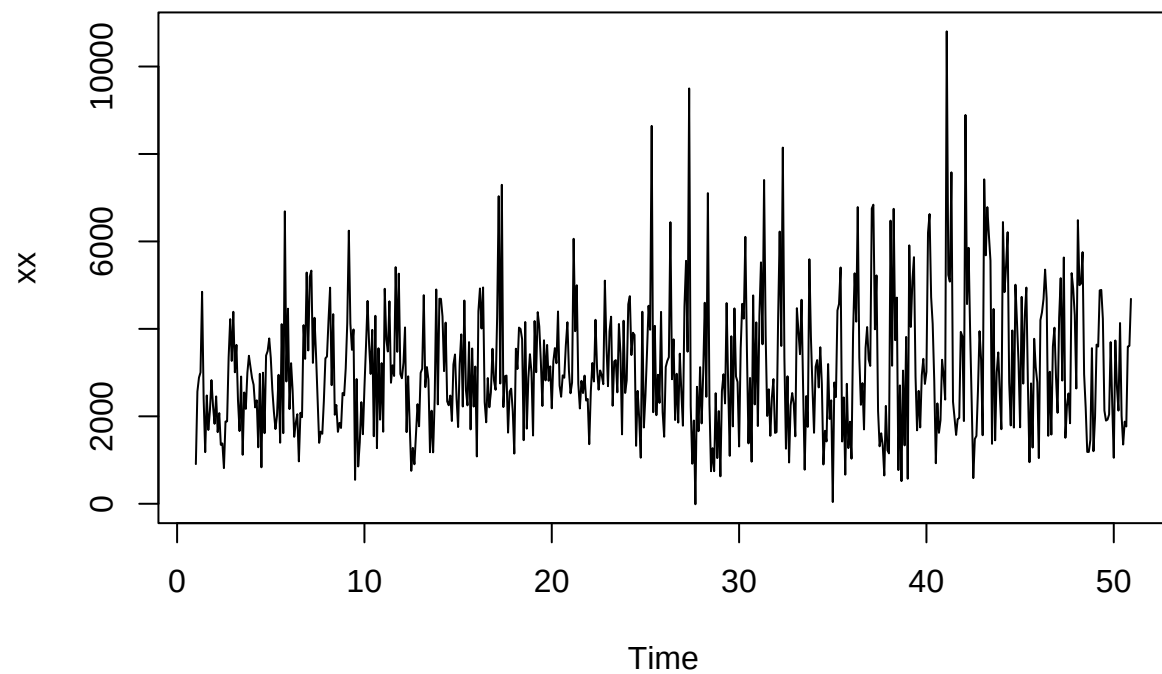
	Estimate	SE	t.value	p.value
ar1	0.2667	0.1446	1.8447	0.0656
ar2	0.6089	0.0459	13.2693	0.0000
ar3	-0.1343	0.0811	-1.6558	0.0983
ar4	-0.2486	0.0442	-5.6252	0.0000
ma1	-0.1801	0.1487	-1.2109	0.2264
sma1	-0.5948	0.0435	-13.6705	0.0000
sma2	-0.2039	0.0469	-4.3443	0.0000

Explanation

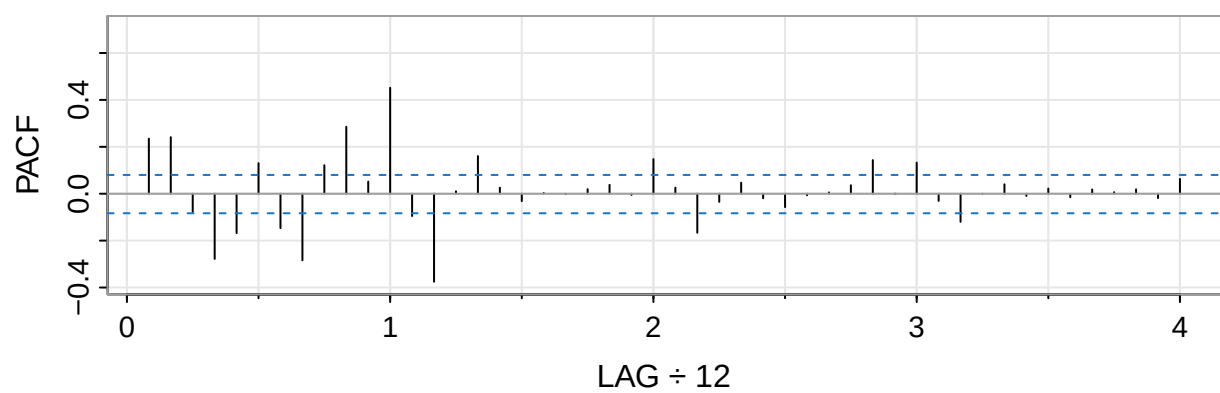
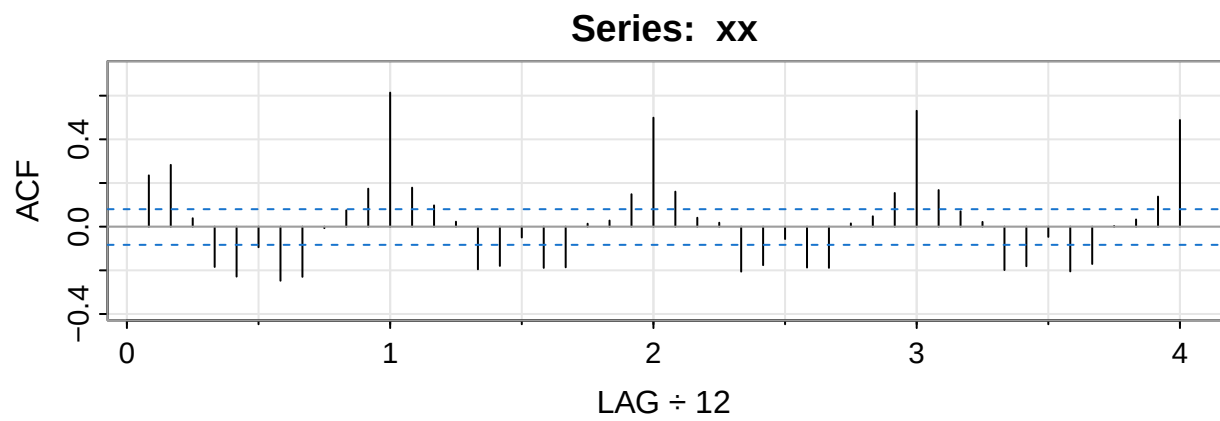
The initial plot and ACF and PACF of X_t suggest that the series is stationary so we don't need to take a non-seasonal difference but there is a seasonality of 12 so we should take a seasonal difference at lag 12. After the seasonal difference, the significant non seasonal lags of the ACF cut off for the most part after lag 2 and those of the PACF cut off after lag 4, while the significant seasonal lags of the ACF cut off for the most part after lag 2 and those of the PACF cut off after lag 2, so we fit different models on the first seasonally differenced data with a maximum p of 4, a maximum q of 2, a maximum P of 2 and a maximum Q of 2. Only the seven models with an AICc multiplied by the length of the series within 5 of the minimum and a BIC multiplied by the length of the series within 7 of the minimum were considered. Of these, the SARMA(4,1)x(0,2) was chosen because it's very close to the best AICc and it has the highest LBP values. So the final model on the original series X_t is SARIMA(4,0,1)x(0,1,2)[12]. The intercept was not included because it was not significant at any level. It's worth noting that there are some notably heavy tails in the Normal QQ plots of the residuals for all models.

Output

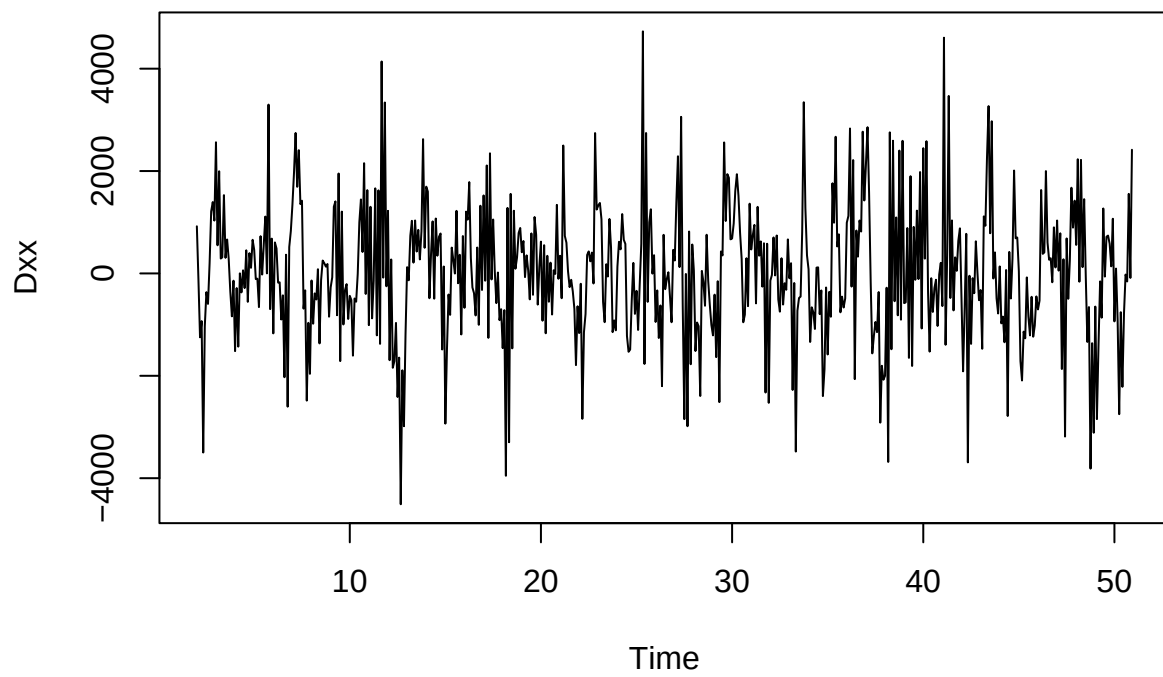
```
load("final-data.rsav")
xx <- dat1
plot(xx)
```



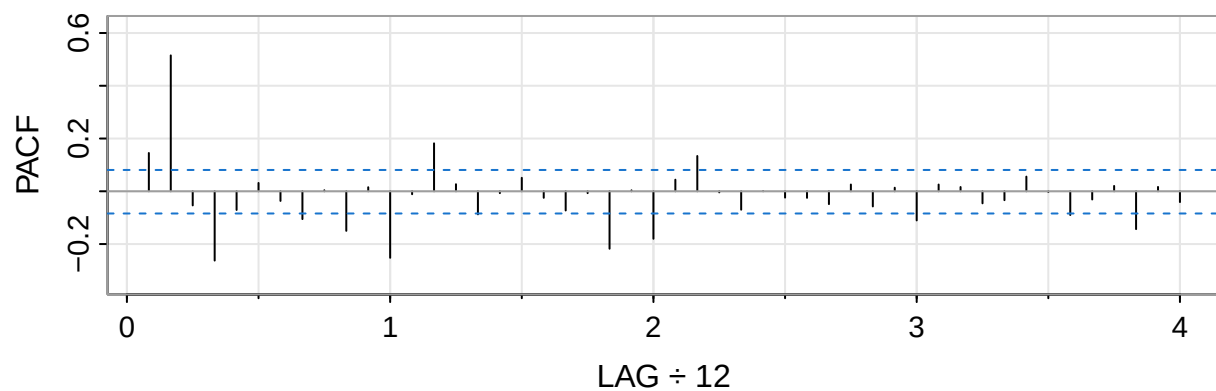
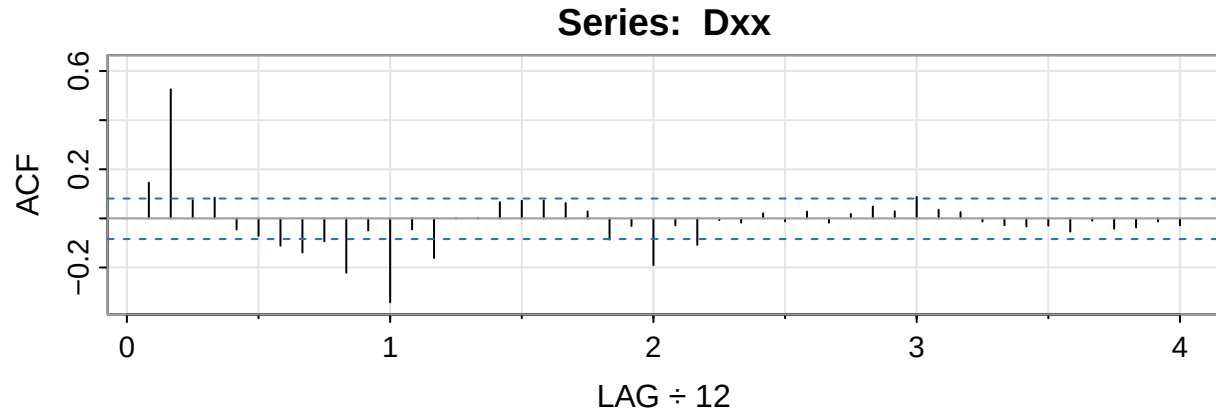
```
library(astsa)
invisible(acf2(xx))
```



```
Dxx <- diff(xx, lag=12)  
plot(Dxx)
```



```
invisible(acf2(Dxx))
```



```

maxAR <- 4+1
maxMA <- 2+1
maxSAR <- 2+1
maxSMA <- 2+1
fits_all <- array(list(), dim = c(maxAR, maxMA, maxSAR, maxSMA))
AICmin <- BICmin <- Inf
for (ii in 1:maxAR) {
  for (jj in 1:maxMA) {
    for (kk in 1:maxSAR) {
      for (ll in 1:maxSMA) {
        #if(!(ii==3 && jj == 3 && kk == 3 && ll == 2)){ #Condition because these models caused an error
        print(ii)
        print(jj)
        print(kk)
        print(ll)
        fits_all[[ii, jj, kk, ll]] <- astsa::sarima(Dxx, p = ii - 1, d = 0,
          q = jj - 1, P = kk - 1, D = 0, Q = ll - 1, S = 12, no.constant = F,
          details = F)
        if (fits_all[[ii, jj, kk, ll]]$ICs["AICc"] < AICmin)
          AICmin <- fits_all[[ii, jj, kk, ll]]$ICs["AICc"]
        if (fits_all[[ii, jj, kk, ll]]$ICs["BIC"] < BICmin)
          BICmin <- fits_all[[ii, jj, kk, ll]]$ICs["BIC"]
      }
    }
  }
}

```

```

## [1] 1
## [1] 1
## [1] 1
## [1] 1
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## xmean    5.9317 54.1579  0.1095  0.9128
##
## sigma^2 estimated as 1724652 on 587 degrees of freedom
##
## AIC = 17.20522  AICc = 17.20523  BIC = 17.2201
##
## [1] 1
## [1] 1
## [1] 1
## [1] 2
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## sma1   -0.7698  0.0374 -20.5708  0.0000
## xmean  13.8490 11.3168   1.2238  0.2215
##
## sigma^2 estimated as 1226366 on 586 degrees of freedom
##
## AIC = 16.88597  AICc = 16.88601  BIC = 16.9083
##
## [1] 1
## [1] 1
## [1] 1
## [1] 3
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## sma1   -0.6263  0.0441 -14.2172  0.0000
## sma2   -0.2147  0.0469  -4.5741  0.0000
## xmean  13.7285  8.3449   1.6451  0.1005
##
## sigma^2 estimated as 1181149 on 585 degrees of freedom
##
## AIC = 16.85603  AICc = 16.8561  BIC = 16.8858
##
## [1] 1
## [1] 1
## [1] 2
## [1] 1
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## sar1   -0.3503  0.039  -8.979  0.0000

```

```

## xmean    9.3679 37.777    0.248  0.8042
##
## sigma^2 estimated as 1513807 on 586 degrees of freedom
##
## AIC = 17.08089  AICc = 17.08093  BIC = 17.10322
##
## [1] 1
## [1] 1
## [1] 2
## [1] 2
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## sar1    0.2252 0.0536  4.2041  0.0000
## sma1   -0.8814 0.0312 -28.2757  0.0000
## xmean   13.7219 8.1674  1.6801  0.0935
##
## sigma^2 estimated as 1187702 on 585 degrees of freedom
##
## AIC = 16.86166  AICc = 16.86173  BIC = 16.89143
##
## [1] 1
## [1] 1
## [1] 2
## [1] 3
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## sar1   -0.3814 0.1575 -2.4211  0.0158
## sma1   -0.2557 0.1456 -1.7565  0.0795
## sma2   -0.4958 0.1058 -4.6870  0.0000
## xmean   13.7334 9.0808  1.5123  0.1310
##
## sigma^2 estimated as 1174574 on 584 degrees of freedom
##
## AIC = 16.85312  AICc = 16.85323  BIC = 16.89033
##
## [1] 1
## [1] 1
## [1] 3
## [1] 1
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## sar1   -0.4850 0.0391 -12.4091  0.0000
## sar2   -0.3709 0.0389  -9.5341  0.0000
## xmean   10.2006 25.7421  0.3963  0.6921
##
## sigma^2 estimated as 1305817 on 585 degrees of freedom
##
## AIC = 16.94259  AICc = 16.94266  BIC = 16.97237

```



```

##
## [1] 1
## [1] 1
## [1] 3
## [1] 2
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## sar1    0.1959 0.0628   3.1188 0.0019
## sar2   -0.0804 0.0548  -1.4673 0.1428
## sma1   -0.8405 0.0511 -16.4332 0.0000
## xmean  13.8743 9.1281   1.5200 0.1291
##
## sigma^2 estimated as 1184813 on 584 degrees of freedom
##
## AIC = 16.86128  AICc = 16.8614  BIC = 16.8985
##
## [1] 1
## [1] 1
## [1] 3
## [1] 3
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## sar1   -0.4188 0.1711  -2.4484 0.0146
## sar2    0.0307 0.0798   0.3843 0.7009
## sma1   -0.2154 0.1653  -1.3027 0.1932
## sma2   -0.5442 0.1544  -3.5242 0.0005
## xmean  13.7094 8.8318   1.5523 0.1211
##
## sigma^2 estimated as 1173891 on 583 degrees of freedom
##
## AIC = 16.85627  AICc = 16.85644  BIC = 16.90093
##
## [1] 1
## [1] 2
## [1] 1
## [1] 1
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ma1      0.0718 0.0289   2.4866 0.0132
## xmean    6.1715 57.7363   0.1069 0.9149
##
## sigma^2 estimated as 1706699 on 586 degrees of freedom
##
## AIC = 17.19816  AICc = 17.1982  BIC = 17.22049
##
## [1] 1
## [1] 2
## [1] 1

```

```

## [1] 2
## <><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ma1      0.0773 0.0293  2.6355 0.0086
## sma1     -0.7718 0.0374 -20.6489 0.0000
## xmean    14.0008 12.0236  1.1644 0.2447
##
## sigma^2 estimated as 1211896 on 585 degrees of freedom
##
## AIC = 16.87767  AICc = 16.87774  BIC = 16.90744
##
## [1] 1
## [1] 2
## [1] 1
## [1] 3
## <><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ma1      0.0785 0.0292  2.6846 0.0075
## sma1     -0.6277 0.0439 -14.2972 0.0000
## sma2     -0.2149 0.0467  -4.6033 0.0000
## xmean    13.8239 8.8676  1.5589 0.1196
##
## sigma^2 estimated as 1166650 on 584 degrees of freedom
##
## AIC = 16.84729  AICc = 16.8474  BIC = 16.8845
##
## [1] 1
## [1] 2
## [1] 2
## [1] 1
## <><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ma1      0.0729 0.029  2.5144 0.0122
## sar1     -0.3502 0.039 -8.9871 0.0000
## xmean     9.7936 40.315  0.2429 0.8081
##
## sigma^2 estimated as 1497727 on 585 degrees of freedom
##
## AIC = 17.07362  AICc = 17.07369  BIC = 17.1034
##
## [1] 1
## [1] 2
## [1] 2
## [1] 2
## <><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value

```

```

## ma1      0.0788 0.0292  2.6960  0.0072
## sar1      0.2262 0.0533  4.2437  0.0000
## sma1     -0.8832 0.0309 -28.5635  0.0000
## xmean    13.8140 8.6678  1.5937  0.1115
##
## sigma^2 estimated as 1172967 on 584 degrees of freedom
##
## AIC = 16.85281  AICc = 16.85293  BIC = 16.89003
##
## [1] 1
## [1] 2
## [1] 2
## [1] 3
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ma1      0.0777 0.0292  2.6584  0.0081
## sar1     -0.3826 0.1612 -2.3736  0.0179
## sma1     -0.2561 0.1490 -1.7185  0.0862
## sma2     -0.4974 0.1085 -4.5838  0.0000
## xmean    13.8466 9.6538  1.4343  0.1520
##
## sigma^2 estimated as 1160451 on 583 degrees of freedom
##
## AIC = 16.8446  AICc = 16.84477  BIC = 16.88926
##
## [1] 1
## [1] 2
## [1] 3
## [1] 1
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ma1      0.0707 0.0291  2.4337  0.0152
## sar1     -0.4842 0.0390 -12.4001  0.0000
## sar2     -0.3694 0.0389  -9.5079  0.0000
## xmean    10.4567 27.4566  0.3808  0.7035
##
## sigma^2 estimated as 1292871 on 584 degrees of freedom
##
## AIC = 16.93598  AICc = 16.9361  BIC = 16.9732
##
## [1] 1
## [1] 2
## [1] 3
## [1] 2
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ma1      0.0784 0.0293  2.6807  0.0076
## sar1      0.1985 0.0619  3.2055  0.0014

```

```

## sar2   -0.0784 0.0544  -1.4424  0.1497
## sma1   -0.8440 0.0500 -16.8955  0.0000
## xmean  13.9711 9.6517   1.4475  0.1483
##
## sigma^2 estimated as 1170256 on 583 degrees of freedom
##
## AIC = 16.85257  AICc = 16.85275  BIC = 16.89723
##
## [1] 1
## [1] 2
## [1] 3
## [1] 3
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ma1      0.0782 0.0292  2.6728  0.0077
## sar1    -0.4295 0.1715 -2.5043  0.0125
## sar2     0.0372 0.0800  0.4654  0.6418
## sma1    -0.2054 0.1657 -1.2394  0.2157
## sma2    -0.5573 0.1550 -3.5947  0.0004
## xmean   13.7928 9.3447  1.4760  0.1405
##
## sigma^2 estimated as 1159556 on 582 degrees of freedom
##
## AIC = 16.84763  AICc = 16.84787  BIC = 16.89973
##
## [1] 1
## [1] 3
## [1] 1
## [1] 1
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ma1      0.0183 0.0299  0.6100  0.5421
## ma2      0.6481 0.0373 17.3837  0.0000
## xmean    9.0200 74.0933  0.1217  0.9031
##
## sigma^2 estimated as 1165587 on 585 degrees of freedom
##
## AIC = 16.82207  AICc = 16.82214  BIC = 16.85184
##
## [1] 1
## [1] 3
## [1] 1
## [1] 2
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ma1      0.0670 0.0366  1.8301  0.0677
## ma2      0.5867 0.0303 19.3491  0.0000
## sma1    -0.7224 0.0402 -17.9804  0.0000

```

```

## xmean 14.3965 18.5457 0.7763 0.4379
##
## sigma^2 estimated as 860123.6 on 584 degrees of freedom
##
## AIC = 16.53639 AICc = 16.53651 BIC = 16.57361
##
## [1] 1
## [1] 3
## [1] 1
## [1] 3
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ma1      0.0612 0.0360  1.7020 0.0893
## ma2      0.5838 0.0312 18.6973 0.0000
## sma1     -0.6032 0.0436 -13.8343 0.0000
## sma2     -0.2029 0.0468  -4.3353 0.0000
## xmean    14.3300 13.4943  1.0619 0.2887
##
## sigma^2 estimated as 831351.7 on 583 degrees of freedom
##
## AIC = 16.50972 AICc = 16.5099 BIC = 16.55438
##
## [1] 1
## [1] 3
## [1] 2
## [1] 1
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ma1      0.0365 0.0332  1.1001 0.2717
## ma2      0.6262 0.0321 19.5132 0.0000
## sar1     -0.3507 0.0390 -8.9851 0.0000
## xmean    11.7809 51.5597  0.2285 0.8193
##
## sigma^2 estimated as 1023197 on 584 degrees of freedom
##
## AIC = 16.69784 AICc = 16.69795 BIC = 16.73505
##
## [1] 1
## [1] 3
## [1] 2
## [1] 2
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ma1      0.0621 0.0360  1.7243 0.0852
## ma2      0.5859 0.0310 18.8785 0.0000
## sar1      0.2441 0.0577  4.2307 0.0000
## sma1     -0.8642 0.0358 -24.1506 0.0000
## xmean    14.3255 12.8151  1.1179 0.2641

```

```

##
## sigma^2 estimated as 833288.4 on 583 degrees of freedom
##
## AIC = 16.51263  AICc = 16.5128  BIC = 16.55729
##
## [1] 1
## [1] 3
## [1] 2
## [1] 3
## <><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ma1      0.0605  0.0360  1.6781  0.0939
## ma2      0.5816  0.0313 18.5670  0.0000
## sar1     -0.3727  0.2123 -1.7554  0.0797
## sma1     -0.2390  0.2006 -1.1914  0.2340
## sma2     -0.4589  0.1344 -3.4150  0.0007
## xmean    14.2173 14.8500  0.9574  0.3388
##
## sigma^2 estimated as 829405.2 on 582 degrees of freedom
##
## AIC = 16.50986  AICc = 16.51011  BIC = 16.56197
##
## [1] 1
## [1] 3
## [1] 3
## [1] 1
## <><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ma1      0.0435  0.0344  1.2642  0.2067
## ma2      0.6024  0.0316 19.0897  0.0000
## sar1     -0.4781  0.0396 -12.0726  0.0000
## sar2     -0.3434  0.0393  -8.7332  0.0000
## xmean    11.4822 35.8248  0.3205  0.7487
##
## sigma^2 estimated as 902602.3 on 583 degrees of freedom
##
## AIC = 16.58089  AICc = 16.58107  BIC = 16.62555
##
## [1] 1
## [1] 3
## [1] 3
## [1] 2
## <><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ma1      0.0624  0.0360  1.7300  0.0842
## ma2      0.5842  0.0312 18.7260  0.0000
## sar1      0.2259  0.0680  3.3217  0.0010
## sar2     -0.0383  0.0580  -0.6601  0.5095

```

```

## sma1    -0.8410  0.0562 -14.9745  0.0000
## xmean   14.4005 13.6101   1.0581  0.2905
##
## sigma^2 estimated as 833291.7 on 582 degrees of freedom
##
## AIC = 16.51527  AICc = 16.51551  BIC = 16.56737
##
## [1] 1
## [1] 3
## [1] 3
## [1] 3
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ma1      0.0614  0.0360  1.7062  0.0885
## ma2      0.5833  0.0313 18.6240  0.0000
## sar1     -0.4902  0.1614 -3.0367  0.0025
## sar2      0.0895  0.0837  1.0686  0.2857
## sma1     -0.1138  0.1543 -0.7375  0.4611
## sma2     -0.6088  0.1437 -4.2358  0.0000
## xmean   14.1578 13.6792  1.0350  0.3011
##
## sigma^2 estimated as 827070.3 on 581 degrees of freedom
##
## AIC = 16.51148  AICc = 16.51181  BIC = 16.57103
##
## [1] 2
## [1] 1
## [1] 1
## [1] 1
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.146  0.0409  3.5697  0.0004
## xmean    6.885 62.7212  0.1098  0.9126
##
## sigma^2 estimated as 1688073 on 586 degrees of freedom
##
## AIC = 17.18722  AICc = 17.18725  BIC = 17.20955
##
## [1] 2
## [1] 1
## [1] 1
## [1] 2
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.1523  0.0409  3.7185  0.0002
## sma1     -0.7729  0.0374 -20.6825  0.0000
## xmean   14.1869 13.0274  1.0890  0.2766
##

```

```

## sigma^2 estimated as 1197865 on 585 degrees of freedom
##
## AIC = 16.86614  AICc = 16.86621  BIC = 16.89592
##
## [1] 2
## [1] 1
## [1] 1
## [1] 3
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1      0.1551 0.0409   3.7960 0.0002
## sma1     -0.6284 0.0437 -14.3633 0.0000
## sma2     -0.2154 0.0464  -4.6386 0.0000
## xmean    13.9477 9.6179   1.4502 0.1475
##
## sigma^2 estimated as 1152493 on 584 degrees of freedom
##
## AIC = 16.83523  AICc = 16.83535  BIC = 16.87245
##
## [1] 2
## [1] 1
## [1] 2
## [1] 1
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1      0.1473 0.0410   3.5933 0.0004
## sar1     -0.3500 0.0389 -8.9906 0.0000
## xmean    10.3852 43.8250   0.2370 0.8128
##
## sigma^2 estimated as 1481244 on 585 degrees of freedom
##
## AIC = 17.06258  AICc = 17.06265  BIC = 17.09235
##
## [1] 2
## [1] 1
## [1] 2
## [1] 2
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1      0.1559 0.0409   3.8122 0.0002
## sar1      0.2277 0.0530   4.2927 0.0000
## sma1     -0.8846 0.0307 -28.8539 0.0000
## xmean    13.9486 9.3847   1.4863 0.1377
##
## sigma^2 estimated as 1158557 on 584 degrees of freedom
##
## AIC = 16.84066  AICc = 16.84077  BIC = 16.87787
##

```



```

## [1] 2
## [1] 1
## [1] 2
## [1] 3
## <><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.1535  0.0409  3.7582  0.0002
## sar1     -0.3833  0.1650 -2.3224  0.0206
## sma1     -0.2562  0.1527 -1.6779  0.0939
## sma2     -0.4987  0.1113 -4.4815  0.0000
## xmean    13.9531 10.4661  1.3332  0.1830
##
## sigma^2 estimated as 1146675 on 583 degrees of freedom
##
## AIC = 16.83279  AICc = 16.83297  BIC = 16.87745
##
## [1] 2
## [1] 1
## [1] 3
## [1] 1
## <><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.1419  0.0409  3.4659  0.0006
## sar1     -0.4832  0.0390 -12.3858  0.0000
## sar2     -0.3677  0.0388  -9.4765  0.0000
## xmean    10.6671 29.7695  0.3583  0.7202
##
## sigma^2 estimated as 1279789 on 584 degrees of freedom
##
## AIC = 16.92577  AICc = 16.92589  BIC = 16.96299
##
## [1] 2
## [1] 1
## [1] 3
## [1] 2
## <><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.1549  0.0409  3.7876  0.0002
## sar1      0.2017  0.0611  3.3011  0.0010
## sar2     -0.0761  0.0540  -1.4095  0.1592
## sma1     -0.8472  0.0488 -17.3581  0.0000
## xmean    14.1050 10.4041  1.3557  0.1757
##
## sigma^2 estimated as 1156055 on 583 degrees of freedom
##
## AIC = 16.84059  AICc = 16.84076  BIC = 16.88525
##
## [1] 2

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```

## [1] 1
## [1] 3
## [1] 3
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.1547  0.0403  3.8396  0.0001
## sar1     -0.4449  0.1297 -3.4315  0.0006
## sar2      0.0459  0.0654  0.7018  0.4831
## sma1     -0.1901  0.1245 -1.5272  0.1272
## sma2     -0.5749  0.1125 -5.1098  0.0000
## xmean    13.9013 10.0776  1.3794  0.1683
##
## sigma^2 estimated as 1145504 on 582 degrees of freedom
##
## AIC = 16.83567  AICc = 16.83591  BIC = 16.88777
##
## [1] 2
## [1] 2
## [1] 1
## [1] 1
## <><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.7447  0.0468 15.9031  0.0000
## ma1     -0.5259  0.0519 -10.1378  0.0000
## xmean     8.4617 95.2742  0.0888  0.9293
##
## sigma^2 estimated as 1557459 on 585 degrees of freedom
##
## AIC = 17.11028  AICc = 17.11035  BIC = 17.14006
##
## [1] 2
## [1] 2
## [1] 1
## [1] 2
## <><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.7171  0.0529 13.5492  0.000
## ma1     -0.5018  0.0585 -8.5798  0.000
## sma1     -0.7609  0.0380 -20.0097  0.000
## xmean    15.1569 19.6576  0.7710  0.441
##
## sigma^2 estimated as 1121101 on 584 degrees of freedom
##
## AIC = 16.80251  AICc = 16.80263  BIC = 16.83973
##
## [1] 2
## [1] 2
## [1] 1

```

```

## [1] 3
## <><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.7242  0.0512  14.1418  0.0000
## ma1     -0.5063  0.0566  -8.9390  0.0000
## sma1    -0.6111  0.0438 -13.9672  0.0000
## sma2    -0.2248  0.0463  -4.8549  0.0000
## xmean   14.6709 14.5834   1.0060  0.3148
##
## sigma^2 estimated as 1075177 on 583 degrees of freedom
##
## AIC = 16.76839  AICc = 16.76857  BIC = 16.81305
##
## [1] 2
## [1] 2
## [1] 2
## [1] 1
## <><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.7321  0.0495  14.7848  0.0000
## ma1     -0.5143  0.0548  -9.3925  0.0000
## sar1    -0.3425  0.0391  -8.7711  0.0000
## xmean   13.6890 65.4671   0.2091  0.8344
##
## sigma^2 estimated as 1374790 on 584 degrees of freedom
##
## AIC = 16.99144  AICc = 16.99155  BIC = 17.02865
##
## [1] 2
## [1] 2
## [1] 2
## [1] 2
## <><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.7221  0.0515  14.0280  0.0000
## ma1     -0.5037  0.0569  -8.8571  0.0000
## sar1     0.2400  0.0533   4.5048  0.0000
## sma1    -0.8815  0.0307 -28.6678  0.0000
## xmean   14.7388 14.1183   1.0439  0.2969
##
## sigma^2 estimated as 1081385 on 583 degrees of freedom
##
## AIC = 16.77435  AICc = 16.77452  BIC = 16.81901
##
## [1] 2
## [1] 2
## [1] 2
## [1] 3

```

```

## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.7265  0.0511 14.2081  0.0000
## ma1     -0.5097  0.0567 -8.9912  0.0000
## sar1    -0.3856  0.1604 -2.4039  0.0165
## sma1    -0.2372  0.1481 -1.6016  0.1098
## sma2    -0.5038  0.1054 -4.7810  0.0000
## xmean   14.5804 15.9979  0.9114  0.3625
##
## sigma^2 estimated as 1069377 on 582 degrees of freedom
##
## AIC = 16.76564  AICc = 16.76589  BIC = 16.81775
##
## [1] 2
## [1] 2
## [1] 3
## [1] 1
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1      0.7330  0.0500 14.6462  0.0000
## ma1     -0.5197  0.0555 -9.3578  0.0000
## sar1    -0.4717  0.0390 -12.0836  0.0000
## sar2    -0.3650  0.0387 -9.4297  0.0000
## xmean   12.0419 44.5214  0.2705  0.7869
##
## sigma^2 estimated as 1189472 on 583 degrees of freedom
##
## AIC = 16.85591  AICc = 16.85609  BIC = 16.90057
##
## [1] 2
## [1] 2
## [1] 3
## [1] 2
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1      0.7231  0.0515 14.0512  0.0000
## ma1     -0.5056  0.0569 -8.8796  0.0000
## sar1     0.2143  0.0613  3.4934  0.0005
## sar2    -0.0757  0.0540 -1.4027  0.1612
## sma1    -0.8438  0.0491 -17.1839  0.0000
## xmean   14.8569 15.6275  0.9507  0.3422
##
## sigma^2 estimated as 1078984 on 582 degrees of freedom
##
## AIC = 16.77431  AICc = 16.77456  BIC = 16.82642
##
## [1] 2
## [1] 2

```

```

## [1] 3
## [1] 3
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.7262  0.0510 14.2482  0.0000
## ma1     -0.5085  0.0565 -9.0013  0.0000
## sar1    -0.4571  0.1543 -2.9629  0.0032
## sar2     0.0587  0.0795  0.7386  0.4604
## sma1    -0.1600  0.1474 -1.0853  0.2782
## sma2    -0.5953  0.1389 -4.2843  0.0000
## xmean   14.4926 15.2564  0.9499  0.3425
##
## sigma^2 estimated as 1067827 on 581 degrees of freedom
##
## AIC = 16.76818  AICc = 16.76851  BIC = 16.82773
##
## [1] 2
## [1] 3
## [1] 1
## [1] 1
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.1392  0.0608  2.2883  0.0225
## ma1     -0.0527  0.0444 -1.1864  0.2359
## ma2      0.6396  0.0374 17.0789  0.0000
## xmean    8.9703 81.5667  0.1100  0.9125
##
## sigma^2 estimated as 1154758 on 584 degrees of freedom
##
## AIC = 16.81613  AICc = 16.81624  BIC = 16.85334
##
## [1] 2
## [1] 3
## [1] 1
## [1] 2
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.1780  0.0865  2.0577  0.0401
## ma1     -0.0744  0.0778 -0.9563  0.3393
## ma2      0.5783  0.0308 18.7698  0.0000
## sma1    -0.7165  0.0403 -17.7881  0.0000
## xmean   14.8613 20.8461  0.7129  0.4762
##
## sigma^2 estimated as 854718.2 on 583 degrees of freedom
##
## AIC = 16.53313  AICc = 16.5333  BIC = 16.57779
##
## [1] 2

```

```

## [1] 3
## [1] 1
## [1] 3
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.1759  0.0805   2.1860  0.0292
## ma1     -0.0719  0.0706  -1.0188  0.3087
## ma2      0.5764  0.0319  18.0942  0.0000
## sma1    -0.5980  0.0432 -13.8287  0.0000
## sma2    -0.2066  0.0468  -4.4168  0.0000
## xmean   14.6587 15.0185   0.9760  0.3295
##
## sigma^2 estimated as 825061.2 on 582 degrees of freedom
##
## AIC = 16.50539  AICc = 16.50563  BIC = 16.55749
##
## [1] 2
## [1] 3
## [1] 2
## [1] 1
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.1589  0.0705   2.2538  0.0246
## ma1     -0.0647  0.0575  -1.1267  0.2603
## ma2      0.6186  0.0326  19.0014  0.0000
## sar1    -0.3494  0.0389  -8.9713  0.0000
## xmean   13.1576 57.0660   0.2306  0.8177
##
## sigma^2 estimated as 1014039 on 583 degrees of freedom
##
## AIC = 16.69222  AICc = 16.69239  BIC = 16.73688
##
## [1] 2
## [1] 3
## [1] 2
## [1] 2
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.1780  0.0806   2.2081  0.0276
## ma1     -0.0730  0.0708  -1.0302  0.3034
## ma2      0.5786  0.0317  18.2703  0.0000
## sar1      0.2517  0.0578   4.3519  0.0000
## sma1    -0.8661  0.0361 -23.9886  0.0000
## xmean   14.6557 14.1649   1.0346  0.3013
##
## sigma^2 estimated as 826777.4 on 582 degrees of freedom
##
## AIC = 16.50817  AICc = 16.50842  BIC = 16.56028

```

```

##
## [1] 2
## [1] 3
## [1] 2
## [1] 3
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.1753  0.0814  2.1540  0.0317
## ma1     -0.0730  0.0716 -1.0192  0.3085
## ma2      0.5738  0.0320 17.9455  0.0000
## sar1    -0.3591  0.2097 -1.7123  0.0874
## sma1    -0.2471  0.1986 -1.2442  0.2139
## sma2    -0.4511  0.1320 -3.4187  0.0007
## xmean   14.5686 16.5283  0.8814  0.3784
##
## sigma^2 estimated as 823364.2 on 581 degrees of freedom
##
## AIC = 16.50575  AICc = 16.50608  BIC = 16.5653
##
## [1] 2
## [1] 3
## [1] 3
## [1] 1
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1      0.1638  0.0790   2.0740  0.0385
## ma1     -0.0728  0.0679  -1.0710  0.2846
## ma2      0.5917  0.0323 18.3103  0.0000
## sar1    -0.4754  0.0395 -12.0349  0.0000
## sar2    -0.3413  0.0393  -8.6781  0.0000
## xmean   12.1603 39.4847   0.3080  0.7582
##
## sigma^2 estimated as 895892.9 on 582 degrees of freedom
##
## AIC = 16.5767  AICc = 16.57695  BIC = 16.62881
##
## [1] 2
## [1] 3
## [1] 3
## [1] 2
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1      0.1771  0.0808   2.1914  0.0288
## ma1     -0.0723  0.0710  -1.0176  0.3093
## ma2      0.5771  0.0318 18.1320  0.0000
## sar1     0.2352  0.0682   3.4501  0.0006
## sar2    -0.0343  0.0583  -0.5873  0.5572
## sma1    -0.8450  0.0565 -14.9672  0.0000

```

```

## xmean 14.7338 14.9746 0.9839 0.3256
##
## sigma^2 estimated as 826899.5 on 581 degrees of freedom
##
## AIC = 16.51097 AICc = 16.5113 BIC = 16.57052
##
## [1] 2
## [1] 3
## [1] 3
## [1] 3
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.1754 0.0807 2.1740 0.0301
## ma1     -0.0715 0.0709 -1.0085 0.3136
## ma2      0.5753 0.0320 17.9585 0.0000
## sar1    -0.4804 0.1622 -2.9628 0.0032
## sar2     0.0945 0.0854 1.1062 0.2691
## sma1    -0.1180 0.1551 -0.7607 0.4471
## sma2    -0.6074 0.1454 -4.1774 0.0000
## xmean  14.5193 15.1277 0.9598 0.3376
##
## sigma^2 estimated as 820855.3 on 580 degrees of freedom
##
## AIC = 16.50724 AICc = 16.50766 BIC = 16.57423
##
## [1] 3
## [1] 1
## [1] 1
## [1] 1
## <><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.0699 0.0353 1.9780 0.0484
## ar2      0.5169 0.0353 14.6398 0.0000
## xmean  13.4007 110.5011 0.1213 0.9035
##
## sigma^2 estimated as 1237204 on 585 degrees of freedom
##
## AIC = 16.88094 AICc = 16.88101 BIC = 16.91071
##
## [1] 3
## [1] 1
## [1] 1
## [1] 2
## <><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.0778 0.0362 2.1493 0.0320
## ar2      0.4871 0.0364 13.4003 0.0000
## sma1    -0.7248 0.0408 -17.7820 0.0000

```



```

## xmean 16.2804 26.3814 0.6171 0.5374
##
## sigma^2 estimated as 920776.4 on 584 degrees of freedom
##
## AIC = 16.60394 AICc = 16.60406 BIC = 16.64116
##
## [1] 3
## [1] 1
## [1] 1
## [1] 3
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.0787 0.0360  2.1835 0.0294
## ar2      0.4910 0.0362 13.5793 0.0000
## sma1     -0.5794 0.0437 -13.2588 0.0000
## sma2     -0.2350 0.0464  -5.0668 0.0000
## xmean    15.5372 18.9270  0.8209 0.4120
##
## sigma^2 estimated as 880443.7 on 583 degrees of freedom
##
## AIC = 16.56709 AICc = 16.56727 BIC = 16.61175
##
## [1] 3
## [1] 1
## [1] 2
## [1] 1
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.0720 0.0357  2.0201 0.0438
## ar2      0.5100 0.0357 14.3041 0.0000
## sar1     -0.3352 0.0393 -8.5331 0.0000
## xmean    17.3551 77.5280  0.2239 0.8229
##
## sigma^2 estimated as 1098988 on 584 degrees of freedom
##
## AIC = 16.76823 AICc = 16.76835 BIC = 16.80545
##
## [1] 3
## [1] 1
## [1] 2
## [1] 2
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.0792 0.0361  2.1966 0.0284
## ar2      0.4911 0.0362 13.5700 0.0000
## sar1      0.2631 0.0546  4.8181 0.0000
## sma1     -0.8717 0.0317 -27.4954 0.0000
## xmean    15.6663 18.2045  0.8606 0.3898

```

```

##
## sigma^2 estimated as 885285.9 on 583 degrees of freedom
##
## AIC = 16.57281  AICc = 16.57298  BIC = 16.61747
##
## [1] 3
## [1] 1
## [1] 2
## [1] 3
## <><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.0778  0.0360  2.1595  0.0312
## ar2      0.4904  0.0362 13.5634  0.0000
## sar1     -0.3544  0.1588 -2.2310  0.0261
## sma1     -0.2364  0.1475 -1.6025  0.1096
## sma2     -0.4788  0.0997 -4.8049  0.0000
## xmean    15.3713 20.6796  0.7433  0.4576
##
## sigma^2 estimated as 876384.3 on 582 degrees of freedom
##
## AIC = 16.56518  AICc = 16.56542  BIC = 16.61728
##
## [1] 3
## [1] 1
## [1] 3
## [1] 1
## <><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.0699  0.0357  1.9571  0.0508
## ar2      0.5038  0.0357 14.0980  0.0000
## sar1     -0.4598  0.0393 -11.7067  0.0000
## sar2     -0.3584  0.0389  -9.2043  0.0000
## xmean    13.6942 52.5178  0.2608  0.7944
##
## sigma^2 estimated as 956873.3 on 583 degrees of freedom
##
## AIC = 16.63877  AICc = 16.63895  BIC = 16.68344
##
## [1] 3
## [1] 1
## [1] 3
## [1] 2
## <><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.0790  0.0361  2.1917  0.0288
## ar2      0.4900  0.0362 13.5285  0.0000
## sar1      0.2432  0.0623  3.9012  0.0001
## sar2     -0.0585  0.0542  -1.0795  0.2808

```

```

## sma1    -0.8422  0.0491 -17.1384  0.0000
## xmean   15.7279 19.5152   0.8059  0.4206
##
## sigma^2 estimated as 884166 on 582 degrees of freedom
##
## AIC = 16.57417  AICc = 16.57442  BIC = 16.62628
##
## [1] 3
## [1] 1
## [1] 3
## [1] 3
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.0786  0.0360  2.1814  0.0296
## ar2      0.4912  0.0361 13.5873  0.0000
## sar1    -0.4232  0.1472 -2.8743  0.0042
## sar2      0.0705  0.0801  0.8802  0.3791
## sma1    -0.1606  0.1399 -1.1479  0.2515
## sma2    -0.5771  0.1304 -4.4255  0.0000
## xmean   15.3345 19.4964  0.7865  0.4319
##
## sigma^2 estimated as 874630.8 on 581 degrees of freedom
##
## AIC = 16.56726  AICc = 16.56759  BIC = 16.62681
##
## [1] 3
## [1] 2
## [1] 1
## [1] 1
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.0311  0.0542  0.5742  0.5660
## ar2      0.5231  0.0355 14.7403  0.0000
## ma1      0.0529  0.0566  0.9347  0.3503
## xmean   12.7263 107.8153  0.1180  0.9061
##
## sigma^2 estimated as 1235381 on 584 degrees of freedom
##
## AIC = 16.88287  AICc = 16.88299  BIC = 16.92009
##
## [1] 3
## [1] 2
## [1] 1
## [1] 2
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.0116  0.0571  0.2039  0.8385
## ar2      0.4983  0.0364 13.7042  0.0000

```

```

## ma1      0.0866  0.0588   1.4727  0.1414
## sma1     -0.7282  0.0409 -17.8239  0.0000
## xmean    16.0554 25.1224   0.6391  0.5230
##
## sigma^2 estimated as 917167.5 on 583 degrees of freedom
##
## AIC = 16.60365  AICc = 16.60383  BIC = 16.64831
##
## [1] 3
## [1] 2
## [1] 1
## [1] 3
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1      0.0168  0.0572   0.2934  0.7693
## ar2      0.5014  0.0362  13.8469  0.0000
## ma1      0.0816  0.0593   1.3758  0.1694
## sma1     -0.5838  0.0437 -13.3568  0.0000
## sma2     -0.2326  0.0462  -5.0357  0.0000
## xmean    15.3901 18.0992   0.8503  0.3955
##
## sigma^2 estimated as 877415.5 on 582 degrees of freedom
##
## AIC = 16.56726  AICc = 16.56751  BIC = 16.61936
##
## [1] 3
## [1] 2
## [1] 2
## [1] 1
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1      0.0192  0.0546   0.3514  0.7254
## ar2      0.5188  0.0357  14.5185  0.0000
## ma1      0.0715  0.0565   1.2652  0.2063
## sar1     -0.3371  0.0393  -8.5847  0.0000
## xmean    16.5780 74.9736   0.2211  0.8251
##
## sigma^2 estimated as 1095914 on 583 degrees of freedom
##
## AIC = 16.76888  AICc = 16.76905  BIC = 16.81354
##
## [1] 3
## [1] 2
## [1] 2
## [1] 2
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1      0.0147  0.0570   0.2573  0.7970

```

```

## ar2      0.5023  0.0362  13.8640  0.0000
## ma1      0.0852  0.0590   1.4436  0.1494
## sar1     0.2608  0.0543   4.8047  0.0000
## sma1    -0.8726  0.0313 -27.8606  0.0000
## xmean   15.4941 17.4052   0.8902  0.3737
##
## sigma^2 estimated as 881950.2 on 582 degrees of freedom
##
## AIC = 16.57265  AICc = 16.5729  BIC = 16.62475
##
## [1] 3
## [1] 2
## [1] 2
## [1] 3
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.0195  0.0574  0.3407  0.7335
## ar2      0.5001  0.0362 13.7986  0.0000
## ma1      0.0768  0.0596  1.2878  0.1983
## sar1    -0.3480  0.1629 -2.1363  0.0331
## sma1    -0.2466  0.1517 -1.6259  0.1045
## sma2    -0.4739  0.1033 -4.5900  0.0000
## xmean   15.2735 19.7559  0.7731  0.4398
##
## sigma^2 estimated as 873720.3 on 581 degrees of freedom
##
## AIC = 16.56575  AICc = 16.56608  BIC = 16.6253
##
## [1] 3
## [1] 2
## [1] 3
## [1] 1
## <><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.0212  0.0556  0.3814  0.7030
## ar2      0.5115  0.0358 14.2691  0.0000
## ma1      0.0652  0.0578  1.1275  0.2600
## sar1    -0.4619  0.0393 -11.7469  0.0000
## sar2    -0.3577  0.0390  -9.1811  0.0000
## xmean   13.9585 50.9636  0.2739  0.7843
##
## sigma^2 estimated as 954810.6 on 582 degrees of freedom
##
## AIC = 16.64003  AICc = 16.64028  BIC = 16.69214
##
## [1] 3
## [1] 2
## [1] 3
## [1] 2
## <><><><><><><><><><><><><>

```

```

##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.0156  0.0572   0.2731  0.7849
## ar2      0.5009  0.0363  13.8056  0.0000
## ma1      0.0835  0.0593   1.4100  0.1591
## sar1     0.2430  0.0611   3.9740  0.0001
## sar2    -0.0556  0.0537  -1.0355  0.3008
## sma1    -0.8455  0.0474 -17.8339  0.0000
## xmean   15.5606 18.5488   0.8389  0.4019
##
## sigma^2 estimated as 880919 on 581 degrees of freedom
##
## AIC = 16.57418  AICc = 16.57451  BIC = 16.63373
##
## [1] 3
## [1] 2
## [1] 3
## [1] 3
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.0200  0.0573   0.3494  0.7269
## ar2      0.5011  0.0362  13.8245  0.0000
## ma1      0.0773  0.0595   1.2998  0.1942
## sar1    -0.4236  0.1506  -2.8134  0.0051
## sar2     0.0722  0.0801   0.9025  0.3672
## sma1    -0.1640  0.1433  -1.1444  0.2529
## sma2    -0.5776  0.1332  -4.3376  0.0000
## xmean   15.2366 18.6272   0.8180  0.4137
##
## sigma^2 estimated as 871915.8 on 580 degrees of freedom
##
## AIC = 16.56777  AICc = 16.56819  BIC = 16.63476
##
## [1] 3
## [1] 3
## [1] 1
## [1] 1
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.1290  0.0624   2.0672  0.0392
## ar2      0.1773  0.0760   2.3312  0.0201
## ma1     -0.0511  0.0530  -0.9646  0.3352
## ma2      0.4964  0.0728   6.8228  0.0000
## xmean   10.8269 91.7693   0.1180  0.9061
##
## sigma^2 estimated as 1145885 on 583 degrees of freedom
##
## AIC = 16.81167  AICc = 16.81184  BIC = 16.85633
##

```

```

## [1] 3
## [1] 3
## [1] 1
## [1] 2
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1      0.1135  0.0724   1.5680  0.1174
## ar2      0.1595  0.0654   2.4378  0.0151
## ma1     -0.0192  0.0655  -0.2924  0.7701
## ma2      0.4850  0.0547   8.8606  0.0000
## sma1    -0.7148  0.0404 -17.6790  0.0000
## xmean   15.2400 22.9760   0.6633  0.5074
##
## sigma^2 estimated as 846610.3 on 582 degrees of freedom
##
## AIC = 16.5269  AICc = 16.52714  BIC = 16.579
##
## [1] 3
## [1] 3
## [1] 1
## [1] 3
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1      0.1183  0.0711   1.6623  0.0970
## ar2      0.1653  0.0686   2.4110  0.0162
## ma1     -0.0235  0.0642  -0.3656  0.7148
## ma2      0.4727  0.0594   7.9580  0.0000
## sma1    -0.5968  0.0432 -13.8082  0.0000
## sma2    -0.2061  0.0467  -4.4131  0.0000
## xmean   14.9633 16.6710   0.8976  0.3698
##
## sigma^2 estimated as 817431.2 on 581 degrees of freedom
##
## AIC = 16.49933  AICc = 16.49966  BIC = 16.55888
##
## [1] 3
## [1] 3
## [1] 2
## [1] 1
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1      0.1258  0.0652   1.9291  0.0542
## ar2      0.1685  0.0673   2.5047  0.0125
## ma1     -0.0420  0.0564  -0.7458  0.4561
## ma2      0.5095  0.0590   8.6370  0.0000
## sar1    -0.3518  0.0389  -9.0510  0.0000
## xmean   14.0513 63.7664   0.2204  0.8257
##

```

```

## sigma^2 estimated as 1003781 on 582 degrees of freedom
##
## AIC = 16.68544  AICc = 16.68568  BIC = 16.73754
##
## [1] 3
## [1] 3
## [1] 2
## [1] 2
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1      0.1191  0.0711   1.6743  0.0946
## ar2      0.1642  0.0680   2.4155  0.0160
## ma1     -0.0233  0.0640  -0.3631  0.7167
## ma2      0.4770  0.0584   8.1710  0.0000
## sar1      0.2504  0.0574   4.3595  0.0000
## sma1     -0.8638  0.0355 -24.3055  0.0000
## xmean    14.9665 15.7896   0.9479  0.3436
##
## sigma^2 estimated as 819196.1 on 581 degrees of freedom
##
## AIC = 16.5021  AICc = 16.50243  BIC = 16.56165
##
## [1] 3
## [1] 3
## [1] 2
## [1] 3
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1      0.1183  0.0713   1.6603  0.0974
## ar2      0.1661  0.0687   2.4184  0.0159
## ma1     -0.0254  0.0644  -0.3942  0.6936
## ma2      0.4692  0.0598   7.8459  0.0000
## sar1     -0.3643  0.2096  -1.7378  0.0828
## sma1     -0.2406  0.1984  -1.2128  0.2257
## sma2     -0.4543  0.1314  -3.4567  0.0006
## xmean    14.8547 18.3417   0.8099  0.4183
##
## sigma^2 estimated as 815639.7 on 580 degrees of freedom
##
## AIC = 16.49961  AICc = 16.50003  BIC = 16.5666
##
## [1] 3
## [1] 3
## [1] 3
## [1] 1
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1      0.1231  0.0680   1.8110  0.0707

```



```

## ar2      0.1782  0.0674   2.6433  0.0084
## ma1     -0.0431  0.0606  -0.7115  0.4771
## ma2      0.4777  0.0596   8.0184  0.0000
## sar1    -0.4766  0.0394 -12.0874  0.0000
## sar2    -0.3424  0.0393  -8.7219  0.0000
## xmean   12.9377 44.2936   0.2921  0.7703
##
## sigma^2 estimated as 885719.4 on 581 degrees of freedom
##
## AIC = 16.56871  AICc = 16.56904  BIC = 16.62826
##
## [1] 3
## [1] 3
## [1] 3
## [1] 2
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1      0.1181  0.0713   1.6574  0.0980
## ar2      0.1642  0.0682   2.4093  0.0163
## ma1     -0.0225  0.0642  -0.3506  0.7260
## ma2      0.4751  0.0588   8.0810  0.0000
## sar1     0.2355  0.0667   3.5331  0.0004
## sar2    -0.0328  0.0574  -0.5720  0.5675
## sma1    -0.8444  0.0543 -15.5625  0.0000
## xmean   15.0510 16.5754   0.9080  0.3642
##
## sigma^2 estimated as 819243.8 on 580 degrees of freedom
##
## AIC = 16.50493  AICc = 16.50535  BIC = 16.57192
##
## [1] 3
## [1] 3
## [1] 3
## [1] 3
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1      0.1191  0.0710   1.6774  0.0940
## ar2      0.1666  0.0686   2.4300  0.0154
## ma1     -0.0246  0.0641  -0.3834  0.7016
## ma2      0.4705  0.0597   7.8859  0.0000
## sar1    -0.4852  0.1616  -3.0035  0.0028
## sar2     0.0954  0.0846   1.1275  0.2600
## sma1    -0.1116  0.1544  -0.7232  0.4699
## sma2    -0.6106  0.1438  -4.2465  0.0000
## xmean   14.7794 16.8262   0.8784  0.3801
##
## sigma^2 estimated as 813094.5 on 579 degrees of freedom
##
## AIC = 16.50099  AICc = 16.50152  BIC = 16.57542
##

```

```

## [1] 4
## [1] 1
## [1] 1
## [1] 1
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.0983   0.0412  2.3821  0.0175
## ar2      0.5207   0.0354 14.7205  0.0000
## ar3     -0.0545   0.0413 -1.3202  0.1873
## xmean    13.0782 104.7636  0.1248  0.9007
##
## sigma^2 estimated as 1233528 on 584 degrees of freedom
##
## AIC = 16.88138  AICc = 16.8815  BIC = 16.9186
##
## [1] 4
## [1] 1
## [1] 1
## [1] 2
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.1212   0.0413  2.9322  0.0035
## ar2      0.4935   0.0363 13.5784  0.0000
## ar3     -0.0892   0.0415 -2.1516  0.0318
## sma1    -0.7321   0.0409 -17.8841  0.0000
## xmean    15.7908 23.5225  0.6713  0.5023
##
## sigma^2 estimated as 913105.7 on 583 degrees of freedom
##
## AIC = 16.59949  AICc = 16.59967  BIC = 16.64415
##
## [1] 4
## [1] 1
## [1] 1
## [1] 3
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.1193   0.0413  2.8887  0.0040
## ar2      0.4967   0.0362 13.7341  0.0000
## ar3     -0.0823   0.0415 -1.9826  0.0479
## sma1    -0.5890   0.0438 -13.4577  0.0000
## sma2    -0.2293   0.0460 -4.9828  0.0000
## xmean    15.2189 17.1137  0.8893  0.3742
##
## sigma^2 estimated as 874221.1 on 582 degrees of freedom
##
## AIC = 16.56383  AICc = 16.56408  BIC = 16.61594
##

```

```

## [1] 4
## [1] 1
## [1] 2
## [1] 1
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.1109  0.0413  2.6851  0.0075
## ar2      0.5155  0.0357 14.4489  0.0000
## ar3     -0.0766  0.0415 -1.8483  0.0651
## sar1    -0.3395  0.0393 -8.6465  0.0000
## xmean   16.1841 71.6112  0.2260  0.8213
##
## sigma^2 estimated as 1092528 on 583 degrees of freedom
##
## AIC = 16.76584  AICc = 16.76601  BIC = 16.8105
##
## [1] 4
## [1] 1
## [1] 2
## [1] 2
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.1216  0.0413  2.9464  0.0033
## ar2      0.4973  0.0362 13.7409  0.0000
## ar3     -0.0864  0.0414 -2.0855  0.0375
## sar1      0.2580  0.0540  4.7742  0.0000
## sma1     -0.8734  0.0310 -28.1992  0.0000
## xmean   15.3038 16.4069  0.9328  0.3513
##
## sigma^2 estimated as 878385 on 582 degrees of freedom
##
## AIC = 16.56884  AICc = 16.56908  BIC = 16.62094
##
## [1] 4
## [1] 1
## [1] 2
## [1] 3
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.1158  0.0414  2.8001  0.0053
## ar2      0.4957  0.0362 13.7012  0.0000
## ar3     -0.0772  0.0417 -1.8522  0.0645
## sar1    -0.3413  0.1682 -2.0295  0.0429
## sma1    -0.2577  0.1570 -1.6419  0.1012
## sma2    -0.4681  0.1078 -4.3425  0.0000
## xmean   15.1542 18.6765  0.8114  0.4175
##
## sigma^2 estimated as 870915.4 on 581 degrees of freedom

```

```

##
## AIC = 16.56276  AICc = 16.56309  BIC = 16.62231
##
## [1] 4
## [1] 1
## [1] 3
## [1] 1
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1      0.1038  0.0413   2.5121  0.0123
## ar2      0.5084  0.0358  14.2136  0.0000
## ar3     -0.0670  0.0415  -1.6137  0.1071
## sar1    -0.4643  0.0394 -11.7901  0.0000
## sar2    -0.3569  0.0390  -9.1561  0.0000
## xmean   13.2104 49.0812   0.2692  0.7879
##
## sigma^2 estimated as 952621.8 on 582 degrees of freedom
##
## AIC = 16.63776  AICc = 16.63801  BIC = 16.68986
##
## [1] 4
## [1] 1
## [1] 3
## [1] 2
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1      0.1205  0.0413   2.9182  0.0037
## ar2      0.4962  0.0362  13.6945  0.0000
## ar3     -0.0844  0.0415  -2.0337  0.0424
## sar1     0.2419  0.0601   4.0240  0.0001
## sar2    -0.0523  0.0532  -0.9838  0.3256
## sma1    -0.8486  0.0459 -18.4944  0.0000
## xmean   15.3629 17.4229   0.8818  0.3783
##
## sigma^2 estimated as 877470.6 on 581 degrees of freedom
##
## AIC = 16.57056  AICc = 16.57089  BIC = 16.63011
##
## [1] 4
## [1] 1
## [1] 3
## [1] 3
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1      0.1170  0.0414   2.8296  0.0048
## ar2      0.4966  0.0362  13.7297  0.0000
## ar3     -0.0780  0.0416  -1.8738  0.0615
## sar1    -0.4185  0.1574  -2.6597  0.0080

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## sar2    0.0717  0.0802  0.8949  0.3712
## sma1   -0.1731  0.1506 -1.1495  0.2508
## sma2   -0.5728  0.1389 -4.1228  0.0000
## xmean  15.1382 17.6201  0.8591  0.3906
##
## sigma^2 estimated as 869074.6 on 580 degrees of freedom
##
## AIC = 16.56472  AICc = 16.56514  BIC = 16.63171
##
## [1] 4
## [1] 2
## [1] 1
## [1] 1
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1      1.0333 0.0342  30.2270  0e+00
## ar2      0.4582 0.0513   8.9262  0e+00
## ar3     -0.5581 0.0344 -16.2472  0e+00
## ma1     -1.0000 0.0060 -165.8754  0e+00
## xmean   13.5033 3.7956   3.5576  4e-04
##
## sigma^2 estimated as 1136893 on 583 degrees of freedom
##
## AIC = 16.81024  AICc = 16.81042  BIC = 16.8549
##
## [1] 4
## [1] 2
## [1] 1
## [1] 2
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1      0.8010 0.0755  10.6046  0.0000
## ar2      0.4499 0.0462   9.7377  0.0000
## ar3     -0.4792 0.0423 -11.3185  0.0000
## ma1     -0.6747 0.0773  -8.7315  0.0000
## sma1    -0.7245 0.0424 -17.0685  0.0000
## xmean   14.9793 16.0715   0.9320  0.3517
##
## sigma^2 estimated as 875959.5 on 582 degrees of freedom
##
## AIC = 16.56142  AICc = 16.56167  BIC = 16.61352
##
## [1] 4
## [1] 2
## [1] 1
## [1] 3
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value

```

```

## ar1      0.8630  0.0784  11.0050   0.000
## ar2      0.4464  0.0477   9.3570   0.000
## ar3     -0.4988  0.0413 -12.0733   0.000
## ma1     -0.7460  0.0833  -8.9522   0.000
## sma1    -0.5808  0.0453 -12.8334   0.000
## sma2    -0.2369  0.0473  -5.0111   0.000
## xmean   14.7641 10.5941   1.3936   0.164
##
## sigma^2 estimated as 837620.8 on 581 degrees of freedom
##
## AIC = 16.52513  AICc = 16.52546  BIC = 16.58468
##
## [1] 4
## [1] 2
## [1] 2
## [1] 1
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1      0.8908  0.0826  10.7806  0.0000
## ar2      0.4672  0.0478   9.7659  0.0000
## ar3     -0.5282  0.0405 -13.0430  0.0000
## ma1     -0.7836  0.0925  -8.4739  0.0000
## sar1    -0.3307  0.0409  -8.0818  0.0000
## xmean   13.5630 40.5511   0.3345  0.7381
##
## sigma^2 estimated as 1040146 on 582 degrees of freedom
##
## AIC = 16.72073  AICc = 16.72097  BIC = 16.77283
##
## [1] 4
## [1] 2
## [1] 2
## [1] 2
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1      0.8498  0.0781  10.8880  0.0000
## ar2      0.4481  0.0474   9.4602  0.0000
## ar3     -0.4966  0.0416 -11.9383  0.0000
## ma1     -0.7295  0.0824  -8.8527  0.0000
## sar1     0.2711  0.0555   4.8852  0.0000
## sma1    -0.8778  0.0312 -28.1122  0.0000
## xmean   14.6628 10.1918   1.4387  0.1508
##
## sigma^2 estimated as 841042.9 on 581 degrees of freedom
##
## AIC = 16.52964  AICc = 16.52997  BIC = 16.58919
##
## [1] 4
## [1] 2
## [1] 2

```

```

## [1] 3
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1      0.8611  0.0785  10.9679  0.0000
## ar2      0.4462  0.0477   9.3565  0.0000
## ar3     -0.4957  0.0416 -11.9198  0.0000
## ma1     -0.7449  0.0830  -8.9746  0.0000
## sar1    -0.3050  0.1869  -1.6321  0.1032
## sma1    -0.2857  0.1754  -1.6285  0.1040
## sma2    -0.4465  0.1191  -3.7496  0.0002
## xmean   14.8656 11.6731   1.2735  0.2034
##
## sigma^2 estimated as 835710 on 580 degrees of freedom
##
## AIC = 16.52541  AICc = 16.52583  BIC = 16.5924
##
## [1] 4
## [1] 2
## [1] 3
## [1] 1
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1      0.8730  0.0772  11.3143  0.0000
## ar2      0.4631  0.0474   9.7757  0.0000
## ar3     -0.5130  0.0408 -12.5590  0.0000
## ma1     -0.7656  0.0827  -9.2560  0.0000
## sar1    -0.4552  0.0413 -11.0213  0.0000
## sar2    -0.3504  0.0390  -8.9816  0.0000
## xmean   13.5026 29.3657   0.4598  0.6458
##
## sigma^2 estimated as 911210.1 on 581 degrees of freedom
##
## AIC = 16.59717  AICc = 16.59749  BIC = 16.65671
##
## [1] 4
## [1] 2
## [1] 3
## [1] 2
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1      0.8535  0.0783  10.9000  0.0000
## ar2      0.4468  0.0475   9.4054  0.0000
## ar3     -0.4963  0.0416 -11.9328  0.0000
## ma1     -0.7342  0.0828  -8.8717  0.0000
## sar1     0.2589  0.0609   4.2475  0.0000
## sar2    -0.0435  0.0531  -0.8179  0.4137
## sma1    -0.8575  0.0451 -19.0136  0.0000
## xmean   14.8026 10.6972   1.3838  0.1670

```

```

##
## sigma^2 estimated as 840570.6 on 580 degrees of freedom
##
## AIC = 16.53189  AICc = 16.53231  BIC = 16.59888
##
## [1] 4
## [1] 2
## [1] 3
## [1] 3
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1      0.8675  0.0783  11.0811  0.0000
## ar2      0.4462  0.0478   9.3280  0.0000
## ar3     -0.4992  0.0414 -12.0622  0.0000
## ma1     -0.7512  0.0830  -9.0451  0.0000
## sar1    -0.4299  0.1562  -2.7526  0.0061
## sar2     0.0983  0.0831   1.1831  0.2372
## sma1    -0.1521  0.1481  -1.0268  0.3050
## sma2    -0.6022  0.1383  -4.3541  0.0000
## xmean   14.7172 10.6822   1.3777  0.1688
##
## sigma^2 estimated as 832991 on 579 degrees of freedom
##
## AIC = 16.52657  AICc = 16.52709  BIC = 16.601
##
## [1] 4
## [1] 3
## [1] 1
## [1] 1
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1      0.1458  0.1067   1.3659  0.1725
## ar2      0.1840  0.0805   2.2872  0.0225
## ar3     -0.0155  0.0784  -0.1976  0.8434
## ma1     -0.0671  0.0982  -0.6833  0.4947
## ma2      0.4884  0.0809   6.0339  0.0000
## xmean   10.7206 91.3011   0.1174  0.9066
##
## sigma^2 estimated as 1145818 on 582 degrees of freedom
##
## AIC = 16.815  AICc = 16.81525  BIC = 16.8671
##
## [1] 4
## [1] 3
## [1] 1
## [1] 2
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value

```



```

## ar1      0.2592  0.1480   1.7512  0.0804
## ar2      0.1652  0.0677   2.4400  0.0150
## ar3     -0.1030  0.0957  -1.0767  0.2820
## ma1     -0.1621  0.1426  -1.1362  0.2564
## ma2      0.4667  0.0618   7.5555  0.0000
## sma1    -0.7174  0.0405 -17.7231  0.0000
## xmean   15.1238 21.7189   0.6963  0.4865
##
## sigma^2 estimated as 845136.9 on 581 degrees of freedom
##
## AIC = 16.52866  AICc = 16.52899  BIC = 16.58821
##
## [1] 4
## [1] 3
## [1] 1
## [1] 3
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1      0.2258  0.1510   1.4955  0.1353
## ar2      0.1736  0.0699   2.4833  0.0133
## ar3     -0.0790  0.0982  -0.8045  0.4214
## ma1     -0.1283  0.1451  -0.8839  0.3771
## ma2      0.4551  0.0667   6.8186  0.0000
## sma1    -0.5999  0.0434 -13.8236  0.0000
## sma2    -0.2046  0.0467  -4.3802  0.0000
## xmean   14.9138 15.9792   0.9333  0.3510
##
## sigma^2 estimated as 816416.3 on 580 degrees of freedom
##
## AIC = 16.5016  AICc = 16.50203  BIC = 16.56859
##
## [1] 4
## [1] 3
## [1] 2
## [1] 1
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1      0.1957  0.1308   1.4962  0.1351
## ar2      0.1811  0.0694   2.6097  0.0093
## ar3     -0.0590  0.0920  -0.6415  0.5214
## ma1     -0.1093  0.1235  -0.8847  0.3767
## ma2      0.4919  0.0671   7.3320  0.0000
## sar1    -0.3535  0.0390  -9.0747  0.0000
## xmean   14.1330 62.0502   0.2278  0.8199
##
## sigma^2 estimated as 1003002 on 581 degrees of freedom
##
## AIC = 16.68806  AICc = 16.68839  BIC = 16.74761
##
## [1] 4

```

```

## [1] 3
## [1] 2
## [1] 2
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.2345  0.1496   1.5677  0.1175
## ar2      0.1728  0.0695   2.4885  0.0131
## ar3     -0.0851  0.0976  -0.8715  0.3838
## ma1     -0.1359  0.1437  -0.9455  0.3448
## ma2      0.4583  0.0660   6.9412  0.0000
## sar1      0.2485  0.0572   4.3426  0.0000
## sma1     -0.8645  0.0353 -24.5112  0.0000
## xmean    14.8847 15.0837   0.9868  0.3241
##
## sigma^2 estimated as 818025.7 on 580 degrees of freedom
##
## AIC = 16.50419  AICc = 16.50461  BIC = 16.57118
##
## [1] 4
## [1] 3
## [1] 2
## [1] 3
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.2173  0.1535   1.4154  0.1575
## ar2      0.1736  0.0700   2.4815  0.0134
## ar3     -0.0721  0.0993  -0.7259  0.4682
## ma1     -0.1218  0.1476  -0.8253  0.4095
## ma2      0.4533  0.0667   6.7985  0.0000
## sar1     -0.3552  0.2115  -1.6792  0.0936
## sma1     -0.2521  0.2007  -1.2560  0.2096
## sma2     -0.4478  0.1336  -3.3527  0.0009
## xmean    14.8317 17.6129   0.8421  0.4001
##
## sigma^2 estimated as 814808.6 on 579 degrees of freedom
##
## AIC = 16.5021  AICc = 16.50263  BIC = 16.57653
##
## [1] 4
## [1] 3
## [1] 3
## [1] 1
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.2234  0.1690   1.3216  0.1868
## ar2      0.1921  0.0703   2.7313  0.0065
## ar3     -0.0752  0.1122  -0.6698  0.5033
## ma1     -0.1406  0.1630  -0.8623  0.3889

```

```

## ma2      0.4557  0.0711  6.4100  0.0000
## sar1     -0.4790  0.0396 -12.0834  0.0000
## sar2     -0.3426  0.0393  -8.7260  0.0000
## xmean    12.9448 42.9406   0.3015  0.7632
##
## sigma^2 estimated as 884931.4 on 580 degrees of freedom
##
## AIC = 16.57122  AICc = 16.57164  BIC = 16.63821
##
## [1] 4
## [1] 3
## [1] 3
## [1] 2
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.2304  0.1503   1.5324  0.1260
## ar2      0.1726  0.0696   2.4811  0.0134
## ar3     -0.0825  0.0979  -0.8427  0.3998
## ma1     -0.1320  0.1445  -0.9136  0.3613
## ma2      0.4571  0.0661   6.9122  0.0000
## sar1      0.2355  0.0656   3.5898  0.0004
## sar2     -0.0300  0.0569  -0.5266  0.5987
## sma1     -0.8472  0.0529 -16.0180  0.0000
## xmean    14.9450 15.7800   0.9471  0.3440
##
## sigma^2 estimated as 818095.2 on 579 degrees of freedom
##
## AIC = 16.50711  AICc = 16.50764  BIC = 16.58154
##
## [1] 4
## [1] 3
## [1] 3
## [1] 3
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.2150  0.1530   1.4056  0.1604
## ar2      0.1743  0.0699   2.4930  0.0129
## ar3     -0.0702  0.0994  -0.7068  0.4800
## ma1     -0.1181  0.1471  -0.8029  0.4224
## ma2      0.4547  0.0668   6.8103  0.0000
## sar1     -0.4740  0.1711  -2.7707  0.0058
## sar2      0.0926  0.0857   1.0803  0.2804
## sma1     -0.1252  0.1647  -0.7601  0.4475
## sma2     -0.6008  0.1523  -3.9461  0.0001
## xmean    14.7653 16.2072   0.9110  0.3627
##
## sigma^2 estimated as 812340.7 on 578 degrees of freedom
##
## AIC = 16.50354  AICc = 16.50418  BIC = 16.58541
##

```

```

## [1] 5
## [1] 1
## [1] 1
## [1] 1
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.0845  0.0398  2.1226  0.0342
## ar2      0.6590  0.0400 16.4953  0.0000
## ar3     -0.0287  0.0400 -0.7172  0.4735
## ar4     -0.2654  0.0398 -6.6605  0.0000
## xmean    12.4194 80.1440  0.1550  0.8769
##
## sigma^2 estimated as 1146493 on 583 degrees of freedom
##
## AIC = 16.81211  AICc = 16.81228  BIC = 16.85677
##
## [1] 5
## [1] 1
## [1] 1
## [1] 2
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1      0.0968  0.0398   2.4304  0.0154
## ar2      0.6313  0.0402  15.6986  0.0000
## ar3     -0.0553  0.0401  -1.3788  0.1685
## ar4     -0.2772  0.0400  -6.9289  0.0000
## sma1    -0.7180  0.0405 -17.7167  0.0000
## xmean    14.8203 18.6796   0.7934  0.4279
##
## sigma^2 estimated as 844506.7 on 582 degrees of freedom
##
## AIC = 16.52436  AICc = 16.5246  BIC = 16.57646
##
## [1] 5
## [1] 1
## [1] 1
## [1] 3
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1      0.0981  0.0399   2.4565  0.0143
## ar2      0.6270  0.0402  15.5818  0.0000
## ar3     -0.0520  0.0402  -1.2918  0.1969
## ar4     -0.2623  0.0402  -6.5300  0.0000
## sma1    -0.5998  0.0434 -13.8168  0.0000
## sma2    -0.2056  0.0466  -4.4135  0.0000
## xmean    14.7285 13.8492   1.0635  0.2880
##
## sigma^2 estimated as 815605.1 on 581 degrees of freedom

```

```

##
## AIC = 16.49715  AICc = 16.49748  BIC = 16.5567
##
## [1] 5
## [1] 1
## [1] 2
## [1] 1
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.0900  0.0397  2.2674  0.0237
## ar2      0.6613  0.0398 16.6045  0.0000
## ar3     -0.0460  0.0399 -1.1510  0.2502
## ar4     -0.2849  0.0399 -7.1464  0.0000
## sar1    -0.3522  0.0389 -9.0517  0.0000
## xmean   13.2547 53.0087  0.2500  0.8026
##
## sigma^2 estimated as 1004519 on 582 degrees of freedom
##
## AIC = 16.686  AICc = 16.68624  BIC = 16.7381
##
## [1] 5
## [1] 1
## [1] 2
## [1] 2
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.0990  0.0399  2.4825  0.0133
## ar2      0.6293  0.0402 15.6507  0.0000
## ar3     -0.0543  0.0402 -1.3525  0.1767
## ar4     -0.2654  0.0400 -6.6276  0.0000
## sar1     0.2468  0.0568  4.3436  0.0000
## sma1    -0.8641  0.0347 -24.9005  0.0000
## xmean   14.6866 13.0964  1.1214  0.2626
##
## sigma^2 estimated as 817592.3 on 581 degrees of freedom
##
## AIC = 16.50011  AICc = 16.50044  BIC = 16.55966
##
## [1] 5
## [1] 1
## [1] 2
## [1] 3
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.0964  0.0400  2.4111  0.0162
## ar2      0.6251  0.0403 15.5104  0.0000
## ar3     -0.0482  0.0404 -1.1950  0.2326
## ar4     -0.2604  0.0403 -6.4662  0.0000

```

```

## sar1    -0.3607   0.2095  -1.7213   0.0857
## sma1    -0.2472   0.1985  -1.2453   0.2135
## sma2    -0.4529   0.1322  -3.4266   0.0007
## xmean   14.7025  15.3040   0.9607   0.3371
##
## sigma^2 estimated as 813781.7 on 580 degrees of freedom
##
## AIC = 16.4974  AICc = 16.49782  BIC = 16.56439
##
## [1] 5
## [1] 1
## [1] 3
## [1] 1
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1      0.0867  0.0399   2.1758  0.0300
## ar2      0.6466  0.0400  16.1572  0.0000
## ar3     -0.0407  0.0401  -1.0154  0.3104
## ar4     -0.2715  0.0401  -6.7721  0.0000
## sar1    -0.4789  0.0395 -12.1191  0.0000
## sar2    -0.3457  0.0391  -8.8418  0.0000
## xmean   12.9208 37.1635   0.3477  0.7282
##
## sigma^2 estimated as 883389.9 on 581 degrees of freedom
##
## AIC = 16.56606  AICc = 16.56639  BIC = 16.62561
##
## [1] 5
## [1] 1
## [1] 3
## [1] 2
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1      0.0987  0.0399   2.4735  0.0137
## ar2      0.6277  0.0403  15.5753  0.0000
## ar3     -0.0533  0.0402  -1.3255  0.1855
## ar4     -0.2641  0.0401  -6.5821  0.0000
## sar1     0.2312  0.0661   3.4973  0.0005
## sar2    -0.0361  0.0569  -0.6346  0.5259
## sma1    -0.8433  0.0535 -15.7550  0.0000
## xmean   14.7575 13.8230   1.0676  0.2861
##
## sigma^2 estimated as 817541.2 on 580 degrees of freedom
##
## AIC = 16.50281  AICc = 16.50323  BIC = 16.5698
##
## [1] 5
## [1] 1
## [1] 3
## [1] 3

```

```

## <><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.0977  0.0400  2.4439  0.0148
## ar2      0.6265  0.0403 15.5474  0.0000
## ar3     -0.0488  0.0403 -1.2109  0.2264
## ar4     -0.2609  0.0402 -6.4857  0.0000
## sar1    -0.4809  0.1650 -2.9146  0.0037
## sar2     0.0912  0.0846  1.0781  0.2815
## sma1    -0.1192  0.1581 -0.7541  0.4511
## sma2    -0.6052  0.1468 -4.1222  0.0000
## xmean   14.6246 14.1094  1.0365  0.3004
##
## sigma^2 estimated as 811453.6 on 579 degrees of freedom
##
## AIC = 16.49897  AICc = 16.4995  BIC = 16.5734
##
## [1] 5
## [1] 2
## [1] 1
## [1] 1
## <><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.3138  0.1515  2.0707  0.0388
## ar2      0.6367  0.0459 13.8701  0.0000
## ar3     -0.1482  0.0874 -1.6949  0.0906
## ar4     -0.2533  0.0438 -5.7785  0.0000
## ma1     -0.2477  0.1572 -1.5751  0.1158
## xmean   11.2891 73.4906  0.1536  0.8780
##
## sigma^2 estimated as 1141053 on 582 degrees of freedom
##
## AIC = 16.8108  AICc = 16.81104  BIC = 16.8629
##
## [1] 5
## [1] 2
## [1] 1
## [1] 2
## <><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.2546  0.1313  1.9390  0.0530
## ar2      0.6131  0.0449 13.6427  0.0000
## ar3     -0.1333  0.0747 -1.7837  0.0750
## ar4     -0.2645  0.0436 -6.0672  0.0000
## ma1     -0.1710  0.1350 -1.2669  0.2057
## sma1    -0.7150  0.0409 -17.4707  0.0000
## xmean   14.7608 17.8177  0.8284  0.4078
##
## sigma^2 estimated as 842324.1 on 581 degrees of freedom

```

```

##
## AIC = 16.52497  AICc = 16.5253  BIC = 16.58452
##
## [1] 5
## [1] 2
## [1] 1
## [1] 3
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1      0.2745  0.1462   1.8773  0.0610
## ar2      0.6063  0.0462  13.1349  0.0000
## ar3     -0.1396  0.0817  -1.7086  0.0881
## ar4     -0.2483  0.0445  -5.5784  0.0000
## ma1     -0.1895  0.1506  -1.2586  0.2087
## sma1    -0.5984  0.0435 -13.7564  0.0000
## sma2    -0.2073  0.0469  -4.4232  0.0000
## xmean   14.7153 13.0233   1.1299  0.2590
##
## sigma^2 estimated as 813296.2 on 580 degrees of freedom
##
## AIC = 16.49772  AICc = 16.49815  BIC = 16.56472
##
## [1] 5
## [1] 2
## [1] 2
## [1] 1
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.2778  0.1242   2.2364  0.0257
## ar2      0.6412  0.0443  14.4707  0.0000
## ar3     -0.1428  0.0738  -1.9360  0.0534
## ar4     -0.2719  0.0431  -6.3038  0.0000
## ma1     -0.2047  0.1277  -1.6031  0.1094
## sar1    -0.3519  0.0389  -9.0439  0.0000
## xmean   13.0946 49.2245   0.2660  0.7903
##
## sigma^2 estimated as 1000042 on 581 degrees of freedom
##
## AIC = 16.68495  AICc = 16.68527  BIC = 16.74449
##
## [1] 5
## [1] 2
## [1] 2
## [1] 2
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1      0.2801  0.1429   1.9604  0.0504
## ar2      0.6073  0.0462  13.1494  0.0000

```



```

## ar3      -0.1445  0.0804  -1.7982  0.0727
## ar4      -0.2503  0.0446  -5.6094  0.0000
## ma1      -0.1944  0.1469  -1.3236  0.1861
## sar1      0.2500  0.0570   4.3836  0.0000
## sma1     -0.8658  0.0349 -24.7985  0.0000
## xmean    14.6214 12.2197   1.1965  0.2320
##
## sigma^2 estimated as 815130.9 on 580 degrees of freedom
##
## AIC = 16.50059  AICc = 16.50101  BIC = 16.56758
##
## [1] 5
## [1] 2
## [1] 2
## [1] 3
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.2798  0.1490  1.8778  0.0609
## ar2      0.6042  0.0463 13.0504  0.0000
## ar3     -0.1392  0.0829 -1.6792  0.0937
## ar4     -0.2466  0.0446 -5.5307  0.0000
## ma1     -0.1969  0.1535 -1.2828  0.2001
## sar1    -0.3600  0.2078 -1.7328  0.0837
## sma1    -0.2459  0.1972 -1.2468  0.2130
## sma2    -0.4530  0.1303 -3.4764  0.0005
## xmean   14.6757 14.4231  1.0175  0.3093
##
## sigma^2 estimated as 811483.8 on 579 degrees of freedom
##
## AIC = 16.4979  AICc = 16.49843  BIC = 16.57233
##
## [1] 5
## [1] 2
## [1] 3
## [1] 1
## <><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.3102  0.1345  2.3069  0.0214
## ar2      0.6236  0.0453 13.7689  0.0000
## ar3     -0.1544  0.0780 -1.9810  0.0481
## ar4     -0.2573  0.0440 -5.8548  0.0000
## ma1     -0.2418  0.1384 -1.7476  0.0811
## sar1    -0.4787  0.0395 -12.1132  0.0000
## sar2    -0.3466  0.0391 -8.8707  0.0000
## xmean   12.9868 34.0577   0.3813  0.7031
##
## sigma^2 estimated as 878792.7 on 580 degrees of freedom
##
## AIC = 16.56428  AICc = 16.56471  BIC = 16.63128
##

```

```

## [1] 5
## [1] 2
## [1] 3
## [1] 2
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.2711  0.1438   1.8850  0.0599
## ar2      0.6074  0.0460  13.1979  0.0000
## ar3     -0.1391  0.0807  -1.7233  0.0854
## ar4     -0.2503  0.0444  -5.6406  0.0000
## ma1     -0.1854  0.1480  -1.2526  0.2108
## sar1     0.2358  0.0662   3.5620  0.0004
## sar2    -0.0324  0.0570  -0.5676  0.5705
## sma1    -0.8469  0.0535 -15.8217  0.0000
## xmean   14.7360 12.9052   1.1419  0.2540
##
## sigma^2 estimated as 815183.2 on 579 degrees of freedom
##
## AIC = 16.50342  AICc = 16.50395  BIC = 16.57786
##
## [1] 5
## [1] 2
## [1] 3
## [1] 3
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.2818  0.1487   1.8956  0.0585
## ar2      0.6052  0.0464  13.0441  0.0000
## ar3     -0.1404  0.0829  -1.6930  0.0910
## ar4     -0.2468  0.0446  -5.5330  0.0000
## ma1     -0.1978  0.1532  -1.2909  0.1973
## sar1    -0.4773  0.1642  -2.9074  0.0038
## sar2     0.0927  0.0852   1.0873  0.2774
## sma1    -0.1207  0.1573  -0.7678  0.4429
## sma2    -0.6054  0.1466  -4.1297  0.0000
## xmean   14.6324 13.2190   1.1069  0.2688
##
## sigma^2 estimated as 809029.5 on 578 degrees of freedom
##
## AIC = 16.49939  AICc = 16.50004  BIC = 16.58127
##
## [1] 5
## [1] 3
## [1] 1
## [1] 1
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.0604  0.0340   1.7792  0.0757

```

```

## ar2      1.4568 0.0342 42.6369 0.0000
## ar3     -0.0698 0.0338 -2.0675 0.0391
## ar4     -0.5809 0.0337 -17.2448 0.0000
## ma1     -0.0140 0.0133 -1.0552 0.2918
## ma2     -0.9860 0.0133 -74.3857 0.0000
## xmean   13.5121 3.6852  3.6666 0.0003
##
## sigma^2 estimated as 1085919 on 581 degrees of freedom
##
## AIC = 16.77307  AICc = 16.7734  BIC = 16.83262
##
## [1] 5
## [1] 3
## [1] 1
## [1] 2
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1      0.2425  0.1471   1.6491 0.0997
## ar2      0.4565  0.2044   2.2331 0.0259
## ar3     -0.1127  0.0933  -1.2082 0.2275
## ar4     -0.1900  0.1088  -1.7471 0.0811
## ma1     -0.1537  0.1510  -1.0177 0.3092
## ma2      0.1738  0.2205   0.7883 0.4309
## sma1    -0.7147  0.0408 -17.5346 0.0000
## xmean   14.8385 19.2371   0.7713 0.4408
##
## sigma^2 estimated as 841131.8 on 580 degrees of freedom
##
## AIC = 16.52705  AICc = 16.52747  BIC = 16.59404
##
## [1] 5
## [1] 3
## [1] 1
## [1] 3
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1      0.2663  0.1574   1.6912 0.0913
## ar2      0.5104  0.2116   2.4122 0.0162
## ar3     -0.1270  0.0958  -1.3263 0.1853
## ar4     -0.2032  0.1094  -1.8572 0.0638
## ma1     -0.1779  0.1624  -1.0955 0.2738
## ma2      0.1060  0.2287   0.4637 0.6430
## sma1    -0.5999  0.0435 -13.7894 0.0000
## sma2    -0.2045  0.0471  -4.3379 0.0000
## xmean   14.7338 13.7217   1.0738 0.2834
##
## sigma^2 estimated as 813020.1 on 579 degrees of freedom
##
## AIC = 16.50071  AICc = 16.50124  BIC = 16.57515
##

```

```

## [1] 5
## [1] 3
## [1] 2
## [1] 1
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.2806  0.1371  2.0471  0.0411
## ar2      0.4716  0.1540  3.0621  0.0023
## ar3     -0.1298  0.0871 -1.4910  0.1365
## ar4     -0.1880  0.0888 -2.1157  0.0348
## ma1     -0.2030  0.1391 -1.4593  0.1450
## ma2      0.1875  0.1617  1.1599  0.2466
## sar1    -0.3550  0.0389 -9.1199  0.0000
## xmean   13.2518 53.1866  0.2492  0.8033
##
## sigma^2 estimated as 997775.9 on 580 degrees of freedom
##
## AIC = 16.68618  AICc = 16.6866  BIC = 16.75317
##
## [1] 5
## [1] 3
## [1] 2
## [1] 2
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1      0.2635  0.1549   1.7010  0.0895
## ar2      0.4891  0.2092   2.3380  0.0197
## ar3     -0.1251  0.0958  -1.3058  0.1922
## ar4     -0.1946  0.1094  -1.7783  0.0759
## ma1     -0.1735  0.1593  -1.0893  0.2765
## ma2      0.1322  0.2259   0.5851  0.5587
## sar1      0.2480  0.0574   4.3192  0.0000
## sma1    -0.8643  0.0354 -24.4137  0.0000
## xmean   14.6772 13.0742   1.1226  0.2621
##
## sigma^2 estimated as 814643.2 on 579 degrees of freedom
##
## AIC = 16.50332  AICc = 16.50385  BIC = 16.57775
##
## [1] 5
## [1] 3
## [1] 2
## [1] 3
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.2739  0.1587  1.7256  0.0849
## ar2      0.5343  0.2039  2.6202  0.0090
## ar3     -0.1303  0.0942 -1.3821  0.1675

```

```

## ar4      -0.2139  0.1048 -2.0398  0.0418
## ma1      -0.1885  0.1640 -1.1494  0.2509
## ma2       0.0773  0.2199  0.3514  0.7254
## sar1     -0.3625  0.2105 -1.7218  0.0856
## sma1     -0.2446  0.1999 -1.2237  0.2216
## sma2     -0.4523  0.1321 -3.4226  0.0007
## xmean    14.7050 14.9978  0.9805  0.3273
##
## sigma^2 estimated as 811351.3 on 578 degrees of freedom
##
## AIC = 16.50107  AICc = 16.50171  BIC = 16.58294
##
## [1] 5
## [1] 3
## [1] 3
## [1] 1
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1      0.3197  0.1458   2.1924  0.0287
## ar2      0.5399  0.1579   3.4193  0.0007
## ar3     -0.1530  0.0859  -1.7815  0.0754
## ar4     -0.2166  0.0879  -2.4647  0.0140
## ma1     -0.2496  0.1489  -1.6756  0.0944
## ma2      0.0912  0.1637   0.5571  0.5777
## sar1    -0.4804  0.0396 -12.1367  0.0000
## sar2    -0.3454  0.0392  -8.8121  0.0000
## xmean   12.8161 35.4039   0.3620  0.7175
##
## sigma^2 estimated as 878376.2 on 579 degrees of freedom
##
## AIC = 16.5672  AICc = 16.56772  BIC = 16.64163
##
## [1] 5
## [1] 3
## [1] 3
## [1] 2
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1      0.2612  0.1562   1.6726  0.0950
## ar2      0.4956  0.2128   2.3296  0.0202
## ar3     -0.1240  0.0962  -1.2889  0.1979
## ar4     -0.1975  0.1107  -1.7842  0.0749
## ma1     -0.1716  0.1608  -1.0667  0.2866
## ma2      0.1239  0.2301   0.5383  0.5906
## sar1      0.2350  0.0663   3.5443  0.0004
## sar2    -0.0295  0.0573  -0.5136  0.6077
## sma1    -0.8472  0.0536 -15.8182  0.0000
## xmean   14.7566 13.6458   1.0814  0.2800
##
## sigma^2 estimated as 814726.5 on 578 degrees of freedom

```

```
##
## AIC = 16.50626  AICc = 16.50691  BIC = 16.58814
##
## [1] 5
## [1] 3
## [1] 3
## [1] 3
## <><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.2750  0.1607  1.7107  0.0877
## ar2      0.5254  0.2081  2.5245  0.0119
## ar3     -0.1300  0.0967 -1.3442  0.1794
## ar4     -0.2093  0.1071 -1.9545  0.0511
## ma1     -0.1879  0.1663 -1.1297  0.2591
## ma2      0.0883  0.2255  0.3913  0.6957
## sar1    -0.4796  0.1665 -2.8801  0.0041
## sar2     0.0936  0.0855  1.0947  0.2741
## sma1    -0.1195  0.1596 -0.7488  0.4543
## sma2    -0.6055  0.1484 -4.0812  0.0001
## xmean   14.6161 13.7944  1.0596  0.2898
##
## sigma^2 estimated as 808867.5 on 577 degrees of freedom
##
## AIC = 16.50254  AICc = 16.50332  BIC = 16.59186
##
```

```
#}
```

```
nn <- length(Dxx)
print("AICcs")
```

```
## [1] "AICcs"
```

```
## [1] "AICcs"
for (ii in 1:maxAR) {
for (jj in 1:maxMA) {
for (kk in 1:maxSAR) {
for (ll in 1:maxSMA) {
##if(!(ii==4 && jj == 5 && kk == 2 && ll == 3)){ #Condition because (3,4)x(1,2) caused an error
AICdelta <- ((fits_all[[ii, jj, kk, ll]])$ICs["AICc"] - AICmin) * nn
if (AICdelta < 5) {
print(paste0("(", ii - 1, ", ", jj - 1, ") x (", kk - 1, ", ",
ll - 1, ")"))
print(AICdelta)
}
}
}
}
}
```

```
## [1] "(1, 2) x (0, 2)"
##      AICc
## 4.793045
## [1] "(2, 2) x (0, 2)"
```

```

##      AICc
## 1.280732
## [1] "(2, 2) x (1, 1)"
##      AICc
## 2.907847
## [1] "(2, 2) x (1, 2)"
##      AICc
## 1.501779
## [1] "(2, 2) x (2, 1)"
##      AICc
## 4.628105
## [1] "(2, 2) x (2, 2)"
##      AICc
## 2.372432
## [1] "(3, 2) x (0, 2)"
##      AICc
## 2.672808
## [1] "(3, 2) x (1, 1)"
##      AICc
## 4.193339
## [1] "(3, 2) x (1, 2)"
##      AICc
## 3.027368
## [1] "(3, 2) x (2, 2)"
##      AICc
## 3.941604
## [1] "(4, 0) x (0, 2)"
## AICc
##      0
## [1] "(4, 0) x (1, 1)"
##      AICc
## 1.740974
## [1] "(4, 0) x (1, 2)"
##      AICc
## 0.2006046
## [1] "(4, 0) x (2, 1)"
##      AICc
## 3.381233
## [1] "(4, 0) x (2, 2)"
##      AICc
## 1.185085
## [1] "(4, 1) x (0, 2)"
##      AICc
## 0.3918796
## [1] "(4, 1) x (1, 1)"
##      AICc
## 2.076963
## [1] "(4, 1) x (1, 2)"
##      AICc
## 0.5568412
## [1] "(4, 1) x (2, 1)"
##      AICc
## 3.805638
## [1] "(4, 1) x (2, 2)"

```

```
##      AICc
## 1.506113
## [1] "(4, 2) x (0, 2)"
##      AICc
## 2.212495
## [1] "(4, 2) x (1, 1)"
##      AICc
## 3.74287
## [1] "(4, 2) x (1, 2)"
##      AICc
## 2.48954
## [1] "(4, 2) x (2, 2)"
##      AICc
## 3.431433
```

```
#}
```

```
print("BICs")
```

```
## [1] "BICs"
```

```
## [1] "BICs"
for (ii in 1:maxAR) {
for (jj in 1:maxMA) {
for (kk in 1:maxSAR) {
for (ll in 1:maxSMA) {
BICdelta <- ((fits_all[[ii, jj, kk, ll]])$ICs["BIC"] - BICmin) * nn
if (BICdelta < 7) {
print(paste0("(", ii - 1, ", ", jj - 1, ") x (", kk - 1, ", ",
ll - 1, ")"))
print(BICdelta)
}
}
}
}
}
```

```
## [1] "(0, 2) x (0, 2)"
## BIC
## 0
## [1] "(0, 2) x (1, 1)"
##      BIC
## 1.708116
## [1] "(0, 2) x (1, 2)"
##      BIC
## 4.45822
## [1] "(1, 2) x (0, 2)"
##      BIC
## 1.826381
## [1] "(1, 2) x (1, 1)"
##      BIC
## 3.464637
## [1] "(1, 2) x (1, 2)"
##      BIC
## 6.419612
## [1] "(2, 2) x (0, 2)"
```



```

##      BIC
## 2.642269
## [1] "(2, 2) x (1, 1)"
##      BIC
## 4.269384
## [1] "(4, 0) x (0, 2)"
##      BIC
## 1.361537
## [1] "(4, 0) x (1, 1)"
##      BIC
## 3.102511
## [1] "(4, 0) x (1, 2)"
##      BIC
## 5.883267
## [1] "(4, 1) x (0, 2)"
##      BIC
## 6.074542

```

```

sarima(Dxx, p = 1, d = 0, q = 2, P = 0, D = 0,
Q = 2, S = 12,no.constant=F,details=T)

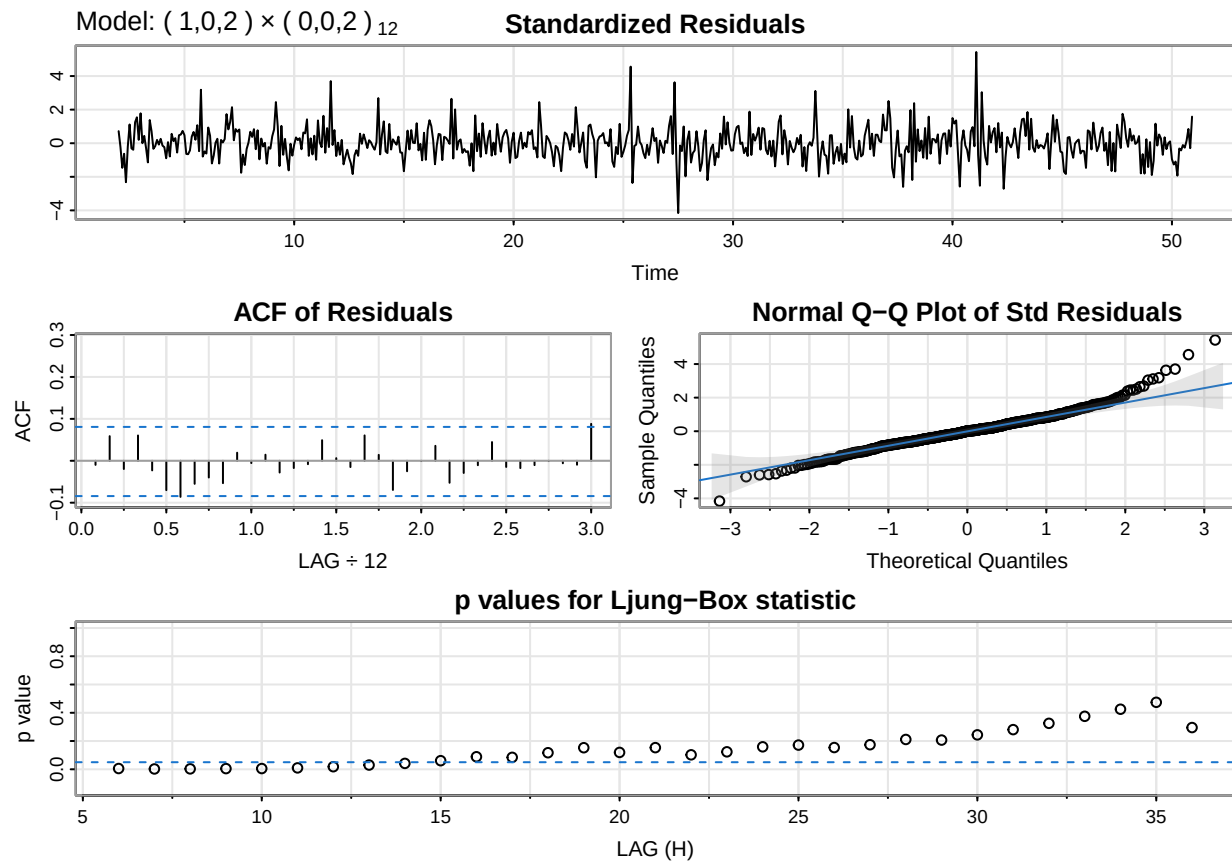
```

```

## initial  value 7.180706
## iter    2 value 6.890383
## iter    3 value 6.850952
## iter    4 value 6.840173
## iter    5 value 6.836862
## iter    6 value 6.830840
## iter    7 value 6.828624
## iter    8 value 6.828127
## iter    9 value 6.827871
## iter   10 value 6.827567
## iter   11 value 6.827549
## iter   12 value 6.827545
## iter   12 value 6.827545
## iter   12 value 6.827545
## final   value 6.827545
## converged
## initial  value 6.822782
## iter    2 value 6.822344
## iter    3 value 6.822036
## iter    4 value 6.821968
## iter    5 value 6.821869
## iter    6 value 6.821858
## iter    7 value 6.821851
## iter    8 value 6.821850
## iter    8 value 6.821850
## iter    8 value 6.821850
## final   value 6.821850
## converged
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1      0.1759  0.0805   2.1860  0.0292
## ma1     -0.0719  0.0706  -1.0188  0.3087

```

```
## ma2      0.5764  0.0319  18.0942  0.0000
## sma1     -0.5980  0.0432 -13.8287  0.0000
## sma2     -0.2066  0.0468  -4.4168  0.0000
## xmean    14.6587 15.0185   0.9760  0.3295
##
## sigma^2 estimated as 825061.2 on 582 degrees of freedom
##
## AIC = 16.50539  AICc = 16.50563  BIC = 16.55749
##
```



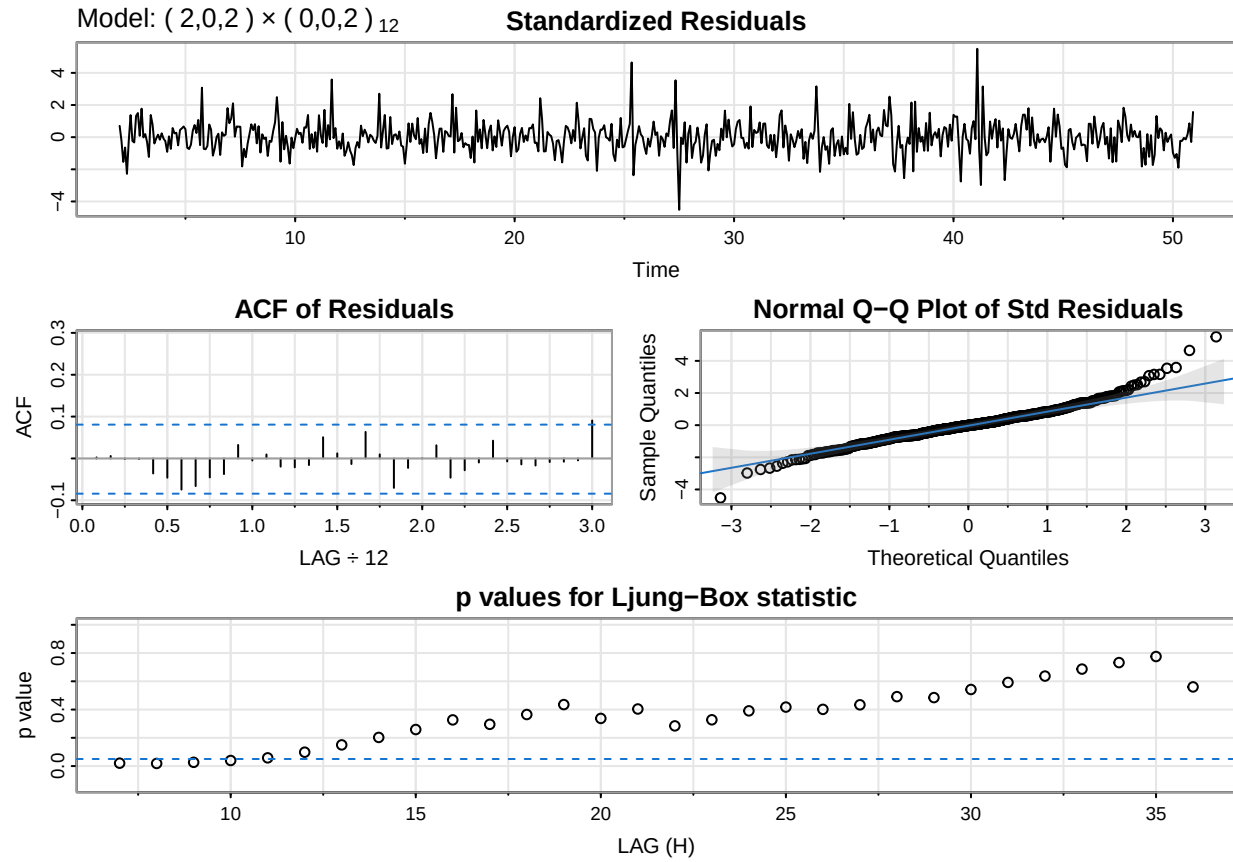
```
sarima(Dxx, p = 2, d = 0, q = 2, P = 0, D = 0,
Q = 2, S = 12,no.constant=F,details=T)
```

```
## initial value 7.181551
## iter 2 value 6.948317
## iter 3 value 6.832951
## iter 4 value 6.830000
## iter 5 value 6.826733
## iter 6 value 6.825579
## iter 7 value 6.824786
## iter 8 value 6.824475
## iter 9 value 6.824433
## iter 10 value 6.824402
## iter 11 value 6.824394
## iter 12 value 6.824392
## iter 13 value 6.824391
## iter 13 value 6.824391
```

```

## iter 13 value 6.824391
## final value 6.824391
## converged
## initial value 6.817929
## iter 2 value 6.817455
## iter 3 value 6.817192
## iter 4 value 6.817157
## iter 5 value 6.817130
## iter 6 value 6.817125
## iter 7 value 6.817122
## iter 8 value 6.817122
## iter 9 value 6.817121
## iter 10 value 6.817121
## iter 10 value 6.817121
## iter 10 value 6.817121
## final value 6.817121
## converged
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1      0.1183 0.0711   1.6623 0.0970
## ar2      0.1653 0.0686   2.4110 0.0162
## ma1     -0.0235 0.0642  -0.3656 0.7148
## ma2      0.4727 0.0594   7.9580 0.0000
## sma1    -0.5968 0.0432 -13.8082 0.0000
## sma2    -0.2061 0.0467  -4.4131 0.0000
## xmean   14.9633 16.6710   0.8976 0.3698
##
## sigma^2 estimated as 817431.2 on 581 degrees of freedom
##
## AIC = 16.49933  AICc = 16.49966  BIC = 16.55888
##

```



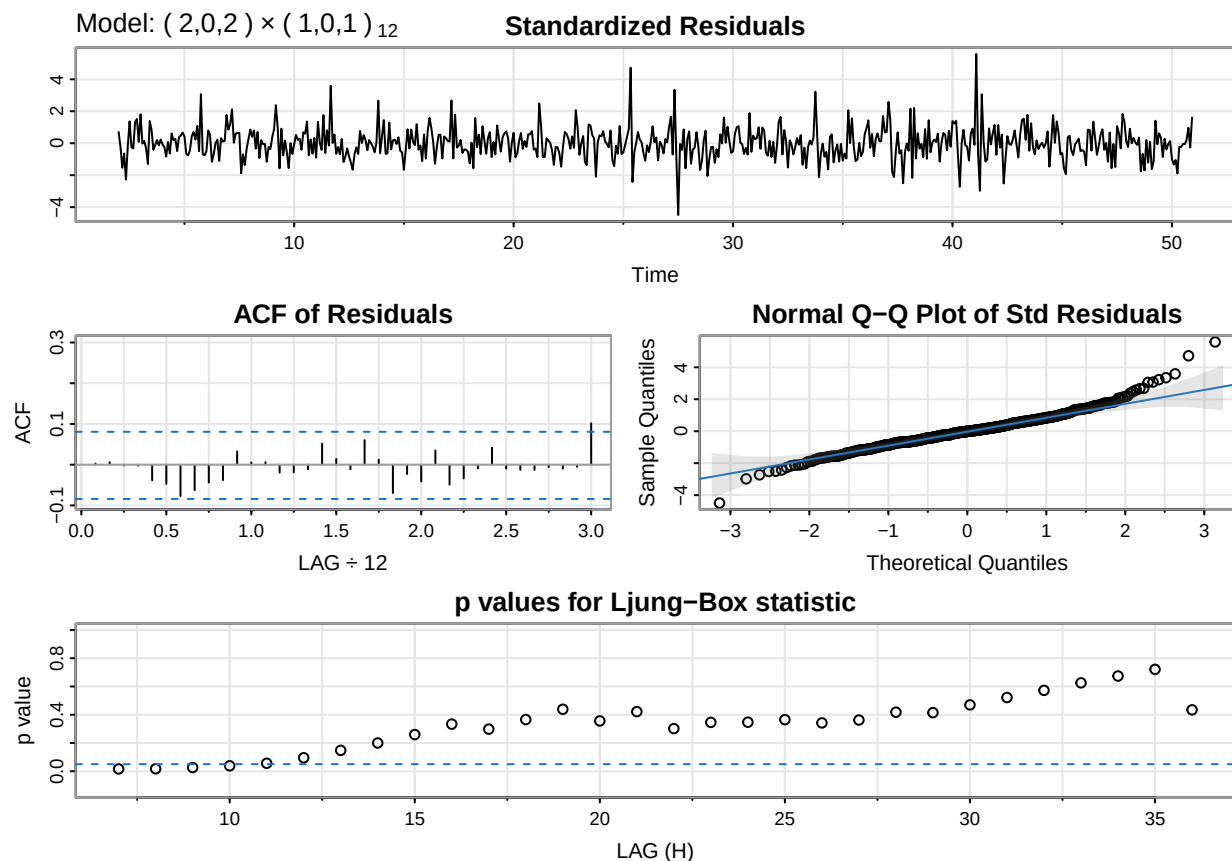
```
sarima(Dxx, p = 2, d = 0, q = 2, P = 1, D = 0,
Q = 1, S = 12,no.constant=F,details=T)
```

```
## initial value 7.178082
## iter 2 value 6.965491
## iter 3 value 6.870518
## iter 4 value 6.853196
## iter 5 value 6.836470
## iter 6 value 6.827709
## iter 7 value 6.825638
## iter 8 value 6.824159
## iter 9 value 6.823944
## iter 10 value 6.823833
## iter 11 value 6.823800
## iter 12 value 6.823783
## iter 13 value 6.823774
## iter 14 value 6.823773
## iter 14 value 6.823773
## iter 14 value 6.823773
## final value 6.823773
## converged
## initial value 6.824842
## iter 2 value 6.821817
## iter 3 value 6.819216
## iter 4 value 6.818794
## iter 5 value 6.818585
```

```

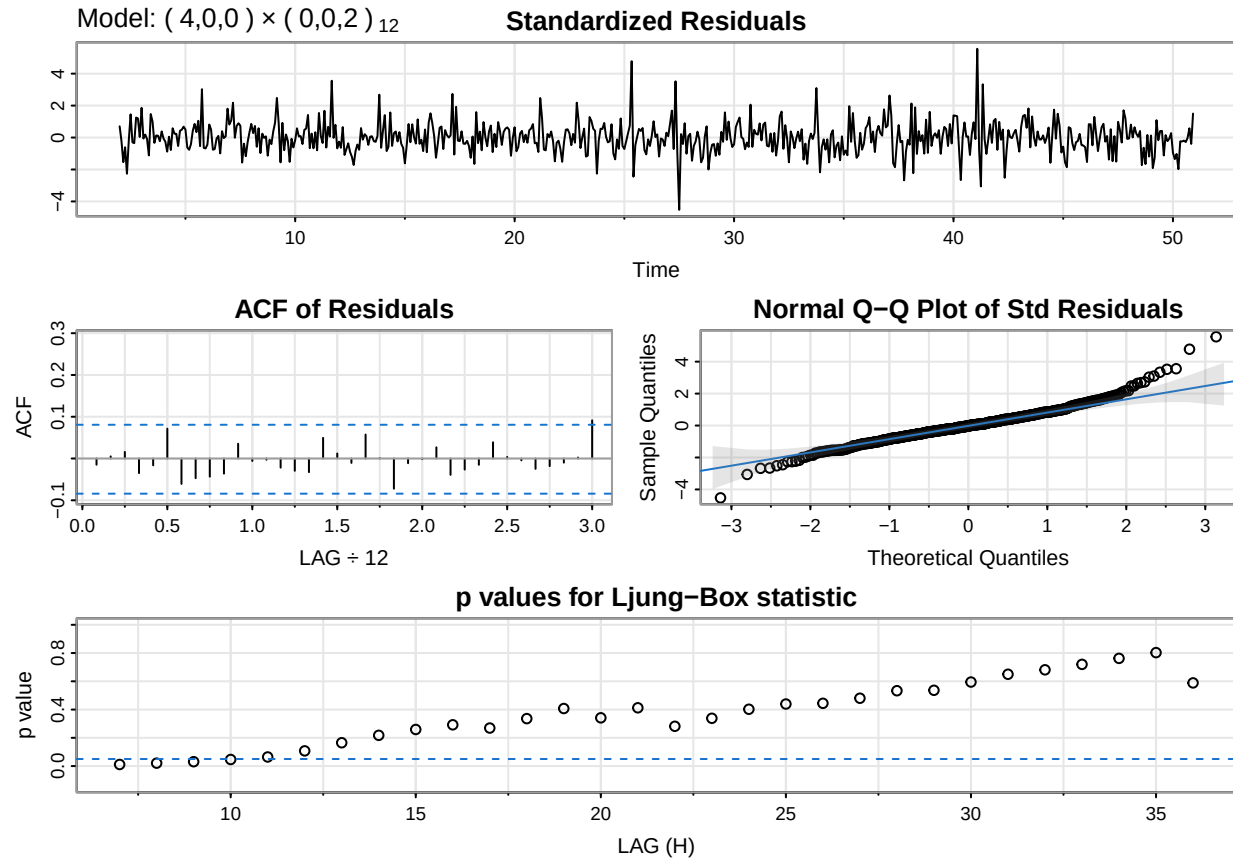
## iter    6 value 6.818510
## iter    7 value 6.818507
## iter    8 value 6.818506
## iter    9 value 6.818505
## iter   10 value 6.818505
## iter   11 value 6.818505
## iter   11 value 6.818505
## iter   11 value 6.818505
## final   value 6.818505
## converged
## <><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1      0.1191  0.0711   1.6743  0.0946
## ar2      0.1642  0.0680   2.4155  0.0160
## ma1     -0.0233  0.0640  -0.3631  0.7167
## ma2      0.4770  0.0584   8.1710  0.0000
## sar1      0.2504  0.0574   4.3595  0.0000
## sma1     -0.8638  0.0355 -24.3055  0.0000
## xmean    14.9665 15.7896   0.9479  0.3436
##
## sigma^2 estimated as 819196.1 on 581 degrees of freedom
##
## AIC = 16.5021  AICc = 16.50243  BIC = 16.56165
##

```



```
sarima(Dxx, p = 4, d = 0, q = 0, P = 0, D = 0,
Q = 2, S = 12,no.constant=F,details=T)
```

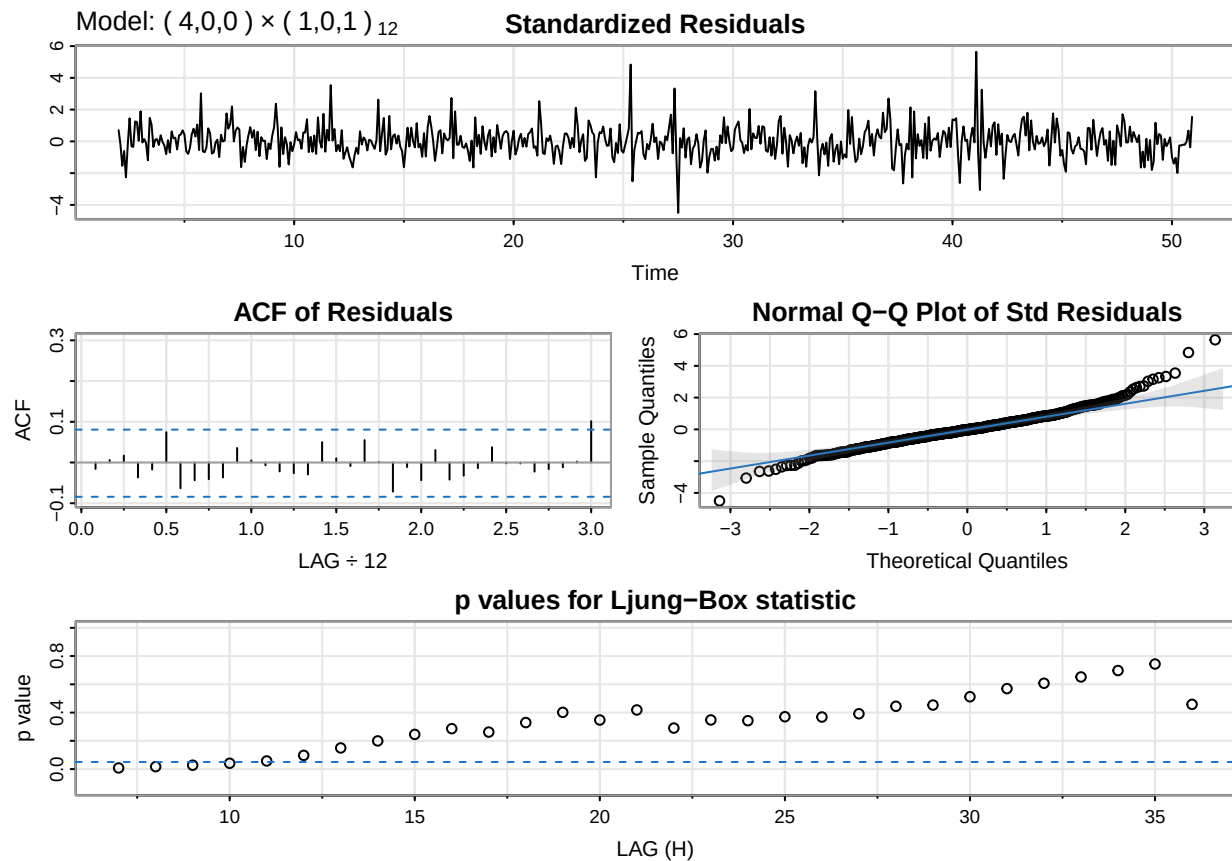
```
## initial value 7.182050
## iter 2 value 6.933936
## iter 3 value 6.890209
## iter 4 value 6.883017
## iter 5 value 6.852329
## iter 6 value 6.851718
## iter 7 value 6.833720
## iter 8 value 6.828888
## iter 9 value 6.825602
## iter 10 value 6.824357
## iter 11 value 6.823465
## iter 12 value 6.823065
## iter 13 value 6.822548
## iter 14 value 6.822520
## iter 15 value 6.822517
## iter 16 value 6.822517
## iter 16 value 6.822517
## final value 6.822517
## converged
## initial value 6.817216
## iter 2 value 6.816476
## iter 3 value 6.816076
## iter 4 value 6.816047
## iter 5 value 6.816038
## iter 6 value 6.816033
## iter 7 value 6.816032
## iter 7 value 6.816032
## iter 7 value 6.816032
## final value 6.816032
## converged
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.0981  0.0399   2.4565  0.0143
## ar2      0.6270  0.0402  15.5818  0.0000
## ar3     -0.0520  0.0402  -1.2918  0.1969
## ar4     -0.2623  0.0402  -6.5300  0.0000
## sma1     -0.5998  0.0434 -13.8168  0.0000
## sma2     -0.2056  0.0466  -4.4135  0.0000
## xmean    14.7285 13.8492   1.0635  0.2880
##
## sigma^2 estimated as 815605.1 on 581 degrees of freedom
##
## AIC = 16.49715 AICc = 16.49748 BIC = 16.5567
##
```



```
sarima(Dxx, p = 4, d = 0, q = 0, P = 1, D = 0,
Q = 1, S = 12,no.constant=F,details=T)
```

```
## initial value 7.177772
## iter 2 value 6.975100
## iter 3 value 6.872229
## iter 4 value 6.844132
## iter 5 value 6.829237
## iter 6 value 6.822942
## iter 7 value 6.822567
## iter 8 value 6.819599
## iter 9 value 6.818489
## iter 10 value 6.817442
## iter 11 value 6.817379
## iter 12 value 6.817377
## iter 12 value 6.817377
## iter 12 value 6.817377
## final value 6.817377
## converged
## initial value 6.820681
## iter 2 value 6.819013
## iter 3 value 6.817726
## iter 4 value 6.817547
## iter 5 value 6.817526
## iter 6 value 6.817513
## iter 6 value 6.817513
```

```
## iter    6 value 6.817513
## final   value 6.817513
## converged
## <><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1      0.0990  0.0399   2.4825 0.0133
## ar2      0.6293  0.0402  15.6507 0.0000
## ar3     -0.0543  0.0402  -1.3525 0.1767
## ar4     -0.2654  0.0400  -6.6276 0.0000
## sar1      0.2468  0.0568   4.3436 0.0000
## sma1     -0.8641  0.0347 -24.9005 0.0000
## xmean    14.6866 13.0964   1.1214 0.2626
##
## sigma^2 estimated as 817592.3 on 581 degrees of freedom
##
## AIC = 16.50011  AICc = 16.50044  BIC = 16.55966
##
```



```
sarima(Dxx, p = 0, d = 0, q = 2, P = 0, D = 0,
Q = 2, S = 12,no.constant=F,details=T)
```

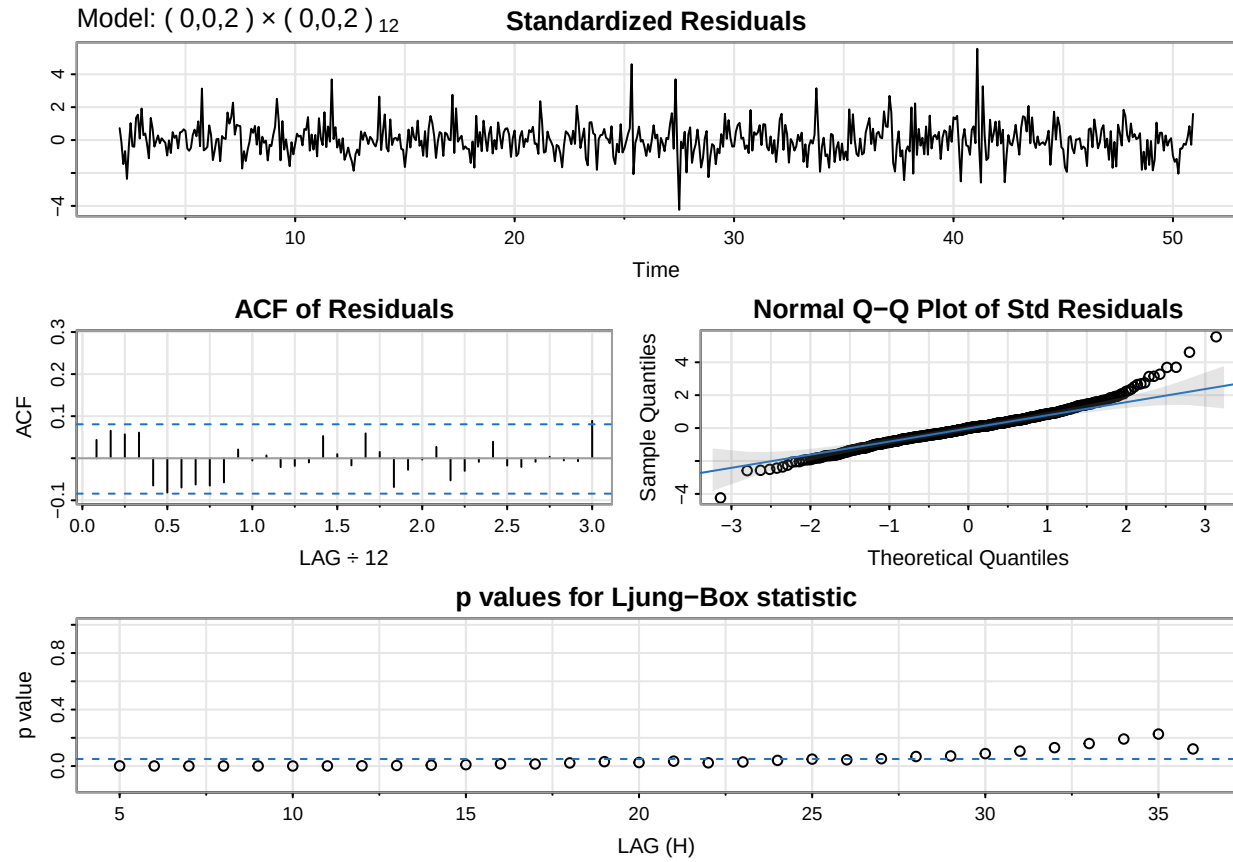
```
## initial value 7.180268
## iter    2 value 6.873812
## iter    3 value 6.848716
## iter    4 value 6.840406
```



```

## iter    5 value 6.835017
## iter    6 value 6.832913
## iter    7 value 6.832425
## iter    8 value 6.832301
## iter    9 value 6.832286
## iter   10 value 6.832286
## iter   10 value 6.832286
## iter   10 value 6.832286
## final   value 6.832286
## converged
## initial  value 6.826296
## iter    2 value 6.825948
## iter    3 value 6.825742
## iter    4 value 6.825725
## iter    5 value 6.825722
## iter    6 value 6.825720
## iter    7 value 6.825719
## iter    7 value 6.825719
## iter    7 value 6.825719
## final   value 6.825719
## converged
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ma1      0.0612  0.0360   1.7020  0.0893
## ma2      0.5838  0.0312  18.6973  0.0000
## sma1     -0.6032  0.0436 -13.8343  0.0000
## sma2     -0.2029  0.0468  -4.3353  0.0000
## xmean    14.3300 13.4943   1.0619  0.2887
##
## sigma^2 estimated as 831351.7 on 583 degrees of freedom
##
## AIC = 16.50972  AICc = 16.5099  BIC = 16.55438
##

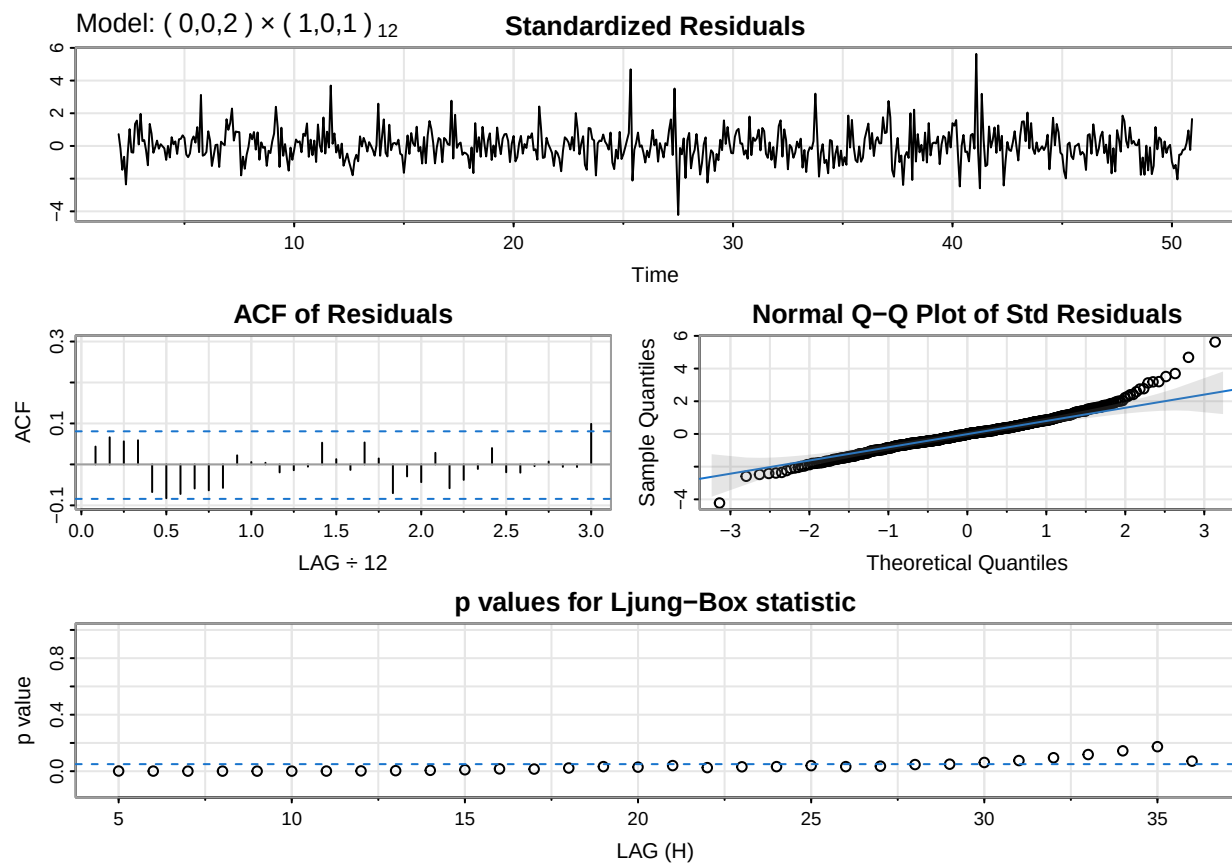
```



```
sarima(Dxx, p = 0, d = 0, q = 2, P = 1, D = 0,
Q = 1, S = 12,no.constant=F,details=T)
```

```
## initial value 7.179797
## iter 2 value 6.902868
## iter 3 value 6.881996
## iter 4 value 6.858252
## iter 5 value 6.838560
## iter 6 value 6.834575
## iter 7 value 6.833208
## iter 8 value 6.832821
## iter 9 value 6.832764
## iter 10 value 6.832753
## iter 11 value 6.832753
## iter 11 value 6.832753
## iter 11 value 6.832753
## final value 6.832753
## converged
## initial value 6.831759
## iter 2 value 6.829681
## iter 3 value 6.827598
## iter 4 value 6.827261
## iter 5 value 6.827195
## iter 6 value 6.827190
## iter 7 value 6.827172
## iter 8 value 6.827172
```

```
## iter    8 value 6.827172
## iter    8 value 6.827172
## final   value 6.827172
## converged
## <><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ma1      0.0621  0.0360   1.7243  0.0852
## ma2      0.5859  0.0310  18.8785  0.0000
## sar1      0.2441  0.0577   4.2307  0.0000
## sma1     -0.8642  0.0358 -24.1506  0.0000
## xmean    14.3255 12.8151   1.1179  0.2641
##
## sigma^2 estimated as 833288.4 on 583 degrees of freedom
##
## AIC = 16.51263  AICc = 16.5128  BIC = 16.55729
##
```



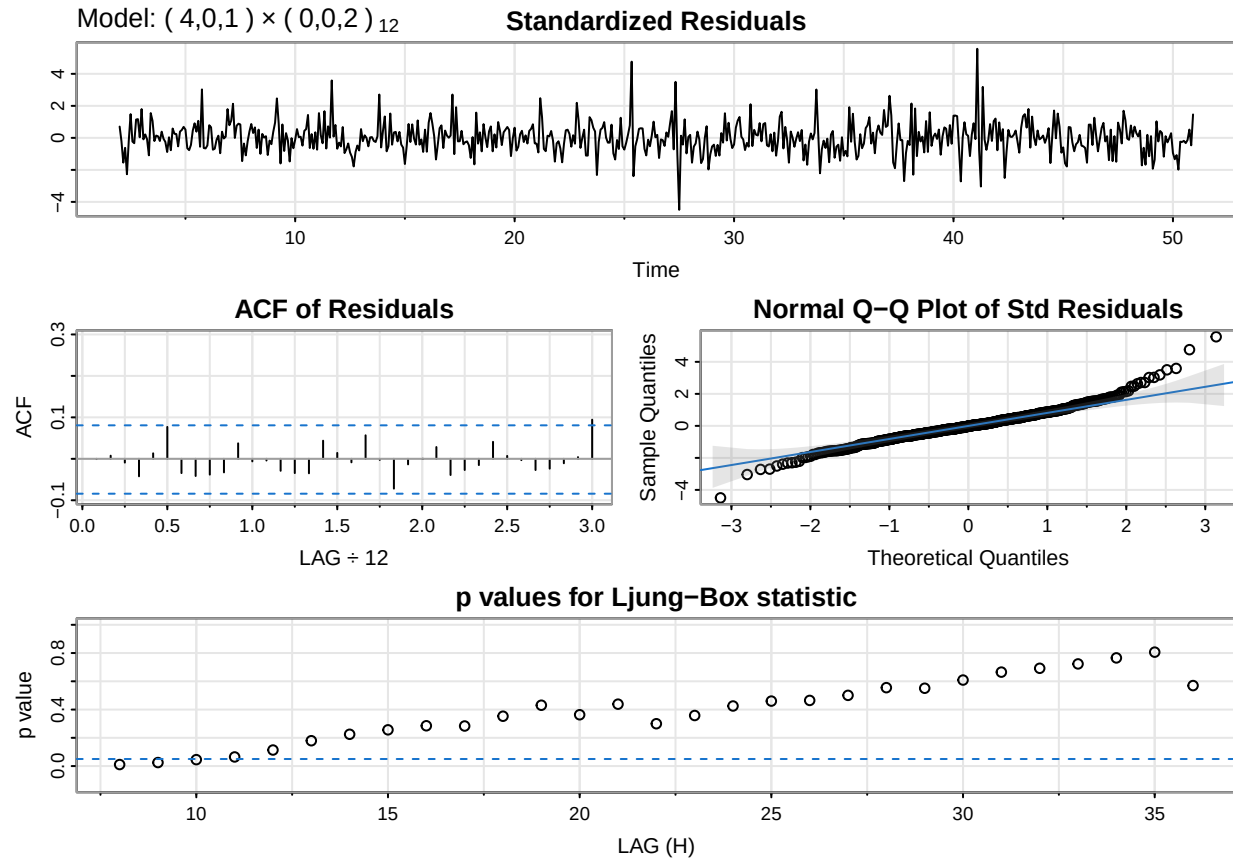
```
sarima(Dxx, p = 4, d = 0, q = 1, P = 0, D = 0,
Q = 2, S = 12,no.constant=F,details=T)
```

```
## initial value 7.182050
## iter    2 value 6.944201
## iter    3 value 6.899067
## iter    4 value 6.860942
## iter    5 value 6.832982
```

```

## iter    6 value 6.824522
## iter    7 value 6.823870
## iter    8 value 6.823328
## iter    9 value 6.822443
## iter   10 value 6.821435
## iter   11 value 6.820389
## iter   12 value 6.819985
## iter   13 value 6.819933
## iter   14 value 6.819924
## iter   15 value 6.819923
## iter   15 value 6.819923
## iter   15 value 6.819923
## final   value 6.819923
## converged
## initial  value 6.815935
## iter    2 value 6.815190
## iter    3 value 6.814741
## iter    4 value 6.814708
## iter    5 value 6.814705
## iter    6 value 6.814693
## iter    7 value 6.814689
## iter    8 value 6.814673
## iter    9 value 6.814653
## iter   10 value 6.814631
## iter   11 value 6.814620
## iter   12 value 6.814618
## iter   13 value 6.814618
## iter   13 value 6.814617
## iter   13 value 6.814617
## final   value 6.814617
## converged
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.2745  0.1462   1.8773  0.0610
## ar2      0.6063  0.0462  13.1349  0.0000
## ar3     -0.1396  0.0817  -1.7086  0.0881
## ar4     -0.2483  0.0445  -5.5784  0.0000
## ma1     -0.1895  0.1506  -1.2586  0.2087
## sma1    -0.5984  0.0435 -13.7564  0.0000
## sma2    -0.2073  0.0469  -4.4232  0.0000
## xmean   14.7153 13.0233   1.1299  0.2590
##
## sigma^2 estimated as 813296.2 on 580 degrees of freedom
##
## AIC = 16.49772  AICc = 16.49815  BIC = 16.56472
##

```



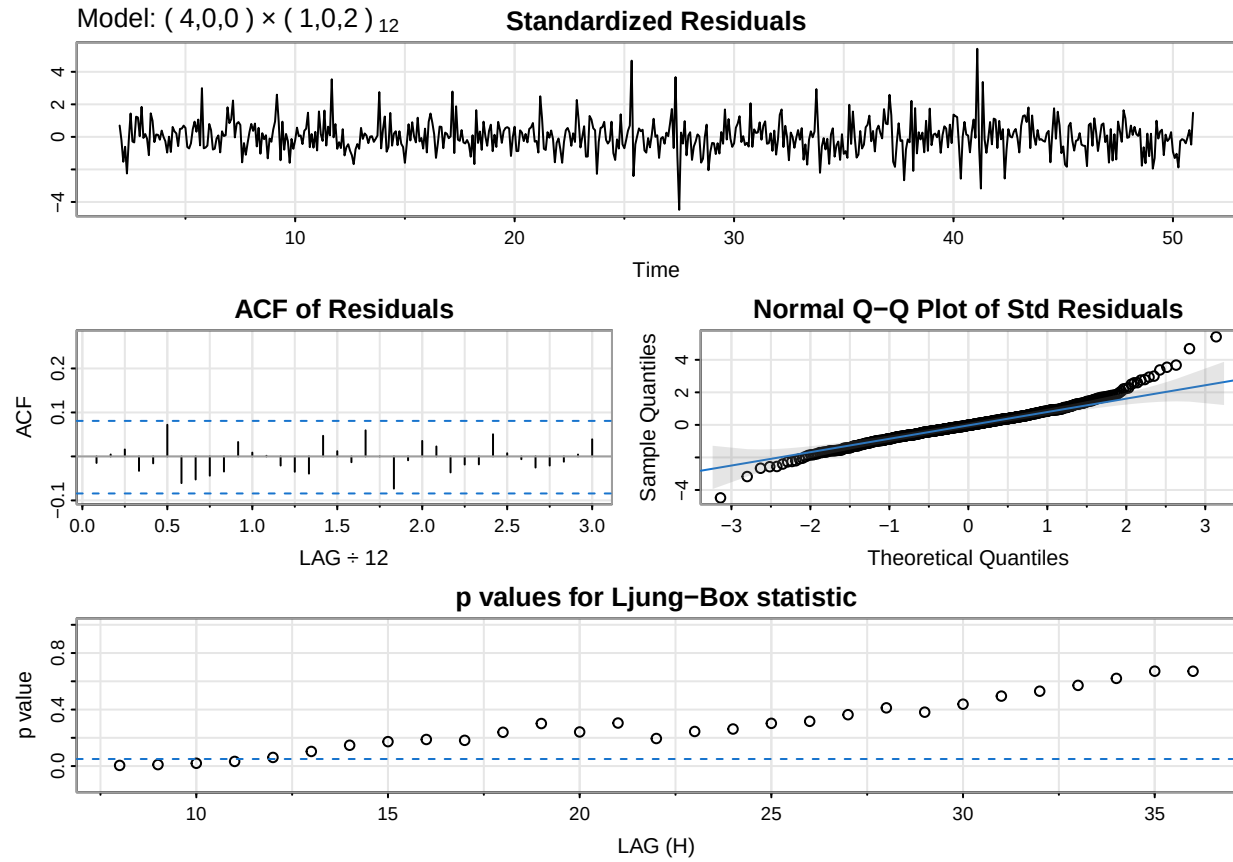
```
sarima(Dxx, p = 4, d = 0, q = 0, P = 1, D = 0,
Q = 2, S = 12,no.constant=F,details=T)
```

```
## initial value 7.177772
## iter 2 value 6.920807
## iter 3 value 6.828443
## iter 4 value 6.813025
## iter 5 value 6.811232
## iter 6 value 6.807737
## iter 7 value 6.807503
## iter 8 value 6.807421
## iter 9 value 6.807394
## iter 10 value 6.807382
## iter 11 value 6.807353
## iter 12 value 6.807304
## iter 13 value 6.807272
## iter 14 value 6.807259
## iter 15 value 6.807257
## iter 15 value 6.807257
## iter 15 value 6.807257
## final value 6.807257
## converged
## initial value 6.814613
## iter 2 value 6.814524
## iter 3 value 6.814484
## iter 4 value 6.814469
```

```

## iter    5 value 6.814466
## iter    6 value 6.814465
## iter    7 value 6.814465
## iter    8 value 6.814463
## iter    9 value 6.814460
## iter   10 value 6.814457
## iter   11 value 6.814455
## iter   12 value 6.814455
## iter   12 value 6.814455
## iter   12 value 6.814455
## final   value 6.814455
## converged
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.0964  0.0400  2.4111  0.0162
## ar2      0.6251  0.0403 15.5104  0.0000
## ar3     -0.0482  0.0404 -1.1950  0.2326
## ar4     -0.2604  0.0403 -6.4662  0.0000
## sar1    -0.3607  0.2095 -1.7213  0.0857
## sma1    -0.2472  0.1985 -1.2453  0.2135
## sma2    -0.4529  0.1322 -3.4266  0.0007
## xmean   14.7025 15.3040  0.9607  0.3371
##
## sigma^2 estimated as 813781.7 on 580 degrees of freedom
##
## AIC = 16.4974  AICc = 16.49782  BIC = 16.56439
##

```



Final model:

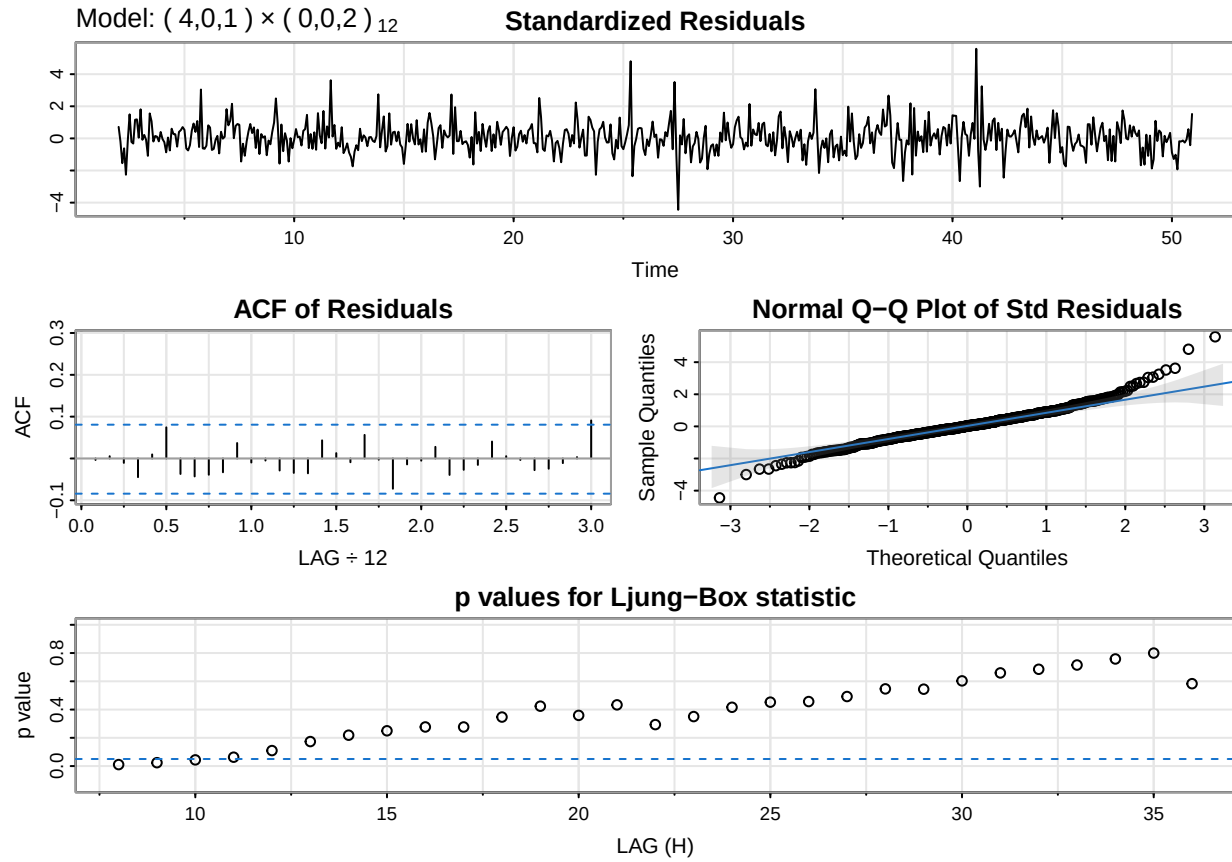
```
sarima(Dxx, p = 4, d = 0, q = 1, P = 0, D = 0,
Q = 2, S = 12,no.constant=T,details=T)
```

```
## initial value 7.182069
## iter 2 value 6.944244
## iter 3 value 6.899142
## iter 4 value 6.861276
## iter 5 value 6.833873
## iter 6 value 6.825166
## iter 7 value 6.824668
## iter 8 value 6.823891
## iter 9 value 6.823006
## iter 10 value 6.822052
## iter 11 value 6.821342
## iter 12 value 6.821133
## iter 13 value 6.821129
## iter 14 value 6.821127
## iter 14 value 6.821127
## iter 14 value 6.821127
## final value 6.821127
## converged
## initial value 6.817017
## iter 2 value 6.816235
## iter 3 value 6.815766
## iter 4 value 6.815734
```

```

## iter    5 value 6.815730
## iter    6 value 6.815720
## iter    7 value 6.815709
## iter    8 value 6.815688
## iter    9 value 6.815670
## iter   10 value 6.815662
## iter   11 value 6.815661
## iter   11 value 6.815661
## iter   11 value 6.815661
## final   value 6.815661
## converged
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1      0.2667 0.1446   1.8447 0.0656
## ar2      0.6089 0.0459  13.2693 0.0000
## ar3     -0.1343 0.0811  -1.6558 0.0983
## ar4     -0.2486 0.0442  -5.6252 0.0000
## ma1     -0.1801 0.1487  -1.2109 0.2264
## sma1    -0.5948 0.0435 -13.6705 0.0000
## sma2    -0.2039 0.0469  -4.3443 0.0000
##
## sigma^2 estimated as 815472.1 on 581 degrees of freedom
##
## AIC = 16.49641  AICc = 16.49674  BIC = 16.55596
##

```

Problem 2

Summary

a.

Frequency 0.027: AR(1) Spectral Density = 8.58566 | CI = [3.51401, 13.45976] | In CI: YES
 Frequency 0.049: AR(1) Spectral Density = 5.49633 | CI = [6.53474, 25.03013] | In CI: NO
 Frequency 0.250: AR(1) Spectral Density = 0.52240 | CI = [0.92678, 3.54986] | In CI: NO

b.

Frequency 0.027: AR(1) Spectral Density = 8.46003 | CI = [3.75334, 14.37649] | In CI: YES

Explanation

For dat2a, of the smooth periodograms tested, the modified Daniell kernel (4,4) was chosen because it was not as jagged as some of the others but still maintained the steepness of the steep portions of the original periodogram as well as any of the others.

The same kernel was selected for dat2b because the plot of different smoothed periodograms is extremely similar.

The added value in part b results in a slightly different confidence interval than that of part a for $\omega = 0.027$ but the true value of the baseline AR(1) is still in this interval. This true AR(1) value for dat2b is slightly different from that of dat2a because the closest index to 0.027 from the smooth periodogram is slightly different for dat2b.

Output

Baseline AR(1):

```
library(TSA) ## ARMAspec
```

```
## Warning: package 'TSA' was built under R version 4.4.2
```

```
##
```

```
## Attaching package: 'TSA'
```

```
## The following objects are masked from 'package:stats':
```

```
##
```

```
##     acf, arima
```

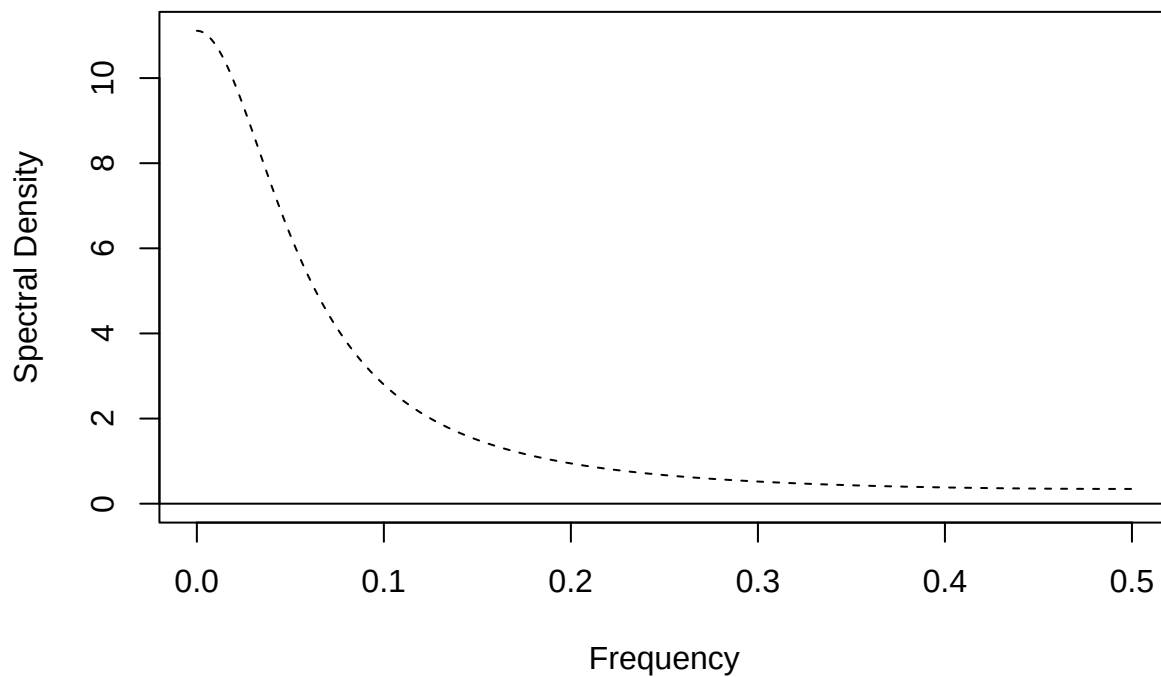
```
## The following object is masked from 'package:utils':
```

```
##
```

```
##     tar
```

```
phi <- .7
```

```
trueARMAspec <- TSA::ARMAspec(model = list(ar=phi),  
                               lty=2) ## function for plotting true
```



Dat2a:

```
xx <- dat2a
```

```
# Reset plot parameters and set smaller margins
```

```
par(mfrow = c(1, 1)) # Single plot layout
```

```
par(mai = c(1, 1, 1, 1)) # Margins: B, L, T, R (reasonable size)
```

```

# Start the plot
plot.new()

# Plot the spectral density with multiple spans
spg <- spec.pgram(x = xx, taper = 0, log = "no", detrend = FALSE,
                  add = FALSE, lwd = 0.5, col = "gray") # First plot (not add=TRUE)

# Overlay additional spectral densities with smoothing
spec.pgram(x = xx, spans = 15, taper = 0, log = "no", detrend = FALSE,
            add = TRUE, col = "red")

spec.pgram(x = xx, spans = 33, taper = 0, log = "no", detrend = FALSE,
            add = TRUE, col = "red")

spec.pgram(x = xx, spans = 71, taper = 0, log = "no", detrend = FALSE,
            add = TRUE, col = "blue")

spec.pgram(x = xx, spans = 171, taper = 0, log = "no", detrend = FALSE,
            add = TRUE, col = "orange")

(myk <- kernel("modified.daniell", m=c(30,30)))

```

```

## mDaniell(30,30)
## coef[-60] = 6.944e-05
## coef[-59] = 2.778e-04
## coef[-58] = 5.556e-04
## coef[-57] = 8.333e-04
## coef[-56] = 1.111e-03
## coef[-55] = 1.389e-03
## coef[-54] = 1.667e-03
## coef[-53] = 1.944e-03
## coef[-52] = 2.222e-03
## coef[-51] = 2.500e-03
## coef[-50] = 2.778e-03
## coef[-49] = 3.056e-03
## coef[-48] = 3.333e-03
## coef[-47] = 3.611e-03
## coef[-46] = 3.889e-03
## coef[-45] = 4.167e-03
## coef[-44] = 4.444e-03
## coef[-43] = 4.722e-03
## coef[-42] = 5.000e-03
## coef[-41] = 5.278e-03
## coef[-40] = 5.556e-03
## coef[-39] = 5.833e-03
## coef[-38] = 6.111e-03
## coef[-37] = 6.389e-03
## coef[-36] = 6.667e-03
## coef[-35] = 6.944e-03
## coef[-34] = 7.222e-03
## coef[-33] = 7.500e-03
## coef[-32] = 7.778e-03
## coef[-31] = 8.056e-03

```

```

## coef[-30] = 8.333e-03
## coef[-29] = 8.611e-03
## coef[-28] = 8.889e-03
## coef[-27] = 9.167e-03
## coef[-26] = 9.444e-03
## coef[-25] = 9.722e-03
## coef[-24] = 1.000e-02
## coef[-23] = 1.028e-02
## coef[-22] = 1.056e-02
## coef[-21] = 1.083e-02
## coef[-20] = 1.111e-02
## coef[-19] = 1.139e-02
## coef[-18] = 1.167e-02
## coef[-17] = 1.194e-02
## coef[-16] = 1.222e-02
## coef[-15] = 1.250e-02
## coef[-14] = 1.278e-02
## coef[-13] = 1.306e-02
## coef[-12] = 1.333e-02
## coef[-11] = 1.361e-02
## coef[-10] = 1.389e-02
## coef[ -9] = 1.417e-02
## coef[ -8] = 1.444e-02
## coef[ -7] = 1.472e-02
## coef[ -6] = 1.500e-02
## coef[ -5] = 1.528e-02
## coef[ -4] = 1.556e-02
## coef[ -3] = 1.583e-02
## coef[ -2] = 1.611e-02
## coef[ -1] = 1.639e-02
## coef[  0] = 1.653e-02
## coef[  1] = 1.639e-02
## coef[  2] = 1.611e-02
## coef[  3] = 1.583e-02
## coef[  4] = 1.556e-02
## coef[  5] = 1.528e-02
## coef[  6] = 1.500e-02
## coef[  7] = 1.472e-02
## coef[  8] = 1.444e-02
## coef[  9] = 1.417e-02
## coef[ 10] = 1.389e-02
## coef[ 11] = 1.361e-02
## coef[ 12] = 1.333e-02
## coef[ 13] = 1.306e-02
## coef[ 14] = 1.278e-02
## coef[ 15] = 1.250e-02
## coef[ 16] = 1.222e-02
## coef[ 17] = 1.194e-02
## coef[ 18] = 1.167e-02
## coef[ 19] = 1.139e-02
## coef[ 20] = 1.111e-02
## coef[ 21] = 1.083e-02
## coef[ 22] = 1.056e-02
## coef[ 23] = 1.028e-02

```

```

## coef[ 24] = 1.000e-02
## coef[ 25] = 9.722e-03
## coef[ 26] = 9.444e-03
## coef[ 27] = 9.167e-03
## coef[ 28] = 8.889e-03
## coef[ 29] = 8.611e-03
## coef[ 30] = 8.333e-03
## coef[ 31] = 8.056e-03
## coef[ 32] = 7.778e-03
## coef[ 33] = 7.500e-03
## coef[ 34] = 7.222e-03
## coef[ 35] = 6.944e-03
## coef[ 36] = 6.667e-03
## coef[ 37] = 6.389e-03
## coef[ 38] = 6.111e-03
## coef[ 39] = 5.833e-03
## coef[ 40] = 5.556e-03
## coef[ 41] = 5.278e-03
## coef[ 42] = 5.000e-03
## coef[ 43] = 4.722e-03
## coef[ 44] = 4.444e-03
## coef[ 45] = 4.167e-03
## coef[ 46] = 3.889e-03
## coef[ 47] = 3.611e-03
## coef[ 48] = 3.333e-03
## coef[ 49] = 3.056e-03
## coef[ 50] = 2.778e-03
## coef[ 51] = 2.500e-03
## coef[ 52] = 2.222e-03
## coef[ 53] = 1.944e-03
## coef[ 54] = 1.667e-03
## coef[ 55] = 1.389e-03
## coef[ 56] = 1.111e-03
## coef[ 57] = 8.333e-04
## coef[ 58] = 5.556e-04
## coef[ 59] = 2.778e-04
## coef[ 60] = 6.944e-05

## plot(myk)
spec.pgram(x=xx, kernel=myk,
            taper=0, log="no", detrend=FALSE,
            add=TRUE, col="green")

(myk <- kernel("modified.daniell", m=c(4,4)))

## mDaniell(4,4)
## coef[-8] = 0.003906
## coef[-7] = 0.015625
## coef[-6] = 0.031250
## coef[-5] = 0.046875
## coef[-4] = 0.062500
## coef[-3] = 0.078125
## coef[-2] = 0.093750
## coef[-1] = 0.109375
## coef[ 0] = 0.117187

```

```

## coef[ 1] = 0.109375
## coef[ 2] = 0.093750
## coef[ 3] = 0.078125
## coef[ 4] = 0.062500
## coef[ 5] = 0.046875
## coef[ 6] = 0.031250
## coef[ 7] = 0.015625
## coef[ 8] = 0.003906

## plot(myk)
spec.pgram(x=xx, kernel=myk,
            taper=0, log="no", detrend=FALSE,
            add=TRUE, col="purple")

(myk <- kernel("modified.daniell", m=c(10, 10)))

## mDaniell(10,10)
## coef[-20] = 0.000625
## coef[-19] = 0.002500
## coef[-18] = 0.005000
## coef[-17] = 0.007500
## coef[-16] = 0.010000
## coef[-15] = 0.012500
## coef[-14] = 0.015000
## coef[-13] = 0.017500
## coef[-12] = 0.020000
## coef[-11] = 0.022500
## coef[-10] = 0.025000
## coef[ -9] = 0.027500
## coef[ -8] = 0.030000
## coef[ -7] = 0.032500
## coef[ -6] = 0.035000
## coef[ -5] = 0.037500
## coef[ -4] = 0.040000
## coef[ -3] = 0.042500
## coef[ -2] = 0.045000
## coef[ -1] = 0.047500
## coef[  0] = 0.048750
## coef[  1] = 0.047500
## coef[  2] = 0.045000
## coef[  3] = 0.042500
## coef[  4] = 0.040000
## coef[  5] = 0.037500
## coef[  6] = 0.035000
## coef[  7] = 0.032500
## coef[  8] = 0.030000
## coef[  9] = 0.027500
## coef[ 10] = 0.025000
## coef[ 11] = 0.022500
## coef[ 12] = 0.020000
## coef[ 13] = 0.017500
## coef[ 14] = 0.015000
## coef[ 15] = 0.012500
## coef[ 16] = 0.010000
## coef[ 17] = 0.007500

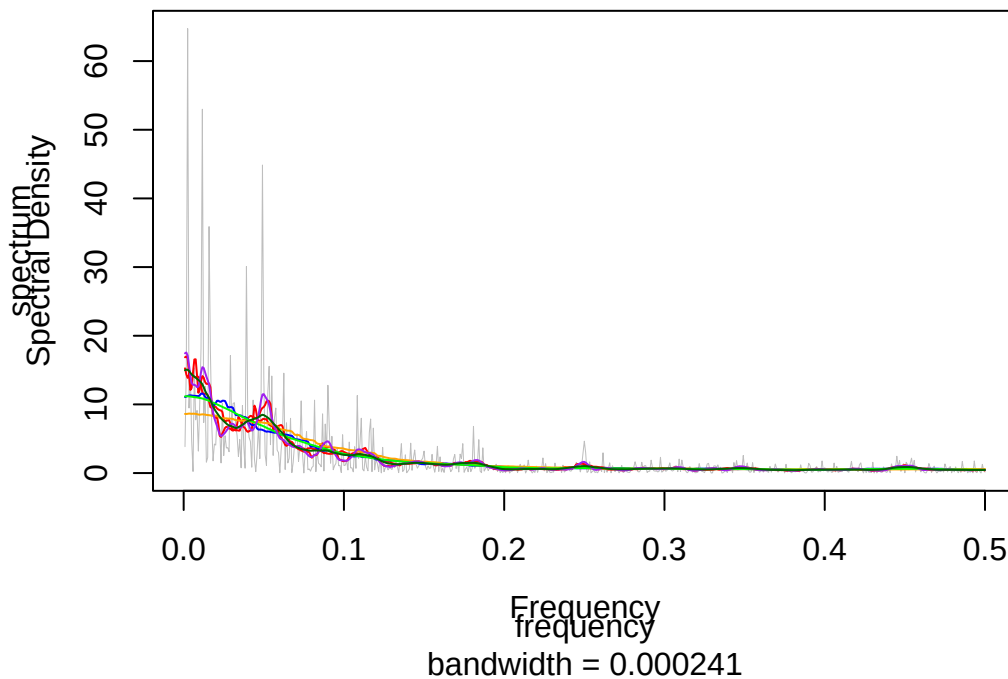
```

```
## coef[ 18] = 0.005000
## coef[ 19] = 0.002500
## coef[ 20] = 0.000625

## plot(myk)
spec.pgram(x=xx, kernel=myk,
            taper=0, log="no", detrend=FALSE,
            add=TRUE, col="darkgreen")

# Add a title and axis labels
title("Spectral Density Estimates with Varying Smoothing Spans")
mtext("Frequency", side = 1, line = 2.5)
mtext("Spectral Density", side = 2, line = 2.5)
```

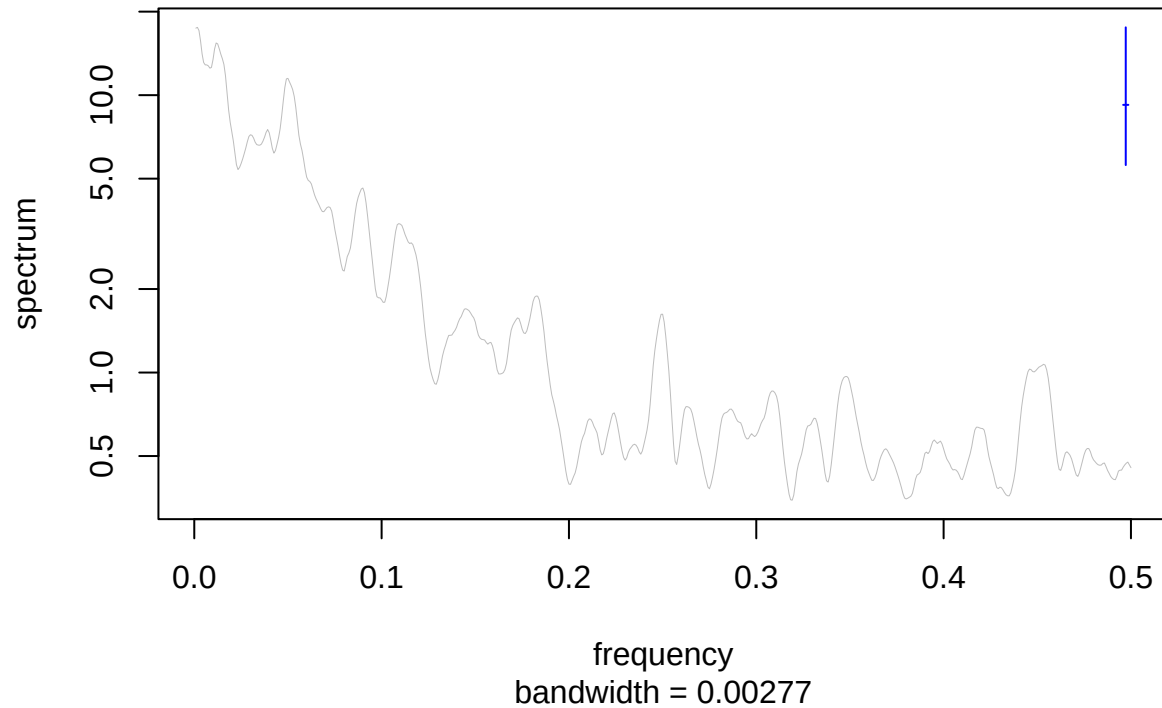
Series: xx Spectral Density Estimates with Varying Smoothing Spans Raw Periodogram



Modified Daniell Kernel (4,4)

```
myk <- kernel("modified.daniell", m=c(4, 4))
spec.pgram(x=xx, kernel=myk,
            taper=0, log="yes", detrend=FALSE,
            lwd=.5, col="gray")
```

Series: xx Smoothed Periodogram



Confidence intervals and tests:

```
# Frequencies of interest
omega <- c(0.027, 0.049, 0.25)

# Spectral density estimate using modified Daniell kernel (m = 4, 4 --> LL = 9)
myk <- kernel("modified.daniell", m = c(4, 4))
spec_est <- spec.pgram(x = xx, kernel = myk, taper = 0, log = "no", detrend = FALSE, plot = FALSE)

# Extract the spectral density estimates at the closest frequencies
closest_indices <- sapply(omega, function(w) which.min(abs(spec_est$freq - w)))
f_hat <- spec_est$spec[closest_indices]

LL <- 9
df <- 2 * LL

# Chi-squared quantiles for 95% CI
q_lower <- qchisq(0.025, df)
q_upper <- qchisq(0.975, df)

# Construct confidence intervals
ci_lower <- f_hat * (df / q_upper)
ci_upper <- f_hat * (df / q_lower)

# Extract true spectral density at the chosen frequencies
true_spectrum <- trueARMAspec$spec[closest_indices]
```



```

# Combine results into a data frame
results <- data.frame(
  Frequency = omega,
  Empirical_Spec = f_hat,
  CI_Lower = ci_lower,
  CI_Upper = ci_upper,
  True_Spec = true_spectrum,
  In_CI = true_spectrum >= ci_lower & true_spectrum <= ci_upper
)

# Print results
print(results)

##   Frequency Empirical_Spec  CI_Lower  CI_Upper True_Spec In_CI
## 1    0.027      6.154660 3.5140057 13.459761  8.585660  TRUE
## 2    0.049     11.445371 6.5347399 25.030134  5.496333 FALSE
## 3    0.250      1.623223 0.9267796  3.549861  0.522398 FALSE

# Display whether the true spectrum falls within the confidence intervals
cat("\nResults:\n")

##
## Results:
for (i in 1:nrow(results)) {
  cat(sprintf("Frequency %.3f: AR(1) Spectral Density = %.5f | CI = [%.5f, %.5f] | In CI: %s\n",
    results$Frequency[i], results$True_Spec[i],
    results$CI_Lower[i], results$CI_Upper[i],
    ifelse(results$In_CI[i], "YES", "NO")))
}

## Frequency 0.027: AR(1) Spectral Density = 8.58566 | CI = [3.51401, 13.45976] | In CI: YES
## Frequency 0.049: AR(1) Spectral Density = 5.49633 | CI = [6.53474, 25.03013] | In CI: NO
## Frequency 0.250: AR(1) Spectral Density = 0.52240 | CI = [0.92678, 3.54986] | In CI: NO

dat2b:
xx <- dat2b

# Reset plot parameters and set smaller margins
par(mfrow = c(1, 1))          # Single plot layout
par(mai = c(1, 1, 1, 1))      # Margins: B, L, T, R (reasonable size)

# Start the plot
plot.new()

# Plot the spectral density with multiple spans
spg <- spec.pgram(x = xx, taper = 0, log = "no", detrend = FALSE,
  add = FALSE, lwd = 0.5, col = "gray") # First plot (not add=TRUE)

# Overlay additional spectral densities with smoothing
spec.pgram(x = xx, spans = 15, taper = 0, log = "no", detrend = FALSE,
  add = TRUE, col = "red")

spec.pgram(x = xx, spans = 33, taper = 0, log = "no", detrend = FALSE,
  add = TRUE, col = "red")

```

```

spec.pgram(x = xx, spans = 71, taper = 0, log = "no", detrend = FALSE,
           add = TRUE, col = "blue")

spec.pgram(x = xx, spans = 171, taper = 0, log = "no", detrend = FALSE,
           add = TRUE, col = "orange")

(myk <- kernel("modified.daniell", m=c(30,30)))

## mDaniell(30,30)
## coef[-60] = 6.944e-05
## coef[-59] = 2.778e-04
## coef[-58] = 5.556e-04
## coef[-57] = 8.333e-04
## coef[-56] = 1.111e-03
## coef[-55] = 1.389e-03
## coef[-54] = 1.667e-03
## coef[-53] = 1.944e-03
## coef[-52] = 2.222e-03
## coef[-51] = 2.500e-03
## coef[-50] = 2.778e-03
## coef[-49] = 3.056e-03
## coef[-48] = 3.333e-03
## coef[-47] = 3.611e-03
## coef[-46] = 3.889e-03
## coef[-45] = 4.167e-03
## coef[-44] = 4.444e-03
## coef[-43] = 4.722e-03
## coef[-42] = 5.000e-03
## coef[-41] = 5.278e-03
## coef[-40] = 5.556e-03
## coef[-39] = 5.833e-03
## coef[-38] = 6.111e-03
## coef[-37] = 6.389e-03
## coef[-36] = 6.667e-03
## coef[-35] = 6.944e-03
## coef[-34] = 7.222e-03
## coef[-33] = 7.500e-03
## coef[-32] = 7.778e-03
## coef[-31] = 8.056e-03
## coef[-30] = 8.333e-03
## coef[-29] = 8.611e-03
## coef[-28] = 8.889e-03
## coef[-27] = 9.167e-03
## coef[-26] = 9.444e-03
## coef[-25] = 9.722e-03
## coef[-24] = 1.000e-02
## coef[-23] = 1.028e-02
## coef[-22] = 1.056e-02
## coef[-21] = 1.083e-02
## coef[-20] = 1.111e-02
## coef[-19] = 1.139e-02
## coef[-18] = 1.167e-02
## coef[-17] = 1.194e-02

```

```
## coef[-16] = 1.222e-02
## coef[-15] = 1.250e-02
## coef[-14] = 1.278e-02
## coef[-13] = 1.306e-02
## coef[-12] = 1.333e-02
## coef[-11] = 1.361e-02
## coef[-10] = 1.389e-02
## coef[ -9] = 1.417e-02
## coef[ -8] = 1.444e-02
## coef[ -7] = 1.472e-02
## coef[ -6] = 1.500e-02
## coef[ -5] = 1.528e-02
## coef[ -4] = 1.556e-02
## coef[ -3] = 1.583e-02
## coef[ -2] = 1.611e-02
## coef[ -1] = 1.639e-02
## coef[  0] = 1.653e-02
## coef[  1] = 1.639e-02
## coef[  2] = 1.611e-02
## coef[  3] = 1.583e-02
## coef[  4] = 1.556e-02
## coef[  5] = 1.528e-02
## coef[  6] = 1.500e-02
## coef[  7] = 1.472e-02
## coef[  8] = 1.444e-02
## coef[  9] = 1.417e-02
## coef[ 10] = 1.389e-02
## coef[ 11] = 1.361e-02
## coef[ 12] = 1.333e-02
## coef[ 13] = 1.306e-02
## coef[ 14] = 1.278e-02
## coef[ 15] = 1.250e-02
## coef[ 16] = 1.222e-02
## coef[ 17] = 1.194e-02
## coef[ 18] = 1.167e-02
## coef[ 19] = 1.139e-02
## coef[ 20] = 1.111e-02
## coef[ 21] = 1.083e-02
## coef[ 22] = 1.056e-02
## coef[ 23] = 1.028e-02
## coef[ 24] = 1.000e-02
## coef[ 25] = 9.722e-03
## coef[ 26] = 9.444e-03
## coef[ 27] = 9.167e-03
## coef[ 28] = 8.889e-03
## coef[ 29] = 8.611e-03
## coef[ 30] = 8.333e-03
## coef[ 31] = 8.056e-03
## coef[ 32] = 7.778e-03
## coef[ 33] = 7.500e-03
## coef[ 34] = 7.222e-03
## coef[ 35] = 6.944e-03
## coef[ 36] = 6.667e-03
## coef[ 37] = 6.389e-03
```

```

## coef[ 38] = 6.111e-03
## coef[ 39] = 5.833e-03
## coef[ 40] = 5.556e-03
## coef[ 41] = 5.278e-03
## coef[ 42] = 5.000e-03
## coef[ 43] = 4.722e-03
## coef[ 44] = 4.444e-03
## coef[ 45] = 4.167e-03
## coef[ 46] = 3.889e-03
## coef[ 47] = 3.611e-03
## coef[ 48] = 3.333e-03
## coef[ 49] = 3.056e-03
## coef[ 50] = 2.778e-03
## coef[ 51] = 2.500e-03
## coef[ 52] = 2.222e-03
## coef[ 53] = 1.944e-03
## coef[ 54] = 1.667e-03
## coef[ 55] = 1.389e-03
## coef[ 56] = 1.111e-03
## coef[ 57] = 8.333e-04
## coef[ 58] = 5.556e-04
## coef[ 59] = 2.778e-04
## coef[ 60] = 6.944e-05

## plot(myk)
spec.pgram(x=xx, kernel=myk,
            taper=0, log="no", detrend=FALSE,
            add=TRUE, col="green")

(myk <- kernel("modified.daniell", m=c(4,4)))

## mDaniell(4,4)
## coef[-8] = 0.003906
## coef[-7] = 0.015625
## coef[-6] = 0.031250
## coef[-5] = 0.046875
## coef[-4] = 0.062500
## coef[-3] = 0.078125
## coef[-2] = 0.093750
## coef[-1] = 0.109375
## coef[ 0] = 0.117187
## coef[ 1] = 0.109375
## coef[ 2] = 0.093750
## coef[ 3] = 0.078125
## coef[ 4] = 0.062500
## coef[ 5] = 0.046875
## coef[ 6] = 0.031250
## coef[ 7] = 0.015625
## coef[ 8] = 0.003906

## plot(myk)
spec.pgram(x=xx, kernel=myk,
            taper=0, log="no", detrend=FALSE,
            add=TRUE, col="purple")

```

```

(myk <- kernel("modified.daniell", m=c(10, 10)))

## mDaniell(10,10)
## coef[-20] = 0.000625
## coef[-19] = 0.002500
## coef[-18] = 0.005000
## coef[-17] = 0.007500
## coef[-16] = 0.010000
## coef[-15] = 0.012500
## coef[-14] = 0.015000
## coef[-13] = 0.017500
## coef[-12] = 0.020000
## coef[-11] = 0.022500
## coef[-10] = 0.025000
## coef[ -9] = 0.027500
## coef[ -8] = 0.030000
## coef[ -7] = 0.032500
## coef[ -6] = 0.035000
## coef[ -5] = 0.037500
## coef[ -4] = 0.040000
## coef[ -3] = 0.042500
## coef[ -2] = 0.045000
## coef[ -1] = 0.047500
## coef[  0] = 0.048750
## coef[  1] = 0.047500
## coef[  2] = 0.045000
## coef[  3] = 0.042500
## coef[  4] = 0.040000
## coef[  5] = 0.037500
## coef[  6] = 0.035000
## coef[  7] = 0.032500
## coef[  8] = 0.030000
## coef[  9] = 0.027500
## coef[ 10] = 0.025000
## coef[ 11] = 0.022500
## coef[ 12] = 0.020000
## coef[ 13] = 0.017500
## coef[ 14] = 0.015000
## coef[ 15] = 0.012500
## coef[ 16] = 0.010000
## coef[ 17] = 0.007500
## coef[ 18] = 0.005000
## coef[ 19] = 0.002500
## coef[ 20] = 0.000625

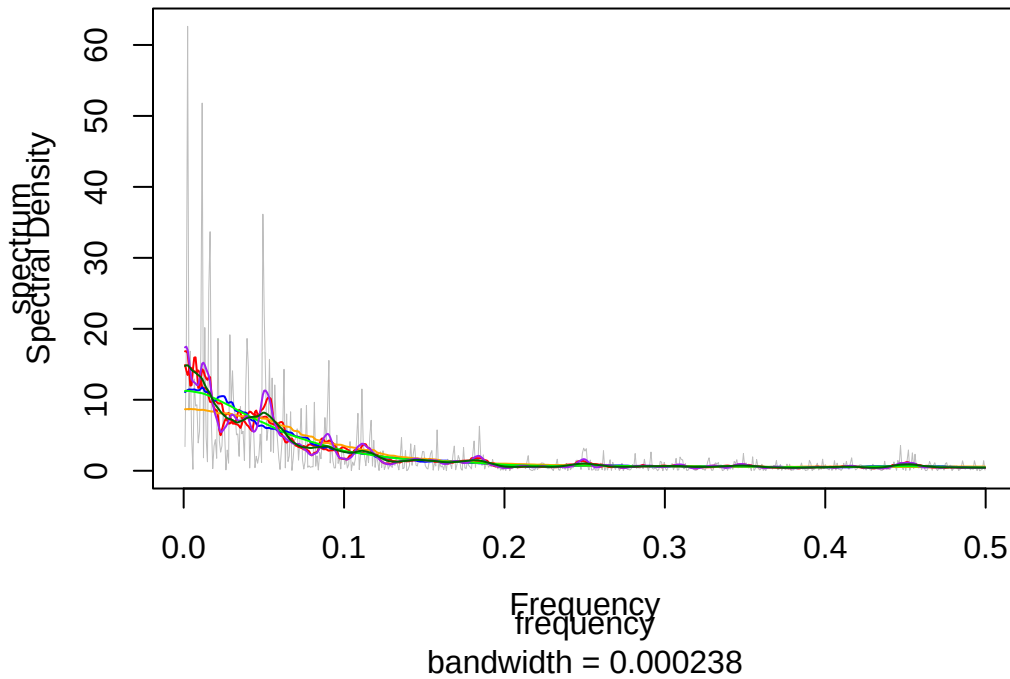
## plot(myk)
spec.pgram(x=xx, kernel=myk,
            taper=0, log="no", detrend=FALSE,
            add=TRUE, col="darkgreen")

# Add a title and axis labels
title("Spectral Density Estimates with Varying Smoothing Spans")
mtext("Frequency", side = 1, line = 2.5)

```

```
mtext("Spectral Density", side = 2, line = 2.5)
```

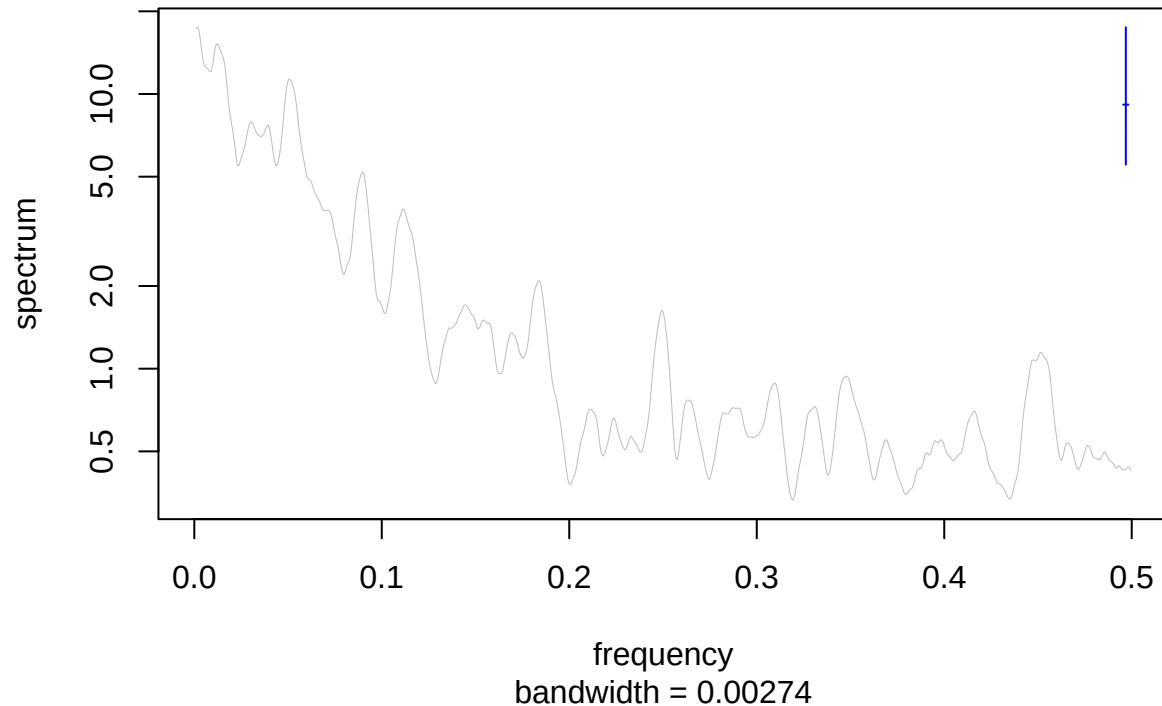
Spectral Density Estimates with Varying Smoothing Spans



Modified Daniell Kernel (4,4)

```
myk <- kernel("modified.daniell", m=c(4, 4))
spec.pgram(x=xx, kernel=myk,
            taper=0, log="yes", detrend=FALSE,
            lwd=.5, col="gray")
```

Series: xx Smoothed Periodogram



Confidence intervals and tests:

```
# Frequencies of interest
omega <- c(0.027)

# Spectral density estimate using modified Daniell kernel (m = 4, 4 --> LL = 9)
myk <- kernel("modified.daniell", m = c(4, 4))
spec_est <- spec.pgram(x = xx, kernel = myk, taper = 0, log = "no", detrend = FALSE, plot = FALSE)

# Extract the spectral density estimates at the closest frequencies
closest_indices <- sapply(omega, function(w) which.min(abs(spec_est$freq - w)))
f_hat <- spec_est$spec[closest_indices]

LL <- 9
df <- 2 * LL

# Chi-squared quantiles for 95% CI
q_lower <- qchisq(0.025, df)
q_upper <- qchisq(0.975, df)

# Construct confidence intervals
ci_lower <- f_hat * (df / q_upper)
ci_upper <- f_hat * (df / q_lower)

# Extract true spectral density at the chosen frequencies
true_spectrum <- trueARMAspec$spec[closest_indices]
```

```

# Combine results into a data frame
results <- data.frame(
  Frequency = omega,
  Empirical_Spec = f_hat,
  CI_Lower = ci_lower,
  CI_Upper = ci_upper,
  True_Spec = true_spectrum,
  In_CI = true_spectrum >= ci_lower & true_spectrum <= ci_upper
)

# Print results
print(results)

##      Frequency Empirical_Spec CI_Lower CI_Upper True_Spec In_CI
## 1      0.027      6.573845  3.75334 14.37649  8.460027  TRUE

# Display whether the true spectrum falls within the confidence intervals
cat("\nResults:\n")

##
## Results:
for (i in 1:nrow(results)) {
  cat(sprintf("Frequency %.3f: AR(1) Spectral Density = %.5f | CI = [%.5f, %.5f] | In CI: %s\n",
    results$Frequency[i], results$True_Spec[i],
    results$CI_Lower[i], results$CI_Upper[i],
    ifelse(results$In_CI[i], "YES", "NO")))
}

```

```
## Frequency 0.027: AR(1) Spectral Density = 8.46003 | CI = [3.75334, 14.37649] | In CI: YES
```

Problem 3

Summary

For the series Y_t I chose an ARMA(1,1) model including the intercept with exogenous regressors X_{t+8} and X_{t+11} .

Coefficients:

	Estimate	SE	t.value	p.value
ar1	0.6878	0.0308	22.3412	0.0000
ma1	0.7379	0.0298	24.7469	0.0000
intercept	94.3697	2.5859	36.4939	0.0000
XL8	-0.2148	0.0990	-2.1701	0.0304
XL11	0.2368	0.0958	2.4711	0.0137

Explanation

Both X_t and Y_t are stationary and do not have changing variance over time or seasonality. The ACF and PACF of X_t show 1 significant lag in the ACF and 5 significant lags in the PACF, so models with a maximum p of 1 and a maximum q of 5 were considered. The ARMA(2,1) model had an AICc and BIC very close to the best and had by far the best LBP values so it was chosen for prewhitening. The intercept was included because it was significant at all levels.

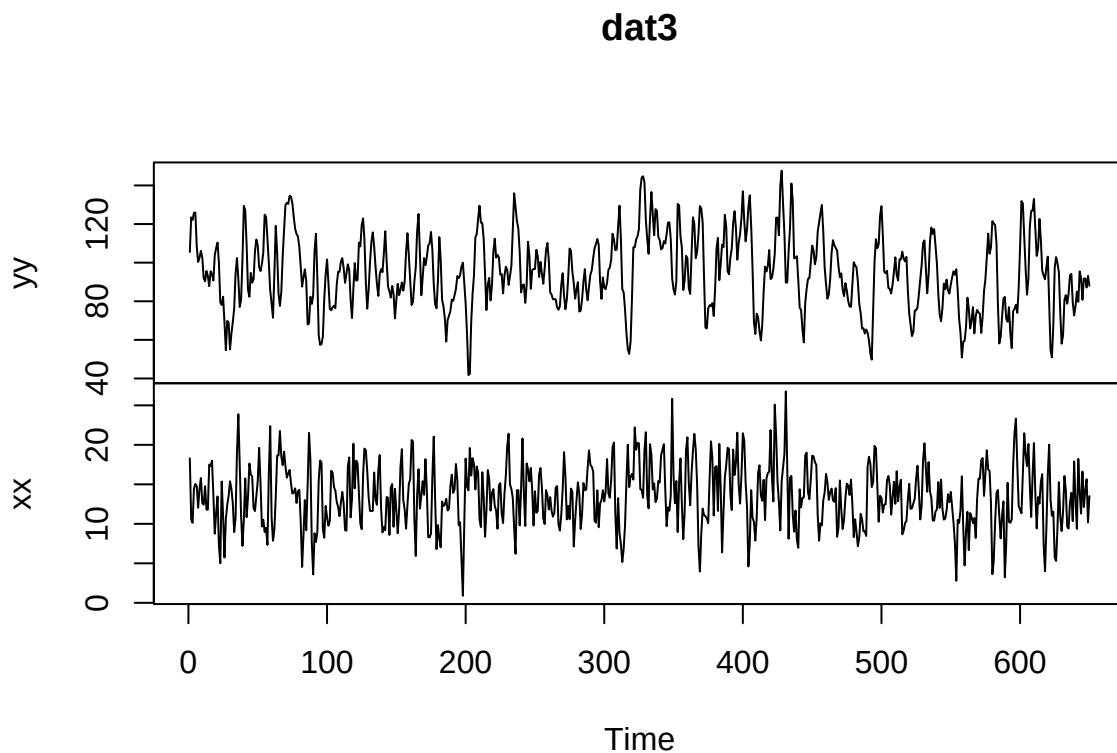
After prewhiteneing both X_t and Y_t with this model, the ccf shows significant correlations at -8, -11, -14, -17, -20, and -23. When testing combinations of these lags, it was found that only -8 and -11 were significant in a regression model. Thus X_{t+8} and X_{t+11} were used in the design matrix with Y_t .

For the model of Y_t including the exogenous regressors X_{t+8} and X_{t+11} , the ACF and PACF show 3 significant lags in the ACF and 7 significant lags in the PACF, so models with a maximum p of 7 and a maximum q of 3 were considered. Of the ones with an AICc and BIC within 10 of the minimum, the ARMA(1,1) had the best BIC and strong diagnostic plots, so it was the chosen model. The intercept was included because it was significant at all levels.

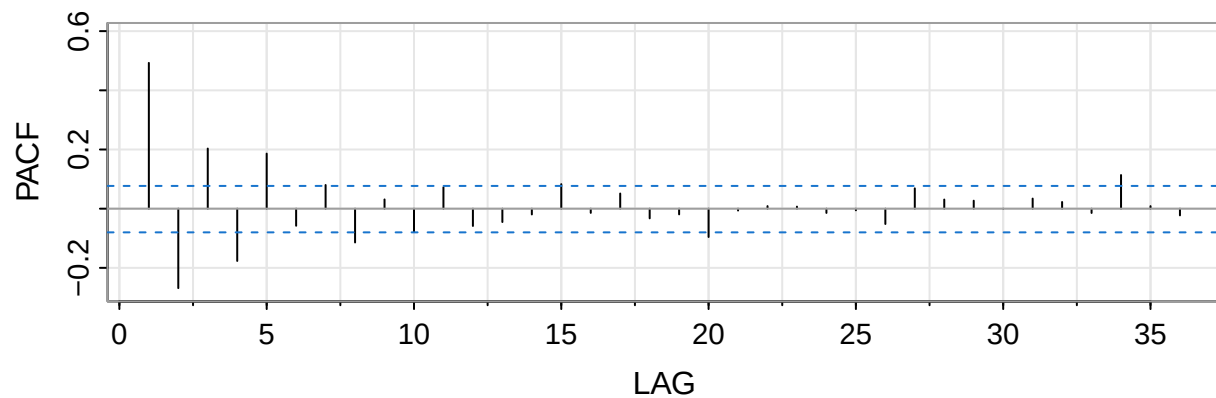
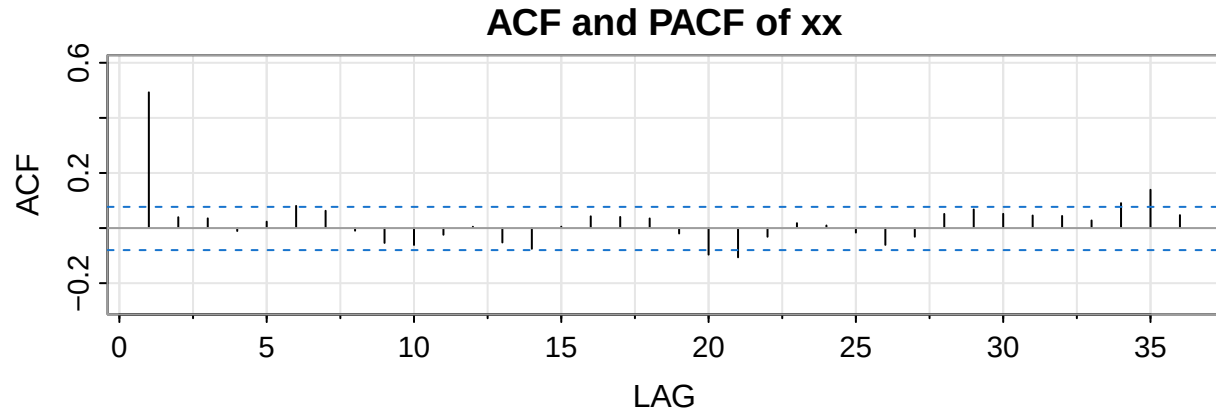
Output

```
# Access the 'xx' column  
xx <- dat3[, "xx"]
```

```
plot(dat3)
```



```
# Compute and plot the ACF and PACF of 'xx'  
invisible(acf2(xx, main = "ACF and PACF of xx"))
```



```
maxAR<-5+1
maxMA<-1+1
fits_all<-vector("list",length=maxAR)
AICmin<-BICmin<-Inf
for (ii in 1:maxAR){
  fits_all[[ii]]<-vector("list",length=maxMA)
  for (jj in 1:maxMA){
    fits_all[[ii]][[jj]]<-invisible(sarima(xx,p=ii-1,d=0,q=jj-1,no.constant=FALSE, details=FALSE))
    if(fits_all[[ii]][[jj]]$ICs["AICc"]<AICmin) AICmin<-fits_all[[ii]][[jj]]$ICs["AICc"]
    if(fits_all[[ii]][[jj]]$ICs["BIC"]<BICmin) BICmin<-fits_all[[ii]][[jj]]$ICs["BIC"]
  }
}
```

```
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## xmean  13.5508 0.1533 88.3772      0
##
## sigma^2 estimated as 15.28137 on 649 degrees of freedom
##
## AIC = 5.570665  AICc = 5.570675  BIC = 5.584441
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
```

```

## ma1      0.8387 0.0245 34.2841      0
## xmean    13.5587 0.2198 61.7004      0
##
## sigma^2 estimated as 9.297686 on 648 degrees of freedom
##
## AIC = 5.078743  AICc = 5.078772  BIC = 5.099406
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.4931 0.0341 14.4597      0
## xmean    13.5582 0.2627 51.6070      0
##
## sigma^2 estimated as 11.56365 on 648 degrees of freedom
##
## AIC = 5.295403  AICc = 5.295431  BIC = 5.316066
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1     -0.0544 0.0482 -1.1273    0.26
## ma1      0.8575 0.0269 31.8826    0.00
## xmean    13.5577 0.2104 64.4481    0.00
##
## sigma^2 estimated as 9.279382 on 647 degrees of freedom
##
## AIC = 5.079883  AICc = 5.07994  BIC = 5.107433
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.6277 0.0378 16.5998      0
## ar2     -0.2715 0.0378 -7.1766      0
## xmean    13.5589 0.1994 68.0128      0
##
## sigma^2 estimated as 10.7122 on 647 degrees of freedom
##
## AIC = 5.222233  AICc = 5.22229  BIC = 5.249784
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1     -0.0756 0.0480 -1.5759  0.1155
## ar2      0.0725 0.0459  1.5813  0.1143
## ma1      0.8816 0.0263 33.5024  0.0000
## xmean    13.5583 0.2235 60.6595  0.0000
##
## sigma^2 estimated as 9.243834 on 646 degrees of freedom
##
## AIC = 5.079151  AICc = 5.079247  BIC = 5.113589

```

```

##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.6838 0.0385 17.7833      0
## ar2     -0.4018 0.0443 -9.0749      0
## ar3      0.2064 0.0385  5.3645      0
## xmean    13.5607 0.2452 55.3155      0
##
## sigma^2 estimated as 10.25636 on 646 degrees of freedom
##
## AIC = 5.182027  AICc = 5.182122  BIC = 5.216465
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1     -0.0737 0.0498 -1.4799  0.1394
## ar2      0.0709 0.0472  1.5044  0.1330
## ar3      0.0069 0.0451  0.1526  0.8788
## ma1      0.8793 0.0308 28.5574  0.0000
## xmean    13.5581 0.2249 60.2978  0.0000
##
## sigma^2 estimated as 9.243515 on 645 degrees of freedom
##
## AIC = 5.082193  AICc = 5.082336  BIC = 5.123518
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1      0.7208 0.0387 18.6475      0
## ar2     -0.4741 0.0463 -10.2473      0
## ar3      0.3301 0.0463  7.1345      0
## ar4     -0.1797 0.0387 -4.6462      0
## xmean    13.5596 0.2050 66.1597      0
##
## sigma^2 estimated as 9.924728 on 645 degrees of freedom
##
## AIC = 5.152438  AICc = 5.152581  BIC = 5.193764
##
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1     -0.0686 0.0540 -1.2706  0.2043
## ar2      0.0676 0.0493  1.3711  0.1708
## ar3      0.0096 0.0466  0.2060  0.8369
## ar4     -0.0122 0.0458 -0.2676  0.7891
## ma1      0.8743 0.0374 23.3641  0.0000
## xmean    13.5576 0.2225 60.9309  0.0000
##
## sigma^2 estimated as 9.242517 on 644 degrees of freedom

```

```

##
## AIC = 5.085159  AICc = 5.08536  BIC = 5.133373
##
## <><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1      0.7552 0.0386  19.5760      0
## ar2     -0.5369 0.0472 -11.3804      0
## ar3      0.4204 0.0490   8.5838      0
## ar4     -0.3177 0.0472  -6.7374      0
## ar5      0.1902 0.0386   4.9327      0
## xmean    13.5614 0.2477  54.7513      0
##
## sigma^2 estimated as 9.564273 on 644 degrees of freedom
##
## AIC = 5.118804  AICc = 5.119005  BIC = 5.167018
##
## <><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE  t.value p.value
## ar1     -0.0726 0.0577  -1.2582  0.2088
## ar2      0.0708 0.0519   1.3630  0.1733
## ar3      0.0075 0.0476   0.1585  0.8741
## ar4     -0.0103 0.0468  -0.2201  0.8259
## ar5     -0.0080 0.0460  -0.1740  0.8619
## ma1      0.8782 0.0422  20.8061  0.0000
## xmean    13.5580 0.2210  61.3437  0.0000
##
## sigma^2 estimated as 9.242063 on 643 degrees of freedom
##
## AIC = 5.08819  AICc = 5.088458  BIC = 5.143291
##
nn <- length(xx)
#look at AICs and BICs
print("check AICcs")

## [1] "check AICcs"
## [1] "check AICcs"
for (ii in 1:maxAR){
  for(jj in 1:maxMA){
    print(paste0(ii-1,"",jj-1))
    print(((fits_all[[ii]][[jj]])$ICs["AICc"] - AICmin) * nn)
  }
}

## [1] "00"
##      AICc
## 319.7372
## [1] "01"
##      AICc
##      0

```

```

## [1] "10"
##      AICc
## 140.8287
## [1] "11"
##      AICc
## 0.7593998
## [1] "20"
##      AICc
## 93.28719
## [1] "21"
##      AICc
## 0.3087791
## [1] "30"
##      AICc
## 67.17777
## [1] "31"
##      AICc
## 2.316835
## [1] "40"
##      AICc
## 47.97615
## [1] "41"
##      AICc
## 4.282729
## [1] "50"
##      AICc
## 26.15193
## [1] "51"
##      AICc
## 6.296258

print("check BICs")

## [1] "check BICs"

## [1] "check BICs"
for (ii in 1:maxAR){
  for(jj in 1:maxMA){
    print(paste0(ii-1,"",jj-1))
    print(((fits_all[[ii]][[jj]])$ICs["BIC"] - BICmin) * nn)
  }
}

## [1] "00"
##      BIC
## 315.2726
## [1] "01"
##      BIC
##      0
## [1] "10"
##      BIC
## 140.8287
## [1] "11"
##      BIC
## 5.217768

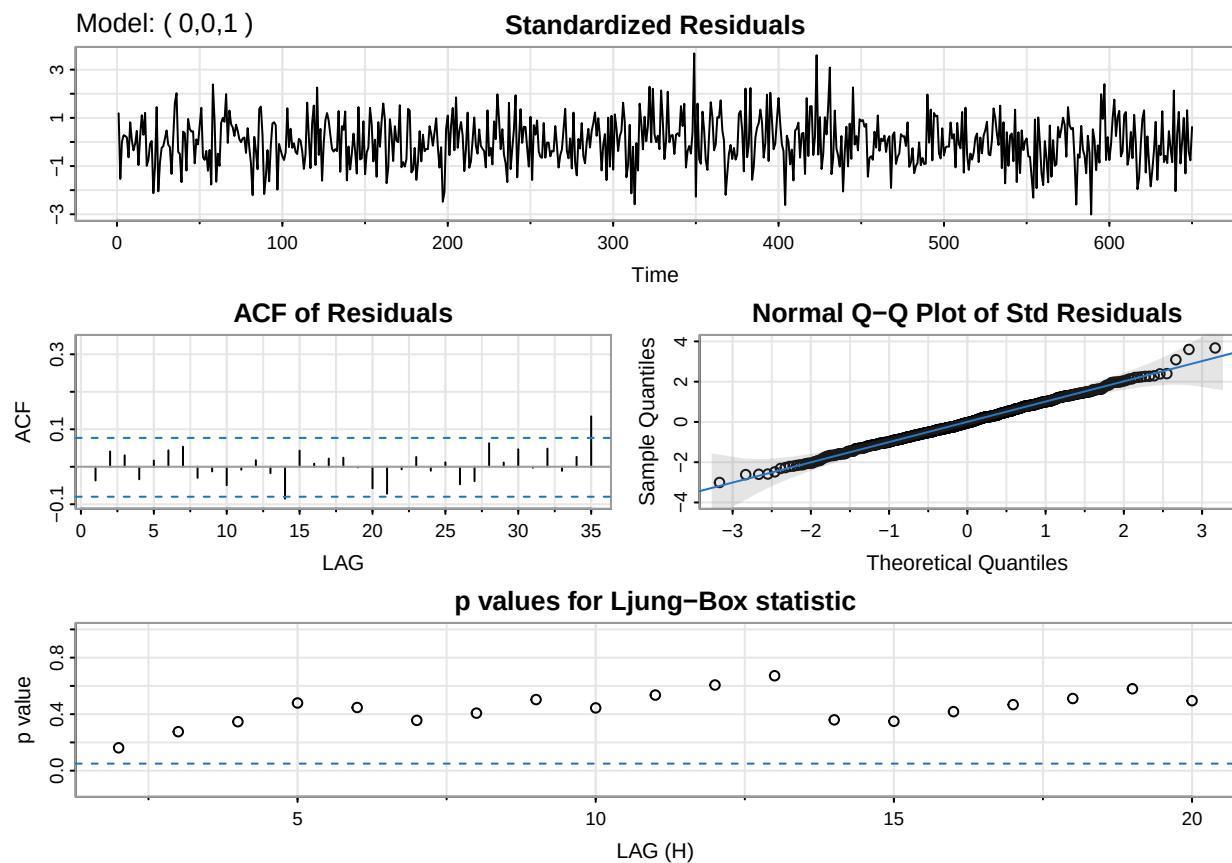
```

```
## [1] "20"
##      BIC
## 97.74556
## [1] "21"
##      BIC
## 9.219255
## [1] "30"
##      BIC
## 76.08825
## [1] "31"
##      BIC
## 15.67313
## [1] "40"
##      BIC
## 61.33244
## [1] "41"
##      BIC
## 22.07853
## [1] "50"
##      BIC
## 43.94773
## [1] "51"
##      BIC
## 28.52521
```

```
sarima(xx,p=0,d=0,q=1,no.constant=FALSE, details=TRUE)
```

```
## initial  value 1.363317
## iter    2 value 1.179235
## iter    3 value 1.149138
## iter    4 value 1.128987
## iter    5 value 1.122043
## iter    6 value 1.122002
## iter    7 value 1.118538
## iter    8 value 1.118384
## iter    9 value 1.118363
## iter   10 value 1.118361
## iter   11 value 1.118358
## iter   12 value 1.118357
## iter   13 value 1.118357
## iter   14 value 1.118356
## iter   15 value 1.118355
## iter   16 value 1.118355
## iter   17 value 1.118354
## iter   18 value 1.118354
## iter   18 value 1.118354
## iter   18 value 1.118354
## final   value 1.118354
## converged
## initial  value 1.115871
## iter    2 value 1.115840
## iter    3 value 1.115827
## iter    4 value 1.115825
## iter    5 value 1.115818
## iter    5 value 1.115818
```

```
## iter    5 value 1.115818
## final   value 1.115818
## converged
## <><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ma1      0.8387 0.0245 34.2841      0
## xmean    13.5587 0.2198 61.7004      0
##
## sigma^2 estimated as 9.297686 on 648 degrees of freedom
##
## AIC = 5.078743  AICc = 5.078772  BIC = 5.099406
##
```



```
sarima(xx,p=1,d=0,q=1,no.constant=FALSE, details=TRUE)
```

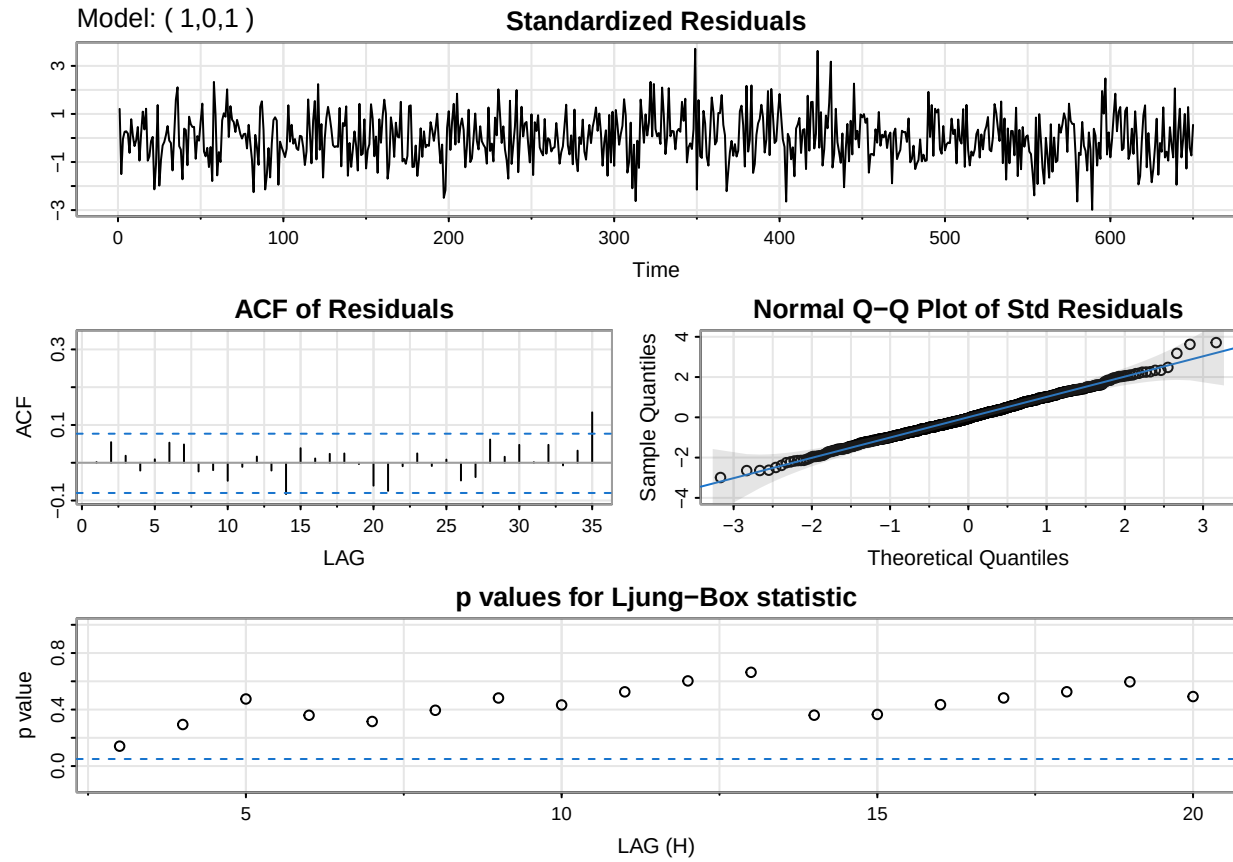
```
## initial value 1.362956
## iter    2 value 1.196330
## iter    3 value 1.158625
## iter    4 value 1.130110
## iter    5 value 1.116376
## iter    6 value 1.113854
## iter    7 value 1.113661
## iter    8 value 1.113432
## iter    9 value 1.113424
## iter   10 value 1.113423
```



```

## iter 11 value 1.113423
## iter 12 value 1.113423
## iter 13 value 1.113422
## iter 14 value 1.113422
## iter 15 value 1.113422
## iter 15 value 1.113422
## final value 1.113422
## converged
## initial value 1.114862
## iter 2 value 1.114853
## iter 3 value 1.114850
## iter 4 value 1.114849
## iter 5 value 1.114849
## iter 6 value 1.114849
## iter 6 value 1.114849
## iter 6 value 1.114849
## final value 1.114849
## converged
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1    -0.0544 0.0482 -1.1273   0.26
## ma1     0.8575 0.0269 31.8826   0.00
## xmean  13.5577 0.2104 64.4481   0.00
##
## sigma^2 estimated as 9.279382 on 647 degrees of freedom
##
## AIC = 5.079883  AICc = 5.07994  BIC = 5.107433
##

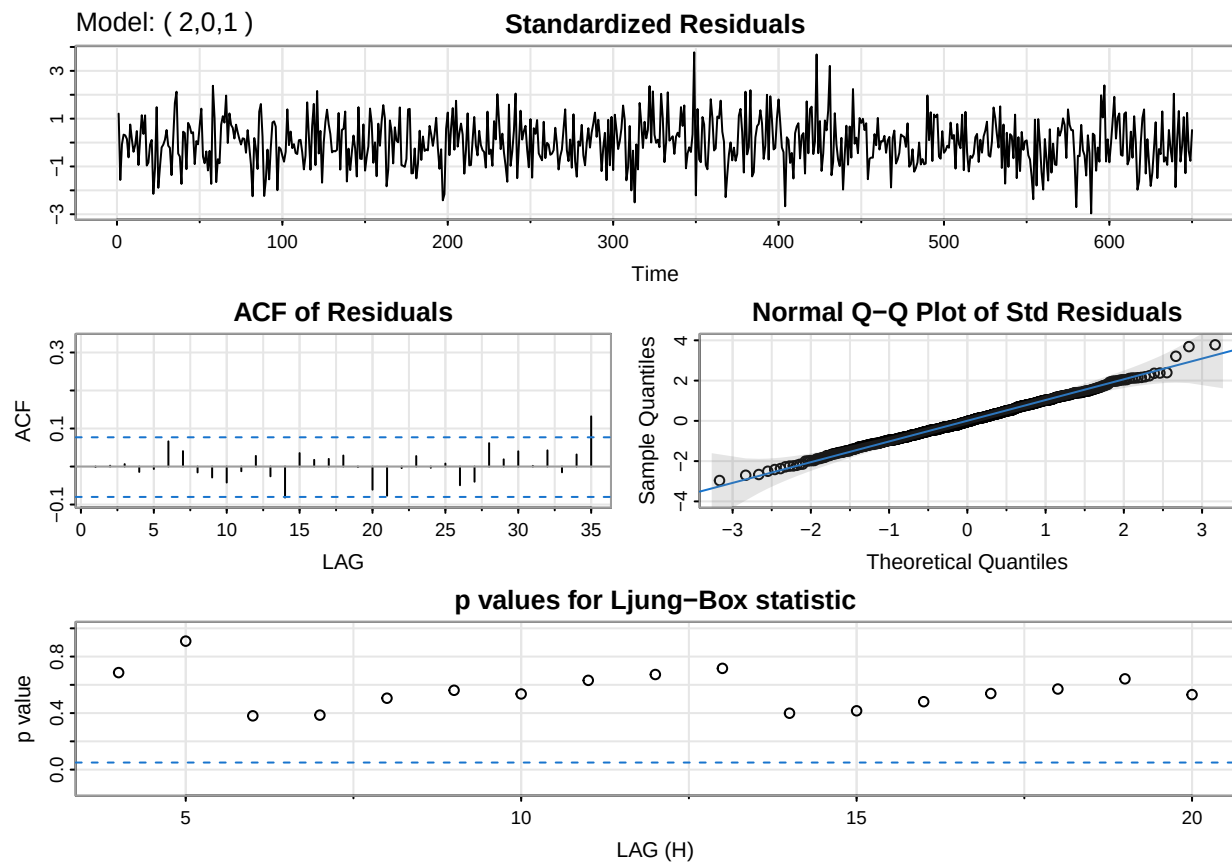
```



```
sarima(xx,p=2,d=0,q=1,no.constant=FALSE, details=TRUE)
```

```
## initial value 1.363278
## iter 2 value 1.204068
## iter 3 value 1.151205
## iter 4 value 1.140462
## iter 5 value 1.130048
## iter 6 value 1.123276
## iter 7 value 1.118764
## iter 8 value 1.118664
## iter 9 value 1.117165
## iter 10 value 1.117153
## iter 11 value 1.117151
## iter 12 value 1.117148
## iter 13 value 1.117143
## iter 14 value 1.117141
## iter 15 value 1.117139
## iter 16 value 1.117138
## iter 17 value 1.117138
## iter 17 value 1.117138
## iter 17 value 1.117138
## final value 1.117138
## converged
## initial value 1.113707
## iter 2 value 1.113470
## iter 3 value 1.113059
```

```
## iter 4 value 1.112975
## iter 5 value 1.112961
## iter 6 value 1.112958
## iter 7 value 1.112950
## iter 8 value 1.112946
## iter 9 value 1.112945
## iter 10 value 1.112945
## iter 10 value 1.112945
## final value 1.112945
## converged
## <><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1    -0.0756 0.0480 -1.5759 0.1155
## ar2     0.0725 0.0459  1.5813 0.1143
## ma1     0.8816 0.0263 33.5024 0.0000
## xmean   13.5583 0.2235 60.6595 0.0000
##
## sigma^2 estimated as 9.243834 on 646 degrees of freedom
##
## AIC = 5.079151  AICc = 5.079247  BIC = 5.113589
##
```



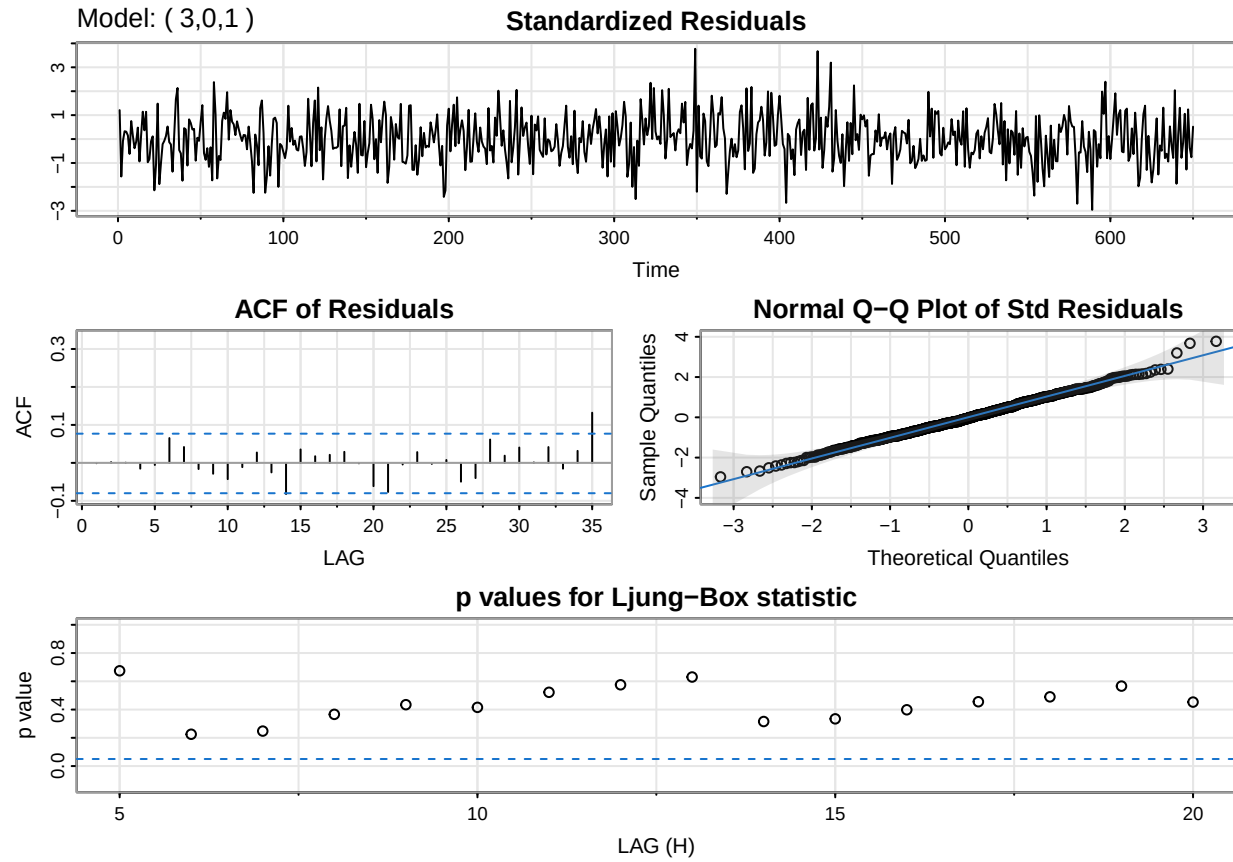
```
sarima(xx,p=3,d=0,q=1,no.constant=FALSE, details=TRUE)
```

```
## initial value 1.363460
```

```

## iter 2 value 1.200588
## iter 3 value 1.142086
## iter 4 value 1.133363
## iter 5 value 1.128999
## iter 6 value 1.122277
## iter 7 value 1.118609
## iter 8 value 1.115966
## iter 9 value 1.111745
## iter 10 value 1.111528
## iter 11 value 1.111247
## iter 12 value 1.111214
## iter 13 value 1.111214
## iter 14 value 1.111214
## iter 15 value 1.111214
## iter 16 value 1.111214
## iter 17 value 1.111214
## iter 18 value 1.111214
## iter 19 value 1.111214
## iter 19 value 1.111214
## iter 19 value 1.111214
## final value 1.111214
## converged
## initial value 1.112934
## iter 2 value 1.112934
## iter 3 value 1.112932
## iter 4 value 1.112927
## iter 5 value 1.112927
## iter 6 value 1.112927
## iter 7 value 1.112927
## iter 8 value 1.112927
## iter 9 value 1.112927
## iter 10 value 1.112927
## iter 10 value 1.112927
## iter 10 value 1.112927
## final value 1.112927
## converged
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1    -0.0737 0.0498 -1.4799 0.1394
## ar2     0.0709 0.0472  1.5044 0.1330
## ar3     0.0069 0.0451  0.1526 0.8788
## ma1     0.8793 0.0308 28.5574 0.0000
## xmean   13.5581 0.2249 60.2978 0.0000
##
## sigma^2 estimated as 9.243515 on 645 degrees of freedom
##
## AIC = 5.082193  AICc = 5.082336  BIC = 5.123518
##

```



```

# Step 1: Pre-whiten yy with the AR(2) part
yy <- dat3[, "yy"]

# Apply AR(2) filter: Use (1 - ar1*B - ar2*B^2)
yy_ar_filtered <- stats::filter(yy, filter = c(1, -ar_coeffs), method = "convolution", sides = 1)

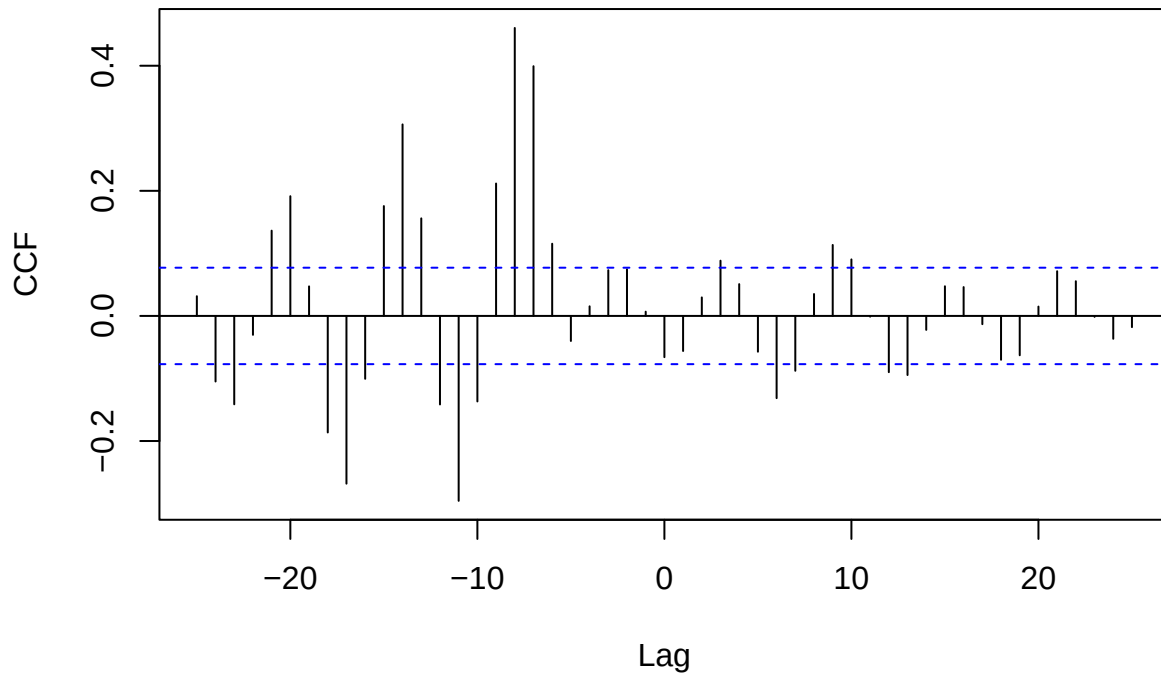
# Step 2: Remove NAs caused by AR filtering
yy_ar_filtered <- na.omit(yy_ar_filtered)
xx_prewhitened <- xx_prewhitened[-c(1, 2)]

# Step 3: Apply MA(1) filter to the AR-filtered series
# Use (1 + ma1*B) where B is the lag operator
yy_prewhitened <- stats::filter(yy_ar_filtered, filter = c(1, -ma_coeff), method = "recursive", sides = 1)

# Plot the cross-correlation function (CCF) of the pre-whitened series
ccf(xx_prewhitened, yy_prewhitened, main = "CCF of Pre-whitened xx and yy",
    ylab = "CCF", na.action = na.omit)

```

CCF of Pre-whitened xx and yy



```

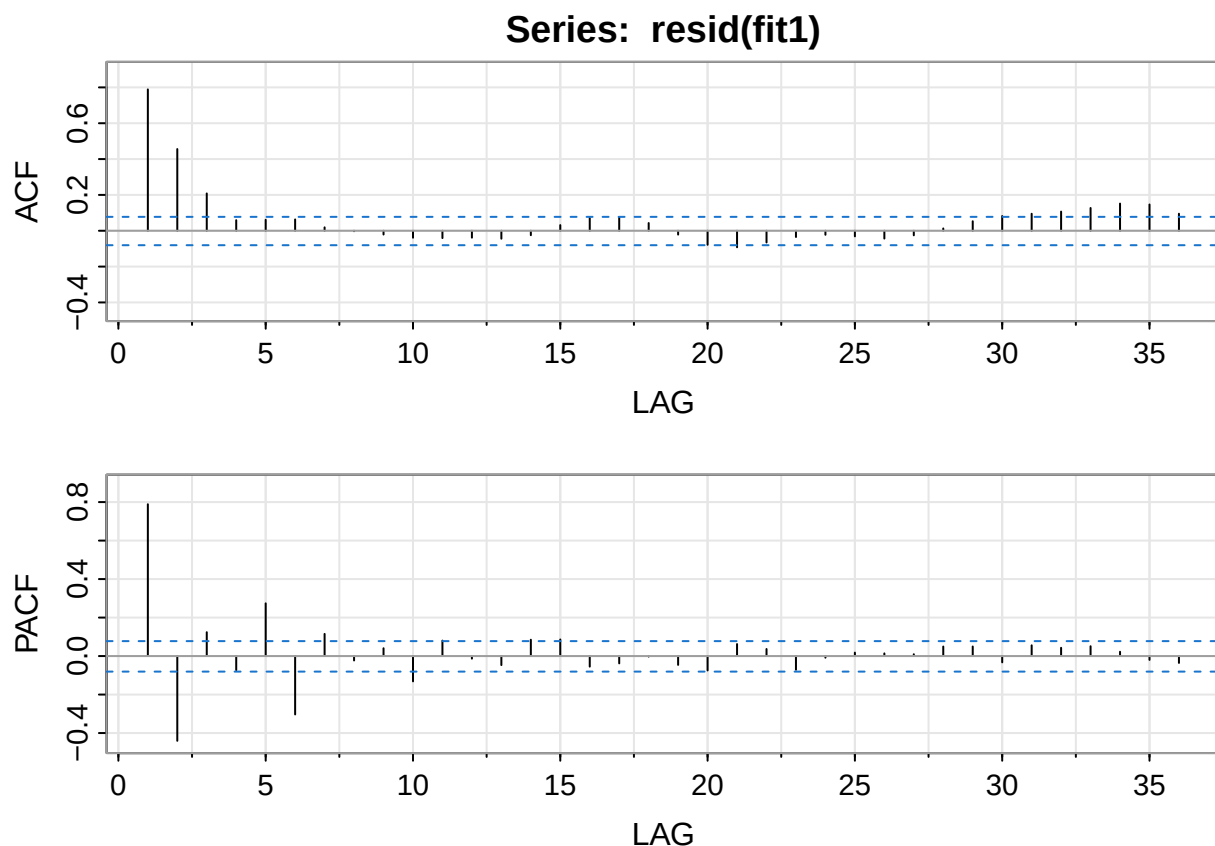
XL8 <- lag(xx, -8)
XL11 <- lag(xx, -11)
design_matrix <- ts.intersect(yy, XL8, XL11)

fit1 <- lm(yy ~ XL8 + XL11, data = design_matrix)
summary(fit1)

```

```
##
```

```
## Call:
## lm(formula = yy ~ XL8 + XL11, data = design_matrix)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -54.921 -12.946  -0.778  13.297  46.093
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   74.9285     3.6115  20.747 < 2e-16 ***
## XL8           0.8998     0.1875   4.799 1.99e-06 ***
## XL11          0.5636     0.1875   3.006 0.00275 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 18.61 on 636 degrees of freedom
## Multiple R-squared:  0.04936,    Adjusted R-squared:  0.04637
## F-statistic: 16.51 on 2 and 636 DF,  p-value: 1.02e-07
invisible(acf2(resid(fit1)))
```



```
maxAR<-7+1
maxMA<-3+1
fits_all<-vector("list",length=maxAR)
AICmin<-BICmin<-Inf
for (ii in 1:maxAR){
  fits_all[[ii]]<-vector("list",length=maxMA)
```

```

for (jj in 1:maxMA){
fits_all[[ii]][[jj]]<-invisible(sarima(design_matrix[, 1],p=ii-1,d=0,q=jj-1,xreg = design_matrix[, 2:3]
if(fits_all[[ii]][[jj]]$ICs["AICc"]<AICmin) AICmin<-fits_all[[ii]][[jj]]$ICs["AICc"]
if(fits_all[[ii]][[jj]]$ICs["BIC"]<BICmin) BICmin<-fits_all[[ii]][[jj]]$ICs["BIC"]
}
}

```

```

## <><><><><><><><><><><><><><>
##
## Coefficients:
##           Estimate      SE t.value p.value
## intercept  74.9285  3.6030 20.7960  0.0000
## XL8         0.8998  0.1871  4.8100  0.0000
## XL11        0.5636  0.1870  3.0132  0.0027
##
## sigma^2 estimated as 344.6042 on 636 degrees of freedom
##
## AIC = 8.692793  AICc = 8.692852  BIC = 8.720711
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##           Estimate      SE t.value p.value
## ma1          0.9331  0.0118 79.1839  0.0000
## intercept  84.4603  2.9398 28.7298  0.0000
## XL8         0.5136  0.1451  3.5392  0.0004
## XL11        0.2445  0.1445  1.6927  0.0910
##
## sigma^2 estimated as 122.0143 on 635 degrees of freedom
##
## AIC = 7.660866  AICc = 7.660965  BIC = 7.695764
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##           Estimate      SE t.value p.value
## ma1          1.3623  0.0317 42.9701  0e+00
## ma2          0.5972  0.0390 15.3096  0e+00
## intercept  95.5820  2.3820 40.1263  0e+00
## XL8        -0.4609  0.1369 -3.3662  8e-04
## XL11        0.3980  0.0993  4.0079  1e-04
##
## sigma^2 estimated as 87.41234 on 634 degrees of freedom
##
## AIC = 7.330708  AICc = 7.330856  BIC = 7.372585
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##           Estimate      SE t.value p.value
## ma1          1.3714  0.0428 32.0272  0.0000
## ma2          0.8108  0.0473 17.1408  0.0000
## ma3          0.3331  0.0444  7.5057  0.0000
## intercept  91.3918  2.3922 38.2034  0.0000

```



```

## XL8          0.0914 0.1364 0.6697 0.5033
## XL11         0.1527 0.1048 1.4568 0.1457
##
## sigma^2 estimated as 79.93234 on 633 degrees of freedom
##
## AIC = 7.244464 AICc = 7.244672 BIC = 7.29332
##
## <><><><><><><><><><><>
##
## Coefficients:
##          Estimate      SE t.value p.value
## ar1          0.8403 0.0214 39.3266 0.0000
## intercept    96.0473 3.3382 28.7725 0.0000
## XL8         -0.3544 0.1132 -3.1320 0.0018
## XL11         0.2527 0.1129 2.2391 0.0255
##
## sigma^2 estimated as 108.225 on 635 degrees of freedom
##
## AIC = 7.539656 AICc = 7.539754 BIC = 7.574553
##
## <><><><><><><><><><><>
##
## Coefficients:
##          Estimate      SE t.value p.value
## ar1          0.6878 0.0308 22.3412 0.0000
## ma1          0.7379 0.0298 24.7469 0.0000
## intercept    94.3697 2.5859 36.4939 0.0000
## XL8         -0.2148 0.0990 -2.1701 0.0304
## XL11         0.2368 0.0958 2.4711 0.0137
##
## sigma^2 estimated as 75.05523 on 634 degrees of freedom
##
## AIC = 7.178399 AICc = 7.178547 BIC = 7.220276
##
## <><><><><><><><><><><>
##
## Coefficients:
##          Estimate      SE t.value p.value
## ar1          0.6767 0.0450 15.0255 0.0000
## ma1          0.7559 0.0598 12.6356 0.0000
## ma2          0.0207 0.0595 0.3478 0.7281
## intercept    94.3923 2.5583 36.8959 0.0000
## XL8         -0.2190 0.0992 -2.2081 0.0276
## XL11         0.2395 0.0957 2.5029 0.0126
##
## sigma^2 estimated as 75.04103 on 633 degrees of freedom
##
## AIC = 7.181338 AICc = 7.181546 BIC = 7.230194
##
## <><><><><><><><><><><>
##
## Coefficients:
##          Estimate      SE t.value p.value
## ar1          0.6845 0.0735 9.3078 0.0000

```

```

## ma1          0.7484 0.0813  9.2013  0.0000
## ma2          0.0077 0.1156  0.0662  0.9472
## ma3         -0.0103 0.0797 -0.1291  0.8973
## intercept    94.5122 2.7048 34.9423  0.0000
## XL8          -0.2291 0.1266 -1.8093  0.0709
## XL11         0.2415 0.0969  2.4935  0.0129
##
## sigma^2 estimated as 75.03954 on 632 degrees of freedom
##
## AIC = 7.184445  AICc = 7.184722  BIC = 7.240281
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##          Estimate      SE  t.value p.value
## ar1          1.2091 0.0353  34.2049  0.0000
## ar2         -0.4472 0.0358 -12.4992  0.0000
## intercept    92.7675 2.2832  40.6311  0.0000
## XL8         -0.0984 0.1033  -0.9519  0.3415
## XL11         0.2415 0.1007   2.3975  0.0168
##
## sigma^2 estimated as 87.16819 on 634 degrees of freedom
##
## AIC = 7.327068  AICc = 7.327216  BIC = 7.368945
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##          Estimate      SE t.value p.value
## ar1          0.7046 0.0570  12.3675  0.0000
## ar2         -0.0191 0.0548  -0.3485  0.7276
## ma1          0.7280 0.0419  17.3931  0.0000
## intercept    94.3852 2.5573  36.9086  0.0000
## XL8         -0.2184 0.0988  -2.2096  0.0275
## XL11         0.2394 0.0957   2.5022  0.0126
##
## sigma^2 estimated as 75.04118 on 633 degrees of freedom
##
## AIC = 7.18134  AICc = 7.181548  BIC = 7.230197
##
## Warning in sqrt(diag(fitit$var.coef)): NaNs produced
## Warning in sqrt(diag(fitit$var.coef)): NaNs produced
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##          Estimate      SE t.value p.value
## ar1          1.0060   NaN    NaN    NaN
## ar2         -0.2268   NaN    NaN    NaN
## ma1          0.4266   NaN    NaN    NaN
## ma2         -0.2226   NaN    NaN    NaN
## intercept    94.3585 2.4893  37.9064  0.0000
## XL8         -0.2141 0.0941  -2.2748  0.0232
## XL11         0.2394 0.0948   2.5250  0.0118

```

```

##
## sigma^2 estimated as 75.0387 on 632 degrees of freedom
##
## AIC = 7.184438  AICc = 7.184716  BIC = 7.240274
##

## Warning in sqrt(diag(fitit$var.coef)): NaNs produced
## Warning in sqrt(diag(fitit$var.coef)): NaNs produced

## <><><><><><><><><><><>
##
## Coefficients:
##           Estimate      SE t.value p.value
## ar1           1.0991    NaN     NaN     NaN
## ar2          -0.2920    NaN     NaN     NaN
## ma1           0.3333    NaN     NaN     NaN
## ma2          -0.2896    NaN     NaN     NaN
## ma3           0.0028 0.0619  0.0459  0.9634
## intercept    94.2773 2.5934 36.3522  0.0000
## XL8          -0.2118 0.1055 -2.0088  0.0450
## XL11          0.2398 0.0933  2.5705  0.0104
##
## sigma^2 estimated as 75.03686 on 631 degrees of freedom
##
## AIC = 7.187544  AICc = 7.187902  BIC = 7.25036
##
## <><><><><><><><><><><>
##
## Coefficients:
##           Estimate      SE t.value p.value
## ar1           1.3241 0.0382 34.6836  0.0000
## ar2          -0.7595 0.0575 -13.2058  0.0000
## ar3           0.2643 0.0388  6.8154  0.0000
## intercept    94.8617 2.6255 36.1314  0.0000
## XL8          -0.2226 0.1030 -2.1620  0.0310
## XL11          0.2073 0.0996  2.0807  0.0379
##
## sigma^2 estimated as 81.28259 on 633 degrees of freedom
##
## AIC = 7.260633  AICc = 7.260841  BIC = 7.30949
##
## <><><><><><><><><><><>
##
## Coefficients:
##           Estimate      SE t.value p.value
## ar1           0.7080 0.0807  8.7698  0.0000
## ar2          -0.0250 0.1123 -0.2231  0.8235
## ar3           0.0042 0.0682  0.0610  0.9514
## ma1           0.7245 0.0708 10.2352  0.0000
## intercept    94.4218 2.6740 35.3109  0.0000
## XL8          -0.2226 0.1202 -1.8513  0.0646
## XL11          0.2401 0.0963  2.4937  0.0129
##
## sigma^2 estimated as 75.04083 on 632 degrees of freedom
##

```

```

## AIC = 7.184464 AICc = 7.184742 BIC = 7.2403
##
## <><><><><><><><><><><>
##
## Coefficients:
##           Estimate      SE t.value p.value
## ar1           0.2857 0.3180  0.8985  0.3692
## ar2           0.2290 0.2297  0.9968  0.3193
## ar3           0.0471 0.1083  0.4351  0.6636
## ma1           1.1468 0.3153  3.6365  0.0003
## ma2           0.3422 0.2526  1.3545  0.1761
## intercept    95.0417 2.9750 31.9470  0.0000
## XL8          -0.2931 0.1816 -1.6145  0.1069
## XL11          0.2656 0.1079  2.4604  0.0141
##
## sigma^2 estimated as 75.00576 on 631 degrees of freedom
##
## AIC = 7.187106 AICc = 7.187464 BIC = 7.249922
##
## <><><><><><><><><><><>
##
## Coefficients:
##           Estimate      SE t.value p.value
## ar1          -0.2612 0.1236 -2.1121  0.0351
## ar2          -0.1630 0.1716 -0.9503  0.3423
## ar3           0.6047 0.1319  4.5841  0.0000
## ma1           1.6897 0.1248 13.5408  0.0000
## ma2           1.5106 0.2842  5.3148  0.0000
## ma3           0.5617 0.1933  2.9064  0.0038
## intercept    97.0079 3.6989 26.2258  0.0000
## XL8          -0.3563 0.1830 -1.9469  0.0520
## XL11          0.1851 0.1045  1.7707  0.0771
##
## sigma^2 estimated as 74.45163 on 630 degrees of freedom
##
## AIC = 7.182971 AICc = 7.183419 BIC = 7.252766
##
## <><><><><><><><><><><>
##
## Coefficients:
##           Estimate      SE t.value p.value
## ar1           1.3906 0.0381 36.5430 0e+00
## ar2          -0.9636 0.0628 -15.3518 0e+00
## ar3           0.6473 0.0687  9.4217 0e+00
## ar4          -0.2747 0.0426 -6.4529 0e+00
## intercept    98.2828 2.4722 39.7551 0e+00
## XL8          -0.5932 0.1085 -5.4656 0e+00
## XL11          0.3283 0.0940  3.4916 5e-04
##
## sigma^2 estimated as 76.61133 on 632 degrees of freedom
##
## AIC = 7.205093 AICc = 7.205371 BIC = 7.260929
##
## <><><><><><><><><><><>

```

```

##
## Coefficients:
##           Estimate      SE t.value p.value
## ar1           0.8729 0.1488  5.8662  0.0000
## ar2          -0.2631 0.2119 -1.2418  0.2148
## ar3           0.2004 0.1615  1.2411  0.2150
## ar4          -0.1009 0.0756 -1.3350  0.1824
## ma1           0.5597 0.1457  3.8427  0.0001
## intercept    95.7875 2.9421 32.5573  0.0000
## XL8          -0.3574 0.1537 -2.3256  0.0204
## XL11          0.2759 0.0945  2.9197  0.0036
##
## sigma^2 estimated as 74.85343 on 631 degrees of freedom
##
## AIC = 7.185096  AICc = 7.185454  BIC = 7.247912
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##           Estimate      SE t.value p.value
## ar1           0.4242 0.2364  1.7941  0.0733
## ar2          -0.0235 0.2021 -0.1161  0.9076
## ar3           0.2977 0.1432  2.0788  0.0380
## ar4          -0.1326 0.0684 -1.9403  0.0528
## ma1           1.0110 0.2348  4.3050  0.0000
## ma2           0.4077 0.1908  2.1367  0.0330
## intercept    97.3952 3.6472 26.7038  0.0000
## XL8          -0.4814 0.1807 -2.6636  0.0079
## XL11          0.2790 0.1032  2.7040  0.0070
##
## sigma^2 estimated as 74.65119 on 630 degrees of freedom
##
## AIC = 7.185597  AICc = 7.186045  BIC = 7.255392
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##           Estimate      SE t.value p.value
## ar1          -0.1283 0.2240 -0.5725  0.5672
## ar2          -0.2181 0.1545 -1.4116  0.1586
## ar3           0.6059 0.0784  7.7243  0.0000
## ar4          -0.0699 0.0789 -0.8864  0.3758
## ma1           1.5661 0.2230  7.0235  0.0000
## ma2           1.4007 0.2484  5.6382  0.0000
## ma3           0.5251 0.1278  4.1091  0.0000
## intercept    95.9173 3.8026 25.2241  0.0000
## XL8          -0.3076 0.1791 -1.7176  0.0864
## XL11          0.2182 0.1108  1.9696  0.0493
##
## sigma^2 estimated as 74.36345 on 629 degrees of freedom
##
## AIC = 7.184902  AICc = 7.18545  BIC = 7.261676
##
## <><><><><><><><><><><><><><>

```

```

##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1          1.4261 0.0393  36.3302  0.0000
## ar2         -1.0435 0.0675 -15.4700  0.0000
## ar3          0.7592 0.0772   9.8357  0.0000
## ar4         -0.4613 0.0676  -6.8289  0.0000
## ar5          0.1488 0.0412   3.6068  0.0003
## intercept    98.0604 2.6984  36.3396  0.0000
## XL8          -0.4858 0.1126  -4.3152  0.0000
## XL11         0.2358 0.0960   2.4560  0.0143
##
## sigma^2 estimated as 75.09768 on 631 degrees of freedom
##
## AIC = 7.188372  AICc = 7.188729  BIC = 7.251187
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1          1.0178 0.2192   4.6432  0.0000
## ar2         -0.4691 0.3124  -1.5019  0.1336
## ar3          0.3507 0.2324   1.5093  0.1317
## ar4         -0.2226 0.1436  -1.5505  0.1215
## ar5          0.0706 0.0664   1.0631  0.2881
## ma1          0.4168 0.2167   1.9232  0.0549
## intercept    95.9368 3.1666  30.2962  0.0000
## XL8          -0.3513 0.1600  -2.1963  0.0284
## XL11         0.2581 0.0967   2.6700  0.0078
##
## sigma^2 estimated as 74.7357 on 630 degrees of freedom
##
## AIC = 7.186665  AICc = 7.187113  BIC = 7.25646
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ar1          0.5091 0.3529   1.4425  0.1496
## ar2         -0.1197 0.3207  -0.3732  0.7091
## ar3          0.3485 0.1840   1.8939  0.0587
## ar4         -0.1844 0.1363  -1.3529  0.1766
## ar5          0.0376 0.0860   0.4373  0.6621
## ma1          0.9270 0.3517   2.6358  0.0086
## ma2          0.3828 0.2419   1.5824  0.1141
## intercept    97.1367 3.7626  25.8161  0.0000
## XL8          -0.4499 0.1973  -2.2798  0.0230
## XL11         0.2670 0.1084   2.4642  0.0140
##
## sigma^2 estimated as 74.62987 on 629 degrees of freedom
##
## AIC = 7.188431  AICc = 7.18898  BIC = 7.265206
##
## <><><><><><><><><><><><><><>

```

```

##
## Coefficients:
##           Estimate      SE t.value p.value
## ar1           0.0345 0.0799  0.4313  0.6664
## ar2          -0.5043 0.0648 -7.7866  0.0000
## ar3           0.6838 0.0623 10.9802  0.0000
## ar4          -0.0811 0.0697 -1.1633  0.2451
## ar5           0.0336 0.0669  0.5025  0.6155
## ma1           1.4076 0.0701 20.0723  0.0000
## ma2           1.4831 0.0513 28.8855  0.0000
## ma3           0.6993 0.0694 10.0758  0.0000
## intercept    92.3516 2.8548 32.3492  0.0000
## XL8          -0.1374 0.1268 -1.0834  0.2790
## XL11          0.3099 0.1011  3.0661  0.0023
##
## sigma^2 estimated as 72.72703 on 628 degrees of freedom
##
## AIC = 7.168607  AICc = 7.169266  BIC = 7.252361
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##           Estimate      SE t.value p.value
## ar1           1.4351 0.0395 36.3147  0.0000
## ar2          -1.0673 0.0694 -15.3877  0.0000
## ar3           0.7918 0.0812  9.7460  0.0000
## ar4          -0.5437 0.0781 -6.9588  0.0000
## ar5           0.2896 0.0813  3.5635  0.0004
## ar6          -0.0925 0.0463 -1.9981  0.0461
## intercept    96.1866 2.8337 33.9442  0.0000
## XL8          -0.3566 0.1379 -2.5867  0.0099
## XL11          0.2460 0.0961  2.5593  0.0107
##
## sigma^2 estimated as 74.61752 on 630 degrees of freedom
##
## AIC = 7.185101  AICc = 7.185548  BIC = 7.254896
##
## <><><><><><><><><><><><><><>
##
## Coefficients:
##           Estimate      SE t.value p.value
## ar1           1.4668 0.6590  2.2259  0.0264
## ar2          -1.1126 0.9418 -1.1814  0.2379
## ar3           0.8250 0.6919  1.1923  0.2336
## ar4          -0.5665 0.4791 -1.1824  0.2375
## ar5           0.3027 0.2823  1.0722  0.2840
## ar6          -0.0967 0.0974 -0.9924  0.3214
## ma1          -0.0319 0.6613 -0.0483  0.9615
## intercept    96.2793 3.3676 28.5903  0.0000
## XL8          -0.3600 0.1529 -2.3544  0.0189
## XL11          0.2427 0.1173  2.0688  0.0390
##
## sigma^2 estimated as 74.61718 on 629 degrees of freedom
##

```

```

## AIC = 7.188227  AICc = 7.188775  BIC = 7.265001
##
## <><><><><><><><><><><>
##
## Coefficients:
##           Estimate      SE t.value p.value
## ar1           0.5559 0.2555  2.1758  0.0299
## ar2          -0.3912 0.3820 -1.0240  0.3062
## ar3           0.6889 0.3660  1.8822  0.0603
## ar4          -0.4067 0.2572 -1.5814  0.1143
## ar5           0.2059 0.1899  1.0842  0.2787
## ar6          -0.1070 0.1063 -1.0059  0.3148
## ma1           0.8806 0.2547  3.4569  0.0006
## ma2           0.5852 0.2260  2.5892  0.0098
## intercept    98.0436 3.5712 27.4536  0.0000
## XL8          -0.5147 0.1631 -3.1562  0.0017
## XL11          0.2688 0.1166  2.3058  0.0214
##
## sigma^2 estimated as 74.4931 on 628 degrees of freedom
##
## AIC = 7.1898  AICc = 7.190459  BIC = 7.273554
##
## <><><><><><><><><><><>
##
## Coefficients:
##           Estimate      SE t.value p.value
## ar1          -0.7291 0.1977 -3.6874  0.0002
## ar2           0.2584 0.1493  1.7306  0.0840
## ar3           0.4626 0.0863  5.3614  0.0000
## ar4           0.1802 0.0886  2.0329  0.0425
## ar5          -0.0818 0.0750 -1.0907  0.2758
## ar6          -0.0585 0.0952 -0.6149  0.5388
## ma1           2.1774 0.1938 11.2361  0.0000
## ma2           1.7938 0.2094  8.5658  0.0000
## ma3           0.5773 0.0944  6.1171  0.0000
## intercept   100.2536 4.2345 23.6754  0.0000
## XL8          -0.6156 0.1945 -3.1643  0.0016
## XL11          0.2019 0.1723  1.1714  0.2419
##
## sigma^2 estimated as 73.8571 on 627 degrees of freedom
##
## AIC = 7.184638  AICc = 7.185418  BIC = 7.275371
##
## <><><><><><><><><><><>
##
## Coefficients:
##           Estimate      SE t.value p.value
## ar1           1.4350 0.0396 36.2321  0.0000
## ar2          -1.0670 0.0698 -15.2915  0.0000
## ar3           0.7913 0.0826  9.5843  0.0000
## ar4          -0.5420 0.0887 -6.1134  0.0000
## ar5           0.2867 0.1087  2.6370  0.0086
## ar6          -0.0891 0.0967 -0.9218  0.3570
## ar7          -0.0020 0.0494 -0.0402  0.9680

```



```

## intercept 96.2496 3.2222 29.8705 0.0000
## XL8       -0.3590 0.1496 -2.4000 0.0167
## XL11      0.2438 0.1105  2.2060 0.0277
##
## sigma^2 estimated as 74.61728 on 629 degrees of freedom
##
## AIC = 7.188228 AICc = 7.188776 BIC = 7.265002
##
## <><><><><><><><><><><>
##
## Coefficients:
##           Estimate      SE t.value p.value
## ar1           0.5138 0.0657  7.8184 0.0000
## ar2           0.2535 0.0881  2.8777 0.0041
## ar3          -0.1860 0.0739 -2.5155 0.0121
## ar4           0.1461 0.0833  1.7535 0.0800
## ar5          -0.1613 0.0823 -1.9592 0.0505
## ar6           0.1479 0.0527  2.8040 0.0052
## ar7          -0.0708 0.0458 -1.5465 0.1225
## ma1           0.9268 0.0523 17.7239 0.0000
## intercept    95.4643 3.0003 31.8178 0.0000
## XL8          -0.2709 0.1628 -1.6643 0.0966
## XL11          0.2130 0.1005  2.1194 0.0344
##
## sigma^2 estimated as 74.30237 on 628 degrees of freedom
##
## AIC = 7.187216 AICc = 7.187875 BIC = 7.27097
##
## <><><><><><><><><><><>
##
## Coefficients:
##           Estimate      SE t.value p.value
## ar1           0.9174 0.4534  2.0232 0.0435
## ar2           0.0418 0.2609  0.1602 0.8728
## ar3          -0.2823 0.1445 -1.9530 0.0513
## ar4           0.2195 0.1308  1.6785 0.0938
## ar5          -0.2224 0.1313 -1.6941 0.0907
## ar6           0.2167 0.0987  2.1966 0.0284
## ar7          -0.1116 0.0544 -2.0515 0.0406
## ma1           0.5219 0.4546  1.1479 0.2514
## ma2          -0.3696 0.4103 -0.9009 0.3680
## intercept    96.0642 2.9157 32.9471 0.0000
## XL8          -0.2788 0.1597 -1.7464 0.0812
## XL11          0.1775 0.1048  1.6936 0.0908
##
## sigma^2 estimated as 74.23641 on 627 degrees of freedom
##
## AIC = 7.189483 AICc = 7.190263 BIC = 7.280216
##
## <><><><><><><><><><><>
##
## Coefficients:
##           Estimate      SE t.value p.value
## ar1          -0.7178 0.2254 -3.1848 0.0015

```

```
## ar2          0.2807 0.2611  1.0749  0.2828
## ar3          0.4463 0.1284  3.4750  0.0005
## ar4          0.1672 0.1410  1.1859  0.2361
## ar5         -0.0819 0.0845 -0.9694  0.3327
## ar6         -0.0582 0.1038 -0.5614  0.5748
## ar7          0.0128 0.0883  0.1445  0.8851
## ma1          2.1661 0.2216  9.7727  0.0000
## ma2          1.7559 0.2930  5.9938  0.0000
## ma3          0.5512 0.2086  2.6421  0.0084
## intercept 100.0439 4.9707 20.1268  0.0000
## XL8          -0.5988 0.2251 -2.6599  0.0080
## XL11         0.2026 0.1930  1.0497  0.2943
##
## sigma^2 estimated as 73.85602 on 626 degrees of freedom
##
## AIC = 7.187736  AICc = 7.188648  BIC = 7.285449
##
```

```
nn <- length(xx)
```

```
#look at AICs and BICs
```

```
print("check AICcs")
```

```
## [1] "check AICcs"
```

```
## [1] "check AICcs"
```

```
for (ii in 1:maxAR){
```

```
  for(jj in 1:maxMA){
```

```
    print(paste0(ii-1,"",jj-1))
```

```
    print(((fits_all[[ii]][[jj]])$ICs["AICc"] - AICmin) * nn)
```

```
  }
```

```
}
```

```
## [1] "00"
```

```
##      AICc
```

```
## 990.3309
```

```
## [1] "01"
```

```
##      AICc
```

```
## 319.6043
```

```
## [1] "02"
```

```
##      AICc
```

```
## 105.0333
```

```
## [1] "03"
```

```
##      AICc
```

```
## 49.01354
```

```
## [1] "10"
```

```
##      AICc
```

```
## 240.8172
```

```
## [1] "11"
```

```
##      AICc
```

```
## 6.032696
```

```
## [1] "12"
```

```
##      AICc
```

```
## 7.98168
```

```
## [1] "13"
```

```
##      AICc
```

```

## 10.04638
## [1] "20"
##      AICc
## 102.6674
## [1] "21"
##      AICc
## 7.983073
## [1] "22"
##      AICc
## 10.04201
## [1] "23"
##      AICc
## 12.11317
## [1] "30"
##      AICc
## 59.5236
## [1] "31"
##      AICc
## 10.05907
## [1] "32"
##      AICc
## 11.8283
## [1] "33"
##      AICc
## 9.199263
## [1] "40"
##      AICc
## 23.46791
## [1] "41"
##      AICc
## 10.52199
## [1] "42"
##      AICc
## 10.90596
## [1] "43"
##      AICc
## 10.5193
## [1] "50"
##      AICc
## 12.65105
## [1] "51"
##      AICc
## 11.6003
## [1] "52"
##      AICc
## 12.81359
## [1] "53"
## AICc
##      0
## [1] "60"
##      AICc
## 10.58333
## [1] "61"
##      AICc

```

```

## 12.68068
## [1] "62"
##      AICc
## 13.77535
## [1] "63"
##      AICc
## 10.49831
## [1] "70"
##      AICc
## 12.68139
## [1] "71"
##      AICc
## 12.09547
## [1] "72"
##      AICc
## 13.64759
## [1] "73"
##      AICc
## 12.59794

print("check BICs")

## [1] "check BICs"
## [1] "check BICs"
for (ii in 1:maxAR){
  for(jj in 1:maxMA){
    print(paste0(ii-1,"",jj-1))
    print(((fits_all[[ii]][[jj]])$ICs["BIC"] - BICmin) * nn)
  }
}

## [1] "00"
##      BIC
## 975.2828
## [1] "01"
##      BIC
## 309.0671
## [1] "02"
##      BIC
## 99.00061
## [1] "03"
##      BIC
## 47.47874
## [1] "10"
##      BIC
## 230.2801
## [1] "11"
##      BIC
##      0
## [1] "12"
##      BIC
## 6.446882
## [1] "13"
##      BIC

```

```
## 13.00291
## [1] "20"
##      BIC
## 96.63473
## [1] "21"
##      BIC
## 6.448275
## [1] "22"
##      BIC
## 12.99854
## [1] "23"
##      BIC
## 19.55442
## [1] "30"
##      BIC
## 57.98881
## [1] "31"
##      BIC
## 13.0156
## [1] "32"
##      BIC
## 19.26955
## [1] "33"
##      BIC
## 21.11861
## [1] "40"
##      BIC
## 26.42443
## [1] "41"
##      BIC
## 17.96324
## [1] "42"
##      BIC
## 22.8253
## [1] "43"
##      BIC
## 26.91007
## [1] "50"
##      BIC
## 20.0923
## [1] "51"
##      BIC
## 23.51964
## [1] "52"
##      BIC
## 29.20435
## [1] "53"
##      BIC
## 20.8555
## [1] "60"
##      BIC
## 22.50268
## [1] "61"
##      BIC
```

```

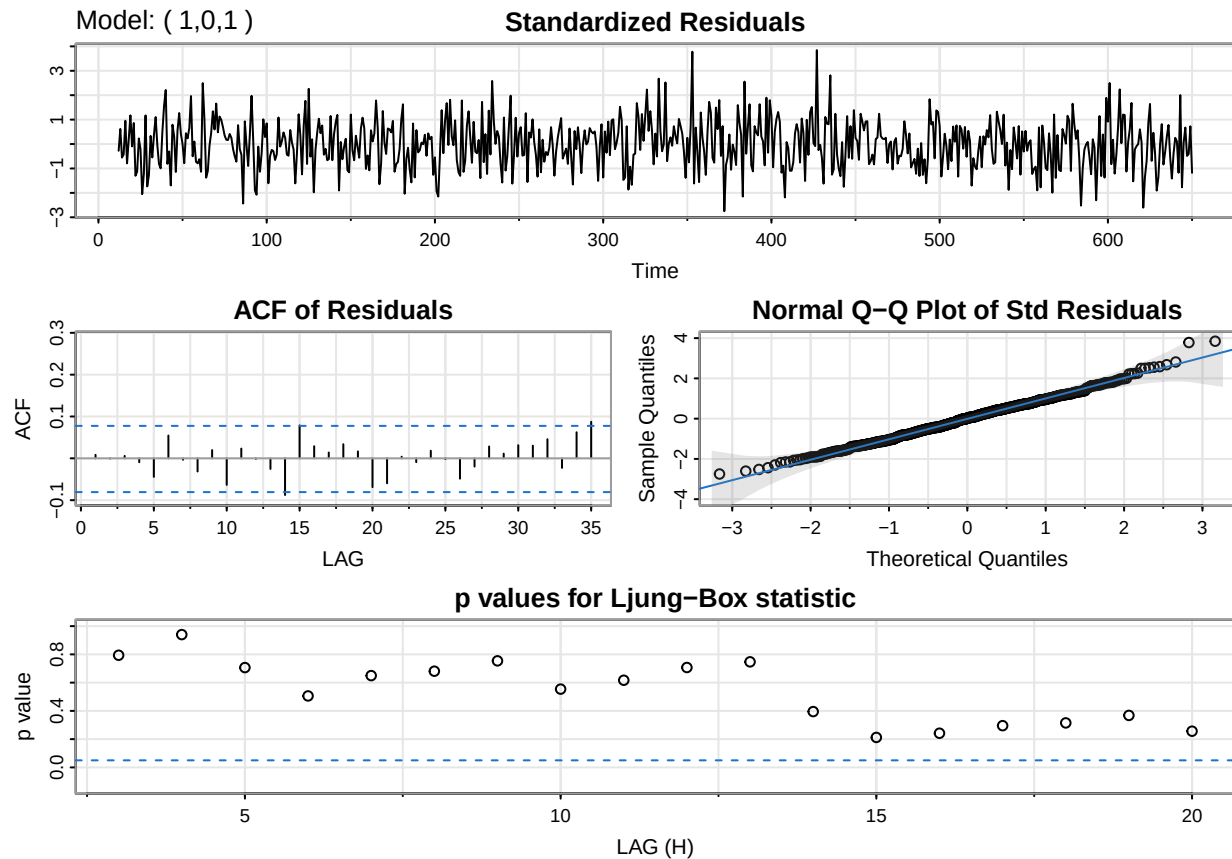
## 29.07145
## [1] "62"
##      BIC
## 34.63084
## [1] "63"
##      BIC
## 35.8118
## [1] "70"
##      BIC
## 29.07215
## [1] "71"
##      BIC
## 32.95096
## [1] "72"
##      BIC
## 38.96108
## [1] "73"
##      BIC
## 42.36266

sarima(design_matrix[, 1],p=1,d=0,q=1,xreg = design_matrix[, 2:3],no.constant=FALSE, details=TRUE)

## initial  value 2.921839
## iter    2 value 2.291314
## iter    3 value 2.202401
## iter    4 value 2.183656
## iter    5 value 2.161894
## iter    6 value 2.160228
## iter    7 value 2.159904
## iter    8 value 2.159854
## iter    9 value 2.159853
## iter   10 value 2.159853
## iter   10 value 2.159853
## iter   10 value 2.159853
## final   value 2.159853
## converged
## initial  value 2.160876
## iter    2 value 2.160874
## iter    3 value 2.160873
## iter    4 value 2.160873
## iter    5 value 2.160873
## iter    6 value 2.160873
## iter    7 value 2.160872
## iter    8 value 2.160872
## iter    9 value 2.160871
## iter    9 value 2.160871
## iter    9 value 2.160871
## final   value 2.160871
## converged
## <><><><><><><><><><><><><><>
##
## Coefficients:
##           Estimate      SE t.value p.value
## ar1           0.6878 0.0308 22.3412 0.0000
## ma1           0.7379 0.0298 24.7469 0.0000

```

```
## intercept  94.3697 2.5859 36.4939  0.0000
## XL8        -0.2148 0.0990 -2.1701  0.0304
## XL11        0.2368 0.0958  2.4711  0.0137
##
## sigma^2 estimated as 75.05523 on 634 degrees of freedom
##
## AIC = 7.178399  AICc = 7.178547  BIC = 7.220276
##
```



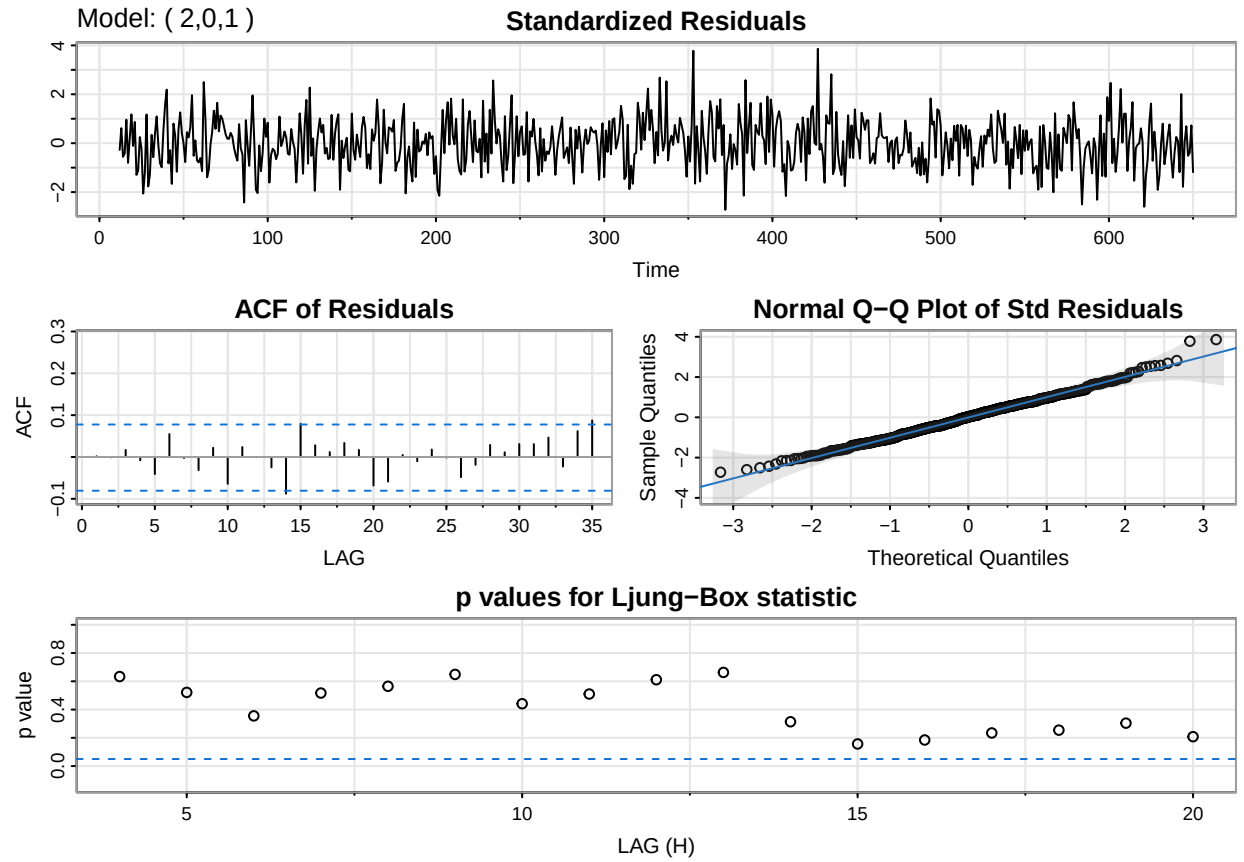
```
sarima(design_matrix[, 1],p=2,d=0,q=1,xreg = design_matrix[, 2:3],no.constant=FALSE, details=TRUE)
```

```
## initial  value 2.922620
## iter    2 value 2.468611
## iter    3 value 2.388819
## iter    4 value 2.273417
## iter    5 value 2.214691
## iter    6 value 2.181330
## iter    7 value 2.166095
## iter    8 value 2.163201
## iter    9 value 2.161562
## iter   10 value 2.160735
## iter   11 value 2.160568
## iter   12 value 2.160561
## iter   13 value 2.160560
## iter   14 value 2.160560
## iter   14 value 2.160560
## final   value 2.160560
```

```

## converged
## initial value 2.160782
## iter 2 value 2.160782
## iter 3 value 2.160779
## iter 4 value 2.160779
## iter 5 value 2.160778
## iter 6 value 2.160778
## iter 7 value 2.160778
## iter 8 value 2.160778
## iter 9 value 2.160777
## iter 10 value 2.160777
## iter 11 value 2.160777
## iter 11 value 2.160777
## iter 11 value 2.160777
## final value 2.160777
## converged
## <><><><><><><><><><><><><>
##
## Coefficients:
##          Estimate      SE t.value p.value
## ar1          0.7046 0.0570 12.3675 0.0000
## ar2         -0.0191 0.0548 -0.3485 0.7276
## ma1          0.7280 0.0419 17.3931 0.0000
## intercept  94.3852 2.5573 36.9086 0.0000
## XL8         -0.2184 0.0988 -2.2096 0.0275
## XL11         0.2394 0.0957 2.5022 0.0126
##
## sigma^2 estimated as 75.04118 on 633 degrees of freedom
##
## AIC = 7.18134  AICc = 7.181548  BIC = 7.230197
##

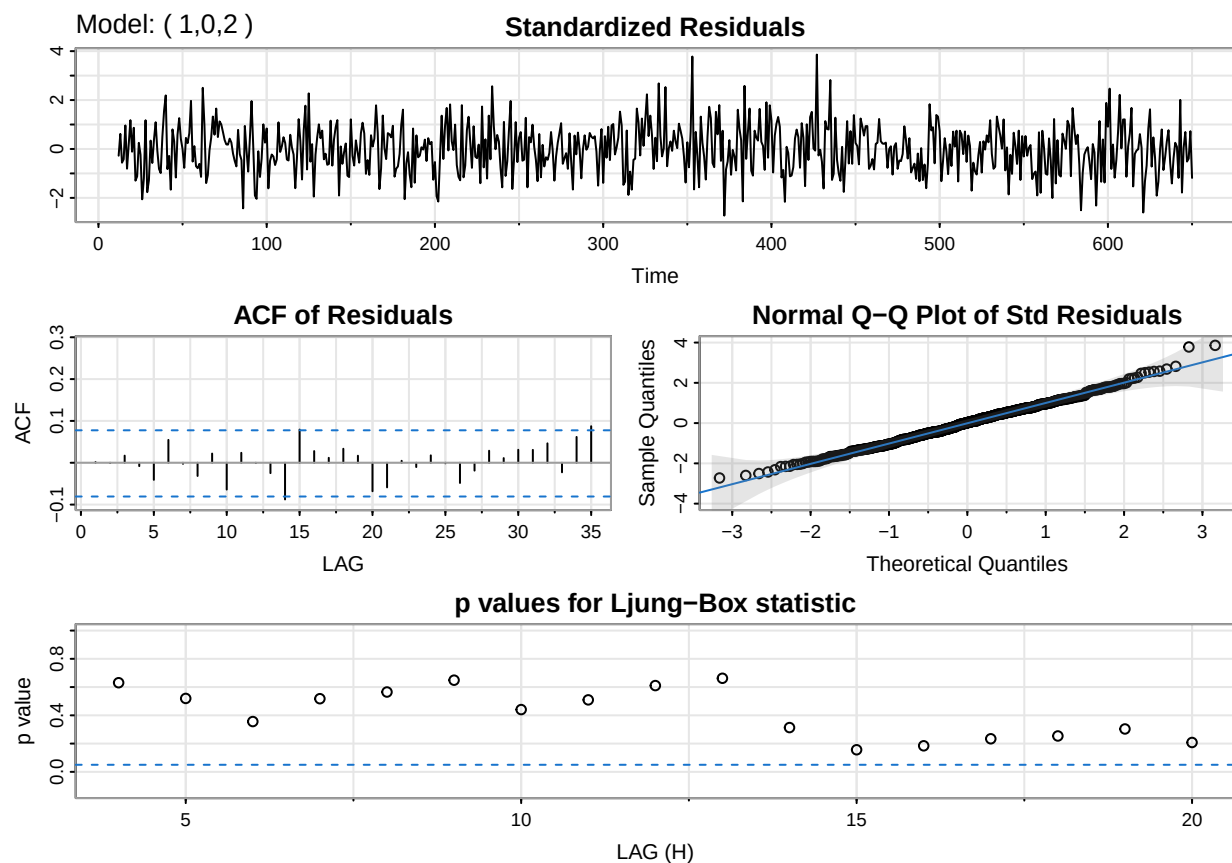
```

```
sarima(design_matrix[, 1],p=1,d=0,q=2,xreg = design_matrix[, 2:3],no.constant=FALSE, details=TRUE)
```

```
## initial value 2.921839
## iter 2 value 2.524236
## iter 3 value 2.195817
## iter 4 value 2.183718
## iter 5 value 2.161653
## iter 6 value 2.160261
## iter 7 value 2.159812
## iter 8 value 2.159776
## iter 9 value 2.159766
## iter 10 value 2.159761
## iter 11 value 2.159756
## iter 12 value 2.159755
## iter 13 value 2.159755
## iter 13 value 2.159755
## iter 13 value 2.159755
## final value 2.159755
## converged
## initial value 2.160780
## iter 2 value 2.160779
## iter 3 value 2.160778
## iter 4 value 2.160778
## iter 5 value 2.160777
## iter 6 value 2.160777
## iter 7 value 2.160776
```

```
## iter    8 value 2.160776
## iter    9 value 2.160776
## iter    9 value 2.160776
## iter    9 value 2.160776
## final   value 2.160776
## converged
## <><><><><><><><><><><><><>
##
## Coefficients:
##           Estimate      SE t.value p.value
## ar1           0.6767 0.0450 15.0255 0.0000
## ma1           0.7559 0.0598 12.6356 0.0000
## ma2           0.0207 0.0595  0.3478 0.7281
## intercept    94.3923 2.5583 36.8959 0.0000
## XL8          -0.2190 0.0992 -2.2081 0.0276
## XL11          0.2395 0.0957  2.5029 0.0126
##
## sigma^2 estimated as 75.04103 on 633 degrees of freedom
##
## AIC = 7.181338  AICc = 7.181546  BIC = 7.230194
##
```



```
sarima(design_matrix[, 1],p=5,d=0,q=3,xreg = design_matrix[, 2:3],no.constant=FALSE, details=TRUE)
```

```
## initial value 2.924892
## iter    2 value 2.610293
## iter    3 value 2.511172
```

```
## iter    4 value 2.416962
## iter    5 value 2.233950
## iter    6 value 2.199928
## iter    7 value 2.179031
## iter    8 value 2.173774
## iter    9 value 2.169595
## iter   10 value 2.166325
## iter   11 value 2.162838
## iter   12 value 2.162124
## iter   13 value 2.161884
## iter   14 value 2.161368
## iter   15 value 2.160803
## iter   16 value 2.160105
## iter   17 value 2.159781
## iter   18 value 2.159662
## iter   19 value 2.159539
## iter   20 value 2.159358
## iter   21 value 2.158944
## iter   22 value 2.158286
## iter   23 value 2.157743
## iter   24 value 2.157393
## iter   25 value 2.157358
## iter   26 value 2.157297
## iter   27 value 2.157243
## iter   28 value 2.157228
## iter   29 value 2.157193
## iter   30 value 2.157172
## iter   31 value 2.157165
## iter   32 value 2.157160
## iter   33 value 2.157117
## iter   34 value 2.157049
## iter   35 value 2.156924
## iter   36 value 2.156678
## iter   37 value 2.156618
## iter   38 value 2.156546
## iter   39 value 2.156431
## iter   40 value 2.156380
## iter   41 value 2.156278
## iter   42 value 2.156105
## iter   43 value 2.155806
## iter   44 value 2.155187
## iter   45 value 2.155096
## iter   46 value 2.155033
## iter   47 value 2.155016
## iter   48 value 2.155013
## iter   49 value 2.155006
## iter   50 value 2.155005
## iter   51 value 2.154996
## iter   52 value 2.154987
## iter   53 value 2.154976
## iter   54 value 2.154969
## iter   55 value 2.154967
## iter   56 value 2.154963
## iter   57 value 2.154958
```

```
## iter 58 value 2.154952
## iter 59 value 2.154948
## iter 60 value 2.154947
## iter 61 value 2.154946
## iter 62 value 2.154946
## iter 63 value 2.154945
## iter 64 value 2.154945
## iter 65 value 2.154944
## iter 66 value 2.154943
## iter 67 value 2.154941
## iter 68 value 2.154940
## iter 69 value 2.154940
## iter 69 value 2.154940
## iter 69 value 2.154940
## final value 2.154940
## converged
## initial value 2.156601
## iter 2 value 2.156535
## iter 3 value 2.156508
## iter 4 value 2.156493
## iter 5 value 2.156444
## iter 6 value 2.156428
## iter 7 value 2.156412
## iter 8 value 2.156407
## iter 9 value 2.156403
## iter 10 value 2.156398
## iter 11 value 2.156387
## iter 12 value 2.156369
## iter 13 value 2.156351
## iter 14 value 2.156338
## iter 15 value 2.156332
## iter 16 value 2.156330
## iter 17 value 2.156328
## iter 18 value 2.156325
## iter 19 value 2.156320
## iter 20 value 2.156311
## iter 21 value 2.156298
## iter 22 value 2.156283
## iter 23 value 2.156265
## iter 24 value 2.156254
## iter 25 value 2.156253
## iter 26 value 2.156249
## iter 27 value 2.156239
## iter 28 value 2.156227
## iter 29 value 2.156196
## iter 30 value 2.156117
## iter 31 value 2.155985
## iter 32 value 2.155165
## iter 33 value 2.155161
## iter 34 value 2.154338
## iter 35 value 2.153685
## iter 36 value 2.153276
## iter 37 value 2.152791
## iter 38 value 2.152458
```

```

## iter 39 value 2.151867
## iter 40 value 2.151280
## iter 41 value 2.150219
## iter 42 value 2.149088
## iter 43 value 2.148391
## iter 44 value 2.147421
## iter 45 value 2.147187
## iter 46 value 2.146966
## iter 47 value 2.146791
## iter 48 value 2.146732
## iter 49 value 2.146723
## iter 50 value 2.146709
## iter 51 value 2.146676
## iter 52 value 2.146660
## iter 53 value 2.146649
## iter 54 value 2.146626
## iter 55 value 2.146606
## iter 56 value 2.146591
## iter 57 value 2.146588
## iter 58 value 2.146587
## iter 59 value 2.146587
## iter 60 value 2.146587
## iter 61 value 2.146587
## iter 62 value 2.146586
## iter 63 value 2.146586
## iter 64 value 2.146586
## iter 64 value 2.146586
## iter 64 value 2.146586
## final value 2.146586
## converged
## <><><><><><><><><><><><><><>
##
## Coefficients:
##           Estimate      SE t.value p.value
## ar1           0.0345 0.0799  0.4313 0.6664
## ar2          -0.5043 0.0648 -7.7866 0.0000
## ar3           0.6838 0.0623 10.9802 0.0000
## ar4          -0.0811 0.0697 -1.1633 0.2451
## ar5           0.0336 0.0669  0.5025 0.6155
## ma1           1.4076 0.0701 20.0723 0.0000
## ma2           1.4831 0.0513 28.8855 0.0000
## ma3           0.6993 0.0694 10.0758 0.0000
## intercept    92.3516 2.8548 32.3492 0.0000
## XL8          -0.1374 0.1268 -1.0834 0.2790
## XL11          0.3099 0.1011  3.0661 0.0023
##
## sigma^2 estimated as 72.72703 on 628 degrees of freedom
##
## AIC = 7.168607  AICc = 7.169266  BIC = 7.252361
##

```

